

Proof

Read the instruction word first — it tells you which of the four proofs to write. Every proof ends with a sentence.



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THE TOPIC ON ONE MAP

Proof

01 — Algebraic proof

$2n, 2n + 1$

consecutive

factor out

02 — Proof by exhaustion

odd / even

$3k, 3k + 1, 3k + 2$

a table of cases

03 — Counter-example

one example

show it fails

04 — Inequalities

complete the square

$(a - b)^2 \geq 0$

discriminant

THE DICTIONARY OF PROOF

even = $2n$ odd = $2n + 1$ consecutive = $n, n + 1$ multiple of k
 $= k(\dots)$

finish with a sentence: "... which is even, so the statement is true for all n "

01 Algebraic Proof: Write It in Symbols

Translate the words into algebra with the dictionary, simplify, then **factor out** the number you need. The last line is a sentence that repeats what you were asked to prove.

WHEN YOU SEE → YOU DO

"use algebra to prove", "for all integers n "

→ write in n , simplify, factor

"consecutive odd numbers"

→ $2n + 1, 2n + 3$

"is a multiple of 8"

→ get $8(\dots)$ — every term must share the 8

"is always even but never a multiple of 4"

→ $2(\text{odd})$: e.g. $4(\dots) + 2$

"prove C has no stationary points"

→ $\frac{dy}{dx}$ has negative discriminant, or is a square + positive

WORKED MODEL • JANUARY 2026

p and q are consecutive odd numbers, $p > q > 0$. Prove $p^2 - q^2$ is a multiple of 8.

1 $q = 2n + 1, p = 2n + 3$

dictionary

2 $p^2 - q^2 = (p - q)(p + q) = 2(4n + 4)$

difference of two squares

ANSWER $8(n + 1)$: a multiple of 8

× **TRAP** Using the same letter for two different odd numbers ($2n+1$ and $2n+1$). Consecutive odd numbers are $2n + 1$ and $2n + 3$.

IDEA 01

Practise: Algebraic Proof: Write It in Symbols

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM The dictionary

P2 · TOPIC 1 3 marks

Prove that the sum of two consecutive odd numbers is a multiple of 4.

Odd numbers: $2n + 1$ and $2n + 3$.

Q2 DR ESLAM Difference of squares

P2 · TOPIC 1 3 marks

Prove that the difference between the squares of two consecutive integers is always odd.

SELECTED PRACTICE

Q3 PRACTICE Practice · Easy

P2 · TOPIC 1 1 mark

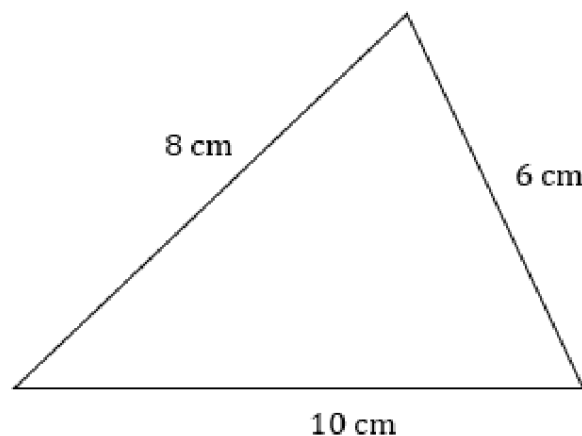
- 1 In a mathematical argument, how are three consecutive integers usually denoted algebraically?

(1 mark)

Q4 PRACTICE Practice · Medium

P2 · TOPIC 1 4 marks

- 9 Prove that a triangle with side lengths of 8 cm, 6 cm and 10 cm must contain a right-angle. You may use the diagram below to help.

Not to
scale

(4 marks)

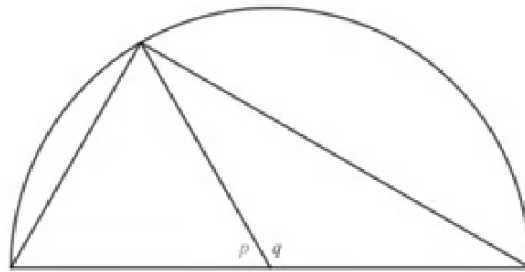
Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 1 4 marks

- 9 Prove the angle at the circumference in a semi-circle is a right angle.

You may use the diagram below to help.



Not to
scale

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · June 2021 · Q3

P2 · TOPIC 1 5 marks

3. (i) Prove that for all single digit prime numbers, p ,

$$p^3 + p \text{ is a multiple of } 10 \quad (2)$$

- (ii) Show, using algebra, that for $n \in \mathbb{N}$

$$(n + 1)^3 - n^3 \text{ is not a multiple of } 3 \quad (3)$$

(Total 5 marks)

8. (i) A student states

“If x and y are irrational numbers, $x \neq y$, then xy is also irrational.”

Show, by counter example, that this statement is not always true.

(1)

(ii) Prove, using algebra, that for all odd integers n , the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4

(4)

(Total for Question 8 is 5 marks)

Q8 PAST PAPER Pearson IAL WMA12 · January 2026 · Q6

P2 · TOPIC 1 8 marks

6. (i) Given that p and q are consecutive odd numbers, where $p > q > 0$, prove that

$$p^2 - q^2$$

is a multiple of 8

(4)

- (ii) The curve C has equation

$$y = x^3 + 12x^2 + 49x + 2$$

Prove that C has no stationary points.

(4)

(Total for Question 6 is 8 marks)

02 Proof by Exhaustion: Every Case

Split *all* the possibilities into a small number of cases, and prove the statement in each one. Say at the start why your cases cover everything, and at the end that every case works.

WHEN YOU SEE → YOU DO

“use proof by exhaustion”, “if n is an integer”

→ cases $n = 2k$ and $n = 2k + 1$

“not divisible by 3”

→ cases $3k, 3k + 1, 3k + 2$ — show the remainder is never 0

“not a multiple of 4”

→ cases $4k, 4k + 1, 4k + 2, 4k + 3$

“integers greater than 0 such that $a + b + c = 10$ ”

→ list every triple in a table

“for all single digit prime numbers”

→ check 2, 3, 5, 7 one by one

ZONE PARITY

2 cases

$2k \quad 2k + 1$

ZONE MOD 3

3 cases

$3k \quad 3k + 1 \quad 3k + 2$

ZONE A LIST

finite set

a table: check each row

WORKED MODEL • JANUARY 2023

Prove that $2n^2 + n + 1$ is not divisible by 3. Case $n = 3k + 1$:

$$1 \quad 2(9k^2 + 6k + 1) + (3k + 1) + 1 = 18k^2 + 15k + 4$$

substitute

$$2 \quad = 3(6k^2 + 5k + 1) + 1$$

a multiple of 3, plus 1

ANSWER remainder 1, so not divisible by 3

× **TRAP** Forgetting a case, or stopping without the concluding sentence “all cases are covered”.

IDEA 02

Practise: Proof by Exhaustion: Every Case

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Odd and even cases

P2 · TOPIC 1 4 marks

Prove that $n^2 + n$ is even for every integer n .

Case $n = 2k$ and case $n = 2k + 1$.

Q10 DR ESLAM A small list

P2 · TOPIC 1 3 marks

Prove by exhaustion that $n^2 + 1$ is not divisible by 3 for $n = 1, 2, 3, 4, 5$.

SELECTED PRACTICE

Q11 **PRACTICE** Practice · Easy

P2 · TOPIC 1 2 marks

7 Prove by exhausting all possible factors that 11 is a prime number.

(2 marks)**Q12** **PRACTICE** Practice · Hard

P2 · TOPIC 1 4 marks

4 The set of numbers S is defined as all positive integers greater than 5 and less than 10.

Prove by exhaustion that the square of all values in S differ from a multiple of 5 by 1.

(4 marks)

Q13**PRACTICE**

Practice · Very Hard

P2 · TOPIC 1 6 marks

- 10** A standard chess board has 64 1×1 - sized squares.
It also has 1 8×8 - sized square.

Prove that there are 204 squares on a standard chess board.

(6 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA12 · January 2022 · Q10

P2 · TOPIC 1 5 marks

10. (i) Prove by counter example that the statement

“if p is a prime number then $2p + 1$ is also a prime number”

is not true.

(1)

- (ii) Use proof by exhaustion to prove that if n is an integer then

$$5n^2 + n + 12$$

is always even.

(4)

Q15 PAST PAPER Pearson IAL WMA12 · October 2022 · Q1

P2 · TOPIC 1 3 marks

1. Given that a , b and c are integers greater than 0 such that

- $c = b + 2$
- $a + b + c = 10$

Prove, by exhaustion, that the product of a , b and c is always even.

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

a	b	c
	1	
	2	

(Total for Question 1 is 3 marks)

Q16 **PAST PAPER** Pearson IAL WMA12 · January 2023 · Q10**P2 · TOPIC 1** **4 marks**

10. A student was asked to prove by exhaustion that

if n is an integer then $2n^2 + n + 1$ is **not** divisible by 3

The start of the student's proof is shown in the box below.

Consider the case when $n = 3k$

$$2n^2 + n + 1 = 18k^2 + 3k + 1 = 3(6k^2 + k) + 1$$

which is not divisible by 3

Complete this proof.

(4)

(Total for Question 10 is 4 marks)

Q17 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q8**P2 · TOPIC 1** **5 marks**

8. (i) A student writes the following statement:

“When a and b are consecutive **prime** numbers, $a^2 + b^2$ is never a multiple of 10”

Prove by counter example that this statement is **not** true.

(2)

- (ii) Given that x and y are even integers greater than 0 and less than 6, prove by exhaustion, that

$$1 < x^2 - \frac{xy}{4} < 15$$

(3)

(Total for Question 8 is 5 marks)

Q18 PAST PAPER Pearson IAL WMA12 · October 2023 · Q1

P2 · TOPIC 1 3 marks

1. Given that a , b and c are integers greater than 0 such that

- $c = 3a + 1$
- $a + b + c = 15$

prove, by exhaustion, that the product abc is always a multiple of 4

You may use the table below to illustrate your answer.

(3)

You may not need to use all rows of this table.

	a	b	c	abc
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(Total for Question 1 is 3 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q8

P2 · TOPIC 1 6 marks

8. (i) Use a counter example to show that the following statement is **false**

“ $n^2 + 3n + 1$ is prime for all $n \in \mathbb{N}$ ”

(2)

- (ii) Use algebra to prove by exhaustion that for all $n \in \mathbb{N}$

“ $n^2 - 2$ is **not** a multiple of 4”

(4)

(Total for Question 8 is 6 marks)

Q20 **PAST PAPER** Pearson IAL WMA12 · October 2024 · Q11**P2 · TOPIC 1** **6 marks**

11. (i) Prove by counter example that the statement

“If n is a prime number then $3^n + 2$ is also a prime number.”

is false.

(2)

(ii) Use proof by exhaustion to prove that if m is an integer that is **not** divisible by 3, then

$$m^2 - 1$$

is divisible by 3

(4)

(Total for Question 11 is 6 marks)

Q21 **PAST PAPER** Pearson IAL WMA12 · June 2025 · Q10**P2 · TOPIC 1** **4 marks****10:** **In this question you must show detailed reasoning.**

Use algebra to prove by exhaustion that,

for all positive integers m that are **not** multiples of 3, the value of

$$m^2 + 3m + 2$$

is always a multiple of 3

(4)**(Total for Question 10 is 4 marks)**

03 Disproof by Counter-Example

To show a statement is **false**, one example is enough. Choose the value, work it out, and say clearly why it breaks the statement.

WHEN YOU SEE → YOU DO

“show that the statement is not true”

→ one value that fails

“is prime for all n ”

→ try small n until you find a composite, show its factors

“is a perfect cube for all n ”

→ find a value between two cubes

“for all real x ” with a square root or a sign

→ try a negative number

“If x and y are irrational then xy is irrational”

→ $\sqrt{2} \times \frac{1}{\sqrt{2}} = 1$

WORKED MODEL • JUNE 2022

Show that “ $(n + 1)^3 - n^3$ is prime for all $n \in \mathbb{N}$ ” is false.

1 $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$

simplify

2 $n = 5: 75 + 15 + 1 = 91$

try values

ANSWER $91 = 7 \times 13$, not prime

× **TRAP** Giving the example without the reason. Write “ $91 = 7 \times 13$, so it is not prime”.

IDEA 03

Practise: Disproof by Counter-Example

My questions first, then selected practice, then real past papers.

Q22 DR ESLAM One example is enough

P2 · TOPIC 1 2 marks

Show that the statement “ $n^2 - n + 11$ is prime for all positive integers n ” is false.

SELECTED PRACTICE**Q23** PRACTICE Practice · Easy

P2 · TOPIC 1 2 marks

6 Use a counter-example to show that $\sqrt{x^2} \neq x$.

(2 marks)

Q24 **PRACTICE** Practice · Hard

P2 · TOPIC 1 2 marks

- 5 Use a counter-example to prove that not all integers of the form $2^n - 1$, where n is an integer, are prime.

(2 marks)

Q25 **PRACTICE** Practice · Very Hard

P2 · TOPIC 1 2 marks

- 5 Give an example to show when the following statement is both true and false.

The square of a positive integer is always greater than doubling it.

(2 marks)

PAST PAPERS

Q26 PAST PAPER Pearson IAL WMA12 · June 2022 · Q3

P2 · TOPIC 1 7 marks

3. (i) Show that the following statement is **false**:

$$“(n + 1)^3 - n^3 \text{ is prime for all } n \in \mathbb{N}” \quad (2)$$

- (ii) Given that the points $A(1, 0)$, $B(3, -10)$ and $C(7, -6)$ lie on a circle, prove that AB is a diameter of this circle. (5)

(Total 7 marks)

04 Proving Inequalities: Complete the Square

Move everything to one side so you need to show “something > 0 ” or “ ≥ 0 ”. Then complete the square: a square is never negative. For two variables, use $(a - b)^2 \geq 0$.

WHEN YOU SEE → YOU DO

“prove $(x + 1)^2 > 6x - 4$ for all real x ”

→ one side: $x^2 - 4x + 5 = (x - 2)^2 + 1 > 0$

“ $\frac{x+y}{2} \geq \sqrt{xy}$ ”

→ $(\sqrt{x} - \sqrt{y})^2 \geq 0$, then expand

“ $3x + 1 \geq 2\sqrt{3x}$ ”

→ $(\sqrt{3x} - 1)^2 \geq 0$

“for all positive x and y ”

→ square roots are allowed only because they are positive

“show that it is not true for all real numbers”

→ counter-example with a negative

Everything on one side

Complete the square

Square ≥ 0

Conclusion

WORKED MODEL • JUNE 2026

Prove that $(x + 1)^2 > 6x - 4$ for all real x .

1 $x^2 + 2x + 1 - 6x + 4 = x^2 - 4x + 5$

one side

2 $= (x - 2)^2 + 1$

complete the square

ANSWER $(x - 2)^2 \geq 0$, so $(x - 2)^2 + 1 \geq 1 > 0$

IDEA 04

Practise: Proving Inequalities: Complete the Square

My questions first, then selected practice, then real past papers.

Q27 **DR ESLAM** Complete the square

P2 · TOPIC 1 3 marks

Prove that $x^2 - 6x + 10 > 0$ for all real x .

Q28 **DR ESLAM** Two variables

P2 · TOPIC 1 3 marks

Prove that $a^2 + b^2 \geq 2ab$ for all real a and b .

SELECTED PRACTICE

Q29

PRACTICE

Practice · Easy

P2 · TOPIC 1 1 mark

4 Explain why $(x - 3)^2 \geq 0$ for all real values of x .

(1 mark)

Q30

PRACTICE

Practice · Medium

P2 · TOPIC 1 2 marks

2 Prove that $x^2 + 2 \geq 2$ for all values of x .

(2 marks)

Q31**PRACTICE**

Practice · Very Hard

P2 · TOPIC 1**4 marks**

7 Prove that, if n is negative, $n^4 - n^3$ is positive.

(4 marks)

PAST PAPERS

Q32**PAST PAPER**

Pearson IAL WMA12 · January 2021 · Q5

P2 · TOPIC 1**4 marks**

5. (i) Use algebra to prove that for all $x \geq 0$

$$3x + 1 \geq 2\sqrt{3x} \quad (3)$$

(ii) Show that the following statement is not true.

“The sum of three consecutive prime numbers is always a multiple of 5”
(1)

(Total 4 marks)

Q33 **PAST PAPER** Pearson IAL WMA12 · October 2021 · Q9**P2 · TOPIC 1** **4 marks**

9. (a) Prove that for all positive values of x and y ,

$$\frac{x + y}{2} \geq \sqrt{xy} \quad (3)$$

- (b) Prove by counter-example that this inequality does not hold when x and y are both negative.

(1)

(Total 4 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June 2024 · Q5

P2 · TOPIC 1 6 marks

5. In this question you must show detailed reasoning.

- (a) Given that x and y are positive numbers such that

$$(x - y)^3 > x^3 - y^3$$

prove that

$$y > x$$

(4)

- (b) Using a counter example, show that the result in part (a) is not true for all real numbers.

(2)

(Total for Question 5 is 6 marks)

Q35**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q4

P2 · TOPIC 1 6 marks

4. **In this question you must show detailed reasoning.**

(i) Show by counterexample that the statement

$$\text{“}f(n) = \frac{5}{2}n^3 - \frac{5}{2}n^2 - 19n + 20 \text{ is a perfect cube for all positive integers } n\text{”}$$

is **not** true.

(3)

(ii) Prove that the following statement is true

$$\text{“}(x + 1)^2 > 6x - 4 \text{ for all real } x\text{”}$$

(3)

(Total for Question 4 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|---|
| 01 | Algebraic | Dictionary; simplify; factor out; sentence. |
| 02 | Exhaustion | Cases that cover everything; prove each; “all cases covered”. |
| 03 | Counter-example | One value; show exactly why it fails. |
| 04 | Inequalities | One side; complete the square; a square ≥ 0 . |

BOTTOM LINE Read the instruction word → choose the proof → write it in symbols → end with a sentence.

Answers

Proof • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ALGEBRAIC PROOF: WRITE IT IN SYMBOLS

- Q1** Dr Eslam • The dictionary $(2n + 1) + (2n + 3) = 4n + 4 = 4(n + 1)$, a multiple of 4
- Q2** Dr Eslam • Difference of squares $(n + 1)^2 - n^2 = 2n + 1$, which is odd
- Q3** Practice • Easy $n, n + 1, n + 2$
- Q4** Practice • Medium $6^2 + 8^2 = 100 = 10^2$, so by the converse of Pythagoras the angle opposite the 10 cm side is 90°
- Q5** Practice • Very Hard join the centre to the point on the circle: two isosceles triangles, $2\alpha + 2\beta = 180^\circ$, so $\alpha + \beta = 90^\circ$
- Q6** Pearson IAL WMA12 • June 2021 • Q3 (i) Proven by checking $p = 2, 3, 5, 7$. (ii) Proven: $(n + 1)^3 - n^3$ leaves remainder 1 on division by 3
- Q7** Pearson IAL WMA12 • January 2025 • Q8 (i) For example, $x = \sqrt{2}$ and $y = \frac{1}{\sqrt{2}}$ give $xy = 1$. (ii) Proven
- Q8** Pearson IAL WMA12 • January 2026 • Q6 (i) Proven. (ii) C has no stationary points

IDEA 02 • PROOF BY EXHAUSTION: EVERY CASE

- Q9** Dr Eslam • Odd and even cases
 $n = 2k$: $4k^2 + 2k = 2(2k^2 + k)$; $n = 2k + 1$:
 $4k^2 + 6k + 2 = 2(2k^2 + 3k + 1)$ — even in both cases
- Q10** Dr Eslam • A small list
 2, 5, 10, 17, 26: none is a multiple of 3
- Q11** Practice • Easy
 11 is not divisible by 2 or 3 (the primes up to $\sqrt{11}$), so it is prime
- Q12** Practice • Hard
 $36 = 35 + 1$, $49 = 50 - 1$, $64 = 65 - 1$, $81 = 80 + 1$
- Q13** Practice • Very Hard
 there are $(9 - k)^2$ squares of size $k \times k$: $64 + 49 + 36 + 25 + 16 + 9 + 4 + 1 = 204$
- Q14** Pearson IAL WMA12 • January 2022 • Q10
 (i) For $p = 7$, $2p + 1 = 15$, not prime. (ii) Proven
- Q15** Pearson IAL WMA12 • October 2022 • Q1
 The possible triples are $(6, 1, 3)$, $(4, 2, 4)$ and $(2, 3, 5)$, giving products 18, 32 and 30. Each product is even, so abc is always even
- Q16** Pearson IAL WMA12 • January 2023 • Q10
 For $n = 3k$, $n = 3k + 1$ and $n = 3k + 2$, the expression has remainders 1, 1 and 2 respectively when divided by 3. Therefore $2n^2 + n + 1$ is never divisible by 3
- Q17** Pearson IAL WMA12 • June 2023 • Q8
 (i) A counterexample is $a = 7$, $b = 11$, since $7^2 + 11^2 = 170$. (ii) Exhaustion proves the inequality for $(x, y) = (2, 2), (2, 4), (4, 2), (4, 4)$
- Q18** Pearson IAL WMA12 • October 2023 • Q1
 The only possible triples are $(1, 10, 4)$, $(2, 6, 7)$ and $(3, 2, 10)$, giving products 40, 84 and 60. Each is a multiple of 4, so abc is always a multiple of 4
- Q19** Pearson IAL WMA12 • January 2024 • Q8
 (i) Counterexample: $n = 6$ gives 55, which is not prime. (ii) Proven by considering $n = 4k, 4k + 1, 4k + 2, 4k + 3$
- Q20** Pearson IAL WMA12 • October 2024 • Q11
 (i) Counterexample: $n = 5$ gives $3^5 + 2 = 245$, not prime. (ii) Proven
- Q21** Pearson IAL WMA12 • June 2025 • Q10
 Proven

IDEA 03 · DISPROOF BY COUNTER-EXAMPLE

- Q22** Dr Eslam · One example is enough
 $n = 11: 121 = 11^2$, not prime
- Q23** Practice · Easy
 $x = -2: \sqrt{(-2)^2} = 2 \neq -2$
- Q24** Practice · Hard
 $n = 4: 2^4 - 1 = 15 = 3 \times 5$, not prime
- Q25** Practice · Very Hard
 true for $n = 3$ ($9 > 6$); false for $n = 1$ ($1 < 2$) or $n = 2$ ($4 = 4$)
- Q26** Pearson IAL WMA12 · June 2022 · Q3
 (i) Counterexample: $n = 5$ gives 91, not prime. (ii) Proven: AB is a diameter

IDEA 04 · PROVING INEQUALITIES: COMPLETE THE SQUARE

- Q27** Dr Eslam · Complete the square
 $(x - 3)^2 + 1 \geq 1 > 0$
- Q28** Dr Eslam · Two variables
 $a^2 + b^2 - 2ab = (a - b)^2 \geq 0$
- Q29** Practice · Easy
 a square is never negative, so $(x - 3)^2 \geq 0$
- Q30** Practice · Medium
 $x^2 \geq 0$, so $x^2 + 2 \geq 2$
- Q31** Practice · Very Hard
 $n^4 - n^3 = n^3(n - 1)$; for $n < 0$ both $n^3 < 0$ and $n - 1 < 0$, so the product is positive
- Q32** Pearson IAL WMA12 · January 2021 · Q5
 (i) Proven. (ii) For example, $5 + 7 + 11 = 23$, which is not a multiple of 5
- Q33** Pearson IAL WMA12 · October 2021 · Q9
 (a) Proven. (b) For example, $x = y = -1$ gives $-1 \not\geq 1$
- Q34** Pearson IAL WMA12 · June 2024 · Q5
 (a) Proven. (b) For example, $x = 1, y = -1$ satisfies the inequality but not $y > x$
- Q35** Pearson IAL WMA12 · June 2026 · Q4
 (i) $f(5) = 175$; $5^3 = 125 < 175 < 216 = 6^3$, so not a cube (ii) $(x+1)^2 - (6x-4) = (x-2)^2 + 1 > 0$

Polynomials: Division, Remainder and Factor Theorems



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To divide by $(x - a)$, put $x = a$ in: the answer is the remainder. If it is zero, $(x - a)$ is a factor — divide it out and factorise the quadratic that is left.

THE TOPIC ON ONE MAP

Polynomials: Division, Remainder and Factor Theorems

01 — Algebraic division

long division

quotient, remainder

missing terms $0x^2$

02 — Remainder theorem

$f(a)$

$f(\frac{b}{a})$ for $(ax - b)$

two equations

03 — Factor theorem

$f(a) = 0$

factorise fully

how many real roots?

ONE NUMBER DOES EVERYTHING

divide by $(x - a) \Rightarrow$ remainder $= f(a)$ divide by
 $(ax - b) \Rightarrow$ remainder $= f\left(\frac{b}{a}\right)$
 remainder 0 $\iff$ factor

01 Algebraic Division: Quotient and Remainder

Long division, one term at a time: divide the first terms, multiply back, subtract, bring down. Write every missing power as $0x^k$ first. The result is $f(x) = (\text{divisor}) \times Q(x) + R$.

WHEN YOU SEE $\rightarrow$ YOU DO

“find the quotient when $f(x)$ is divided by $(x - 2)$ ”

$\rightarrow$ long division (or compare coefficients)

“in the form $(x - 2)(ax^2 + bx + c) + d$ ”

$\rightarrow$ quotient gives a, b, c ; remainder is d

a missing term, e.g. $x^3 - 19x - 30$

$\rightarrow$ write $x^3 + 0x^2 - 19x - 30$

“hence, or otherwise”

$\rightarrow$ the remainder theorem gives R in one line

WORKED MODEL · JUNE 2026

$f(x) = 2x^3 - 7x^2 + 5x + 4$. Remainder and quotient when divided by $(x - 2)$.

1 $f(2) = 16 - 28 + 10 + 4 = 2$

remainder theorem

2 $2x^3 - 7x^2 + 5x + 4 = (x - 2)(2x^2 - 3x - 1) + 2$

long division

ANSWER $Q(x) = 2x^2 - 3x - 1, \quad R = 2$

IDEA 01

Practise: Algebraic Division: Quotient and Remainder

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Long division

P2 · TOPIC 2 3 marks

Divide $2x^3 + 3x^2 - 5x + 6$ by $(x + 2)$. State the quotient and the remainder.

Q2 DR ESLAM Write in the form

P2 · TOPIC 2 3 marks

Write $x^3 - 7x + 3$ in the form $(x - 2)(ax^2 + bx + c) + d$.

A missing x^2 term: write it as $0x^2$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 2 4 marks

2 Divide $x^3 + 6x^2 + 11x + 6$ by $(x + 2)$.

(4 marks)**Q4****PRACTICE**

Practice · Medium

P2 · TOPIC 2 5 marks

6 Consider the function $f(x) = 4x^3 + 6x^2 - 7x + 2$.

(i) Find the quotient and remainder when $4x^3 + 6x^2 - 7x + 2$ is divided by $(x - 2)$.

(ii) Hence write $4x^3 + 6x^2 - 7x + 2$ in the form $(x - 2)(ax^2 + bx + c) + d$, where a , b , c and d are constants to be determined.

(5 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 2 3 marks

1 Divide $4x^4 - 37x^2 + 9$ by $(2x - 1)$.

(3 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · January 2026 · Q3

P2 · TOPIC 2 7 marks

3. (i) Given that

- $f(x) = 4x^3 + 6x + k$, where k is a constant
- $(x + 2)$ is a factor of $f(x)$

find the remainder when $f(x)$ is divided by $(x - 5)$

(4)

(ii) Find the remainder, R , and the quotient, $Q(x)$, when

$$6x^3 - 15x^2 - 21x + 8$$

is divided by $(2x + 3)$

(3)

(Total for Question 3 is 7 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q1

P2 · TOPIC 2 4 marks**1.**

$$f(x) = 2x^3 - 7x^2 + 5x + 4$$

- (a) Use the remainder theorem to find the remainder when $f(x)$ is divided by $(x - 2)$ (2)
- (b) Hence, or otherwise, find the quotient when $f(x)$ is divided by $(x - 2)$ (2)

(Total for Question 1 is 4 marks)

02 The Remainder Theorem: Find the Unknowns

Each fact about a remainder or a factor gives one equation: put the root of the divisor into $f(x)$ and set it equal to the remainder. Two unknowns need two facts — then solve simultaneously.

WHEN YOU SEE → YOU DO

“when divided by $(x + 1)$ the remainder is 4”

$$\rightarrow f(-1) = 4$$

“ $(3x - 2)$ is a factor”

$$\rightarrow f\left(\frac{2}{3}\right) = 0$$

“show that $11A + 9B = 83$ ”

→ one fact → tidy into the given equation

“find the value of A and the value of B ”

→ two equations, solve simultaneously

“the remainder is k ” and “ $-10k$ ”

→ eliminate k between the two

Root of the divisor

Substitute into f

Set = remainder

Solve simultaneously

WORKED MODEL · JANUARY 2021

$f(x) = x^4 + ax^3 - 3x^2 + bx + 5$. Remainder 4 on $(x + 1)$ and -23 on $(x - 2)$. Find a and b .

$$1 \quad f(-1) = 1 - a - 3 - b + 5 = 4 \Rightarrow a + b = -1$$

first fact

$$2 \quad f(2) = 16 + 8a - 12 + 2b + 5 = -23 \Rightarrow 4a + b = -16$$

second fact

ANSWER $a = -5, \quad b = 4$

× **TRAP** Using $f(3)$ for the factor $(2x - 3)$. The root of $2x - 3 = 0$ is $x = \frac{3}{2}$.

IDEA 02

Practise: The Remainder Theorem: Find the Unknowns

My questions first, then selected practice, then real past papers.

Q8 **DR ESLAM** Put the root in

P2 · TOPIC 2 2 marks

Find the remainder when $x^3 - 4x + 7$ is divided by (a) $(x - 3)$ (b) $(2x - 1)$.

Q9 **DR ESLAM** Two facts, two equations

P2 · TOPIC 2 5 marks

$f(x) = x^3 + ax^2 + bx - 6$. $(x - 1)$ is a factor, and the remainder on division by $(x - 2)$ is 18. Find a and b .

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 2 3 marks

6 Given that $(x - 4)$ is a factor of $x^3 - kx^2 - 4x + 16$, find the value of k .

(3 marks)

Q11**PRACTICE**

Practice · Medium

P2 · TOPIC 2 9 marks

5 (a) $f(x) = x^3 + 9x^2 + rx + s$. Given that $f(2) = 0$ and $f(-1) = -54$:
find the values of r and s .

(6 marks)

(b) Factorise $f(x)$ completely.

(3 marks)

Q12**PRACTICE**

Practice · Hard

P2 · TOPIC 2**6 marks**

- 6 Given that $(x + 4)$ is a factor of the function $f(x) = px^3 + (5p + 1)x^2 + 5qx - 2q - 2$ and that the remainder when $f(x)$ is divided by $(x + 1)$ is -12 , find the values of the constants p and q .

(6 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2021 · Q1

P2 · TOPIC 2 6 marks

1. $f(x) = x^4 + ax^3 - 3x^2 + bx + 5$

where a and b are constants.

When $f(x)$ is divided by $(x + 1)$, the remainder is 4

(a) Show that $a + b = -1$

(2)

When $f(x)$ is divided by $(x - 2)$, the remainder is -23

(b) Find the value of a and the value of b .

(4)

(Total 6 marks)

Q14 PAST PAPER Pearson IAL WMA12 · January 2022 · Q5

P2 · TOPIC 2 8 marks

5.
$$f(x) = 3x^3 + Ax^2 + Bx - 10$$

where A and B are integers.

Given that

- when $f(x)$ is divided by $(x - 1)$ the remainder is k
- when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
- k is a constant

(a) show that

$$11A + 9B = 83 \quad (3)$$

Given also that $(3x - 2)$ is a factor of $f(x)$,

(b) find the value of A and the value of B . (3)

(c) Hence find the quadratic expression $g(x)$ such that

$$f(x) = (3x - 2)g(x) \quad (2)$$

(Total 8 marks)

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q1

P2 · TOPIC 2 3 marks

1. $f(x) = ax^3 + 3x^2 - 8x + 2$ where a is a constant

Given that when $f(x)$ is divided by $(x - 2)$ the remainder is 3, find the value of a .

(3)

(Total for Question 1 is 3 marks)

Q16**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q3

P2 · TOPIC 2**7 marks****3.**

$$f(x) = 2x^3 - x^2 + Ax + B$$

where A and B are integers.

Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

(a) show that

$$3A - B = -118$$

(2)

Given also that $(2x - 5)$ is a factor of $f(x)$,

(b) find the value of A and the value of B .

(3)

(c) Hence find the quotient when $f(x)$ is divided by $(x - 7)$

(2)

(Total for Question 3 is 7 marks)

Q17 PAST PAPER Pearson IAL WMA12 · June 2025 · Q4

P2 · TOPIC 2 9 marks

4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

(a) show that

$$b = 3a + 2 \quad (2)$$

Given further that when $f(x)$ is divided by $(2x - 1)$ the remainder is $\frac{7}{4}$

(b) find the value of a and the value of b . (4)

(c) Using algebra, find the quotient and the remainder when $f(x)$ is divided by $(x - 2)$ (3)

(Total for Question 4 is 9 marks)

03 The Factor Theorem: Factorise Fully and Count the Roots

Show $f(a) = 0$, so $(x - a)$ is a factor. Divide it out to get a quadratic, then factorise that (calculator first, then write the brackets). If the quadratic has $b^2 - 4ac < 0$, it gives no more real roots.

WHEN YOU SEE → YOU DO

“use the factor theorem to show $(2x + 3)$ is a factor”

→ $f(-\frac{3}{2}) = 0$, and say so

“hence fully factorise $f(x)$ ”

→ divide, then factorise the quadratic

“show that $f(x) = 0$ has only one real root”

→ quadratic factor: $b^2 - 4ac < 0$

“solve $6 \tan^3 \theta + 17 \tan^2 \theta + \dots = 0$ ”

→ same cubic in $\tan \theta$: roots, then angles

“using algebra”, “not calculator technology”

→ show the division and the factorised form

WORKED MODEL · JUNE 2023

$f(x) = 4x^3 - 8x^2 + 5x - 3$ with factor $(2x - 3)$. Show $f(x) = 0$ has only one real root.

1 $f(x) = (2x - 3)(2x^2 - x + 1)$

divide

2 $2x^2 - x + 1$: $b^2 - 4ac = 1 - 8 = -7 < 0$

no real roots

ANSWER only real root $x = \frac{3}{2}$

IDEA 03

Practise: The Factor Theorem: Factorise Fully and Count the Roots

My questions first, then selected practice, then real past papers.

Q18 **DR ESLAM** Factor, then factorise

P2 · TOPIC 2 4 marks

Show that $(x + 3)$ is a factor of $2x^3 + 5x^2 - 4x - 3$ and hence factorise it fully.

Q19 **DR ESLAM** Only one real root

P2 · TOPIC 2 3 marks

Show that $(x - 2)$ is a factor of $x^3 - 2x^2 + 2x - 4$, and that $x^3 - 2x^2 + 2x - 4 = 0$ has only one real root.

SELECTED PRACTICE

Q20

PRACTICE

Practice · Easy

P2 · TOPIC 2 2 marks

- 1 Use the factor theorem to verify that $(x - 2)$ is a factor of $x^3 - x^2 - 14x + 24$.

(2 marks)

Q21

PRACTICE

Practice · Medium

P2 · TOPIC 2 8 marks

- 4 (a) $f(x) = 4x^3 + 4x^2 - 23x - 30$

Use the factor theorem to show that $(x + 2)$ is a factor of $f(x)$.

(2 marks)

- (b) Factorise $f(x)$ completely.

(4 marks)

- (c) Write down all the real roots of the equation $f(x) = 0$.

(2 marks)

Q22**PRACTICE**

Practice · Very Hard

P2 · TOPIC 2 8 marks

5 (a) $f(x) = 2x^4 - 15x^3 - 10x^2 + 105x + 98$

Show that $f(-1) = 0$ and $f(-2) = 0$.

(1 mark)

(b) Hence, solve $f(x) = 0$.

(7 marks)

PAST PAPERS

Q23 PAST PAPER Pearson IAL WMA12 · October 2019 · Q4

P2 · TOPIC 2 8 marks

4. $f(x) = (x - 3)(3x^2 + x + a) - 35$ where a is a constant

(a) State the remainder when $f(x)$ is divided by $(x - 3)$. (1)

Given $(3x - 2)$ is a factor of $f(x)$,

(b) show that $a = -17$ (2)

(c) Using algebra and showing each step of your working, fully factorise $f(x)$. (5)

(Total 8 marks)

Q24 PAST PAPER Pearson IAL WMA12 · January 2020 · Q3

P2 · TOPIC 2 8 marks

3. $f(x) = 6x^3 + 17x^2 + 4x - 12$

(a) Use the factor theorem to show that $(2x + 3)$ is a factor of $f(x)$. (2)

(b) Hence, using algebra, write $f(x)$ as a product of three linear factors. (4)

(c) Solve, for $\frac{\pi}{2} < \theta < \pi$, the equation

$$6 \tan^3 \theta + 17 \tan^2 \theta + 4 \tan \theta - 12 = 0$$

giving your answers to 3 significant figures. (2)

(Total 8 marks)

Q25**PAST PAPER**

Pearson IAL WMA12 · October 2021 · Q4

P2 · TOPIC 2 8 marks

4. $f(x) = (x^2 - 2)(2x - 3) - 21$

(a) State the value of the remainder when $f(x)$ is divided by $(2x - 3)$ (1)

(b) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$ (2)

(c) Hence,

(i) factorise $f(x)$

(ii) show that the equation $f(x) = 0$ has only one real root. (5)

(Total 8 marks)

Q26**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q5

P2 · TOPIC 2 9 marks

5.
$$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that

$$9p + q = -51 \quad (2)$$

Given also that when $f(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0 \quad (2)$$

(c) Hence find the value of p and the value of q . (3)

(d) Hence find a quadratic expression $g(x)$ such that

$$f(x) = (x - 3)g(x) \quad (2)$$

(Total for Question 5 is 9 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · June 2023 · Q2

P2 · TOPIC 2 6 marks

2. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where a is a constant.

Given that $(2x - 3)$ is a factor of $f(x)$,

(a) use the factor theorem to show that $a = -3$ (2)

(b) Hence show that the equation $f(x) = 0$ has only one real root. (4)

(Total for Question 2 is 6 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q4

P2 · TOPIC 2**9 marks**

4. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of $f(x)$,

(a) show that

$$a + 4b = -56 \quad (2)$$

Given also that when $f(x)$ is divided by $(x - 2)$ the remainder is -25

(b) find a second simplified equation linking a and b . (2)

(c) Hence, using algebra and showing your working,

(i) find the value of a and the value of b ,

(ii) fully factorise $f(x)$. (5)

(Total for Question 4 is 9 marks)

Q29**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q4

P2 · TOPIC 2 8 marks

4.
$$f(x) = (x - 2)(2x^2 + 5x + k) + 21$$

where k is a constant.

- (a) State the remainder when $f(x)$ is divided by $(x - 2)$ (1)

Given that $(2x - 1)$ is a factor of $f(x)$

- (b) show that $k = 11$ (2)

- (c) Hence

- (i) fully factorise $f(x)$,
(ii) find the number of real solutions of the equation

$$f(x) = 0$$

giving a reason for your answer. (5)

(Total for Question 4 is 8 marks)

Q30**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q5

P2 · TOPIC 2 9 marks**5.**

$$f(x) = 3x^3 + ax^2 - 10x + b$$

where a and b are constants.

Given that $(3x - 4)$ is a factor of $f(x)$,

(a) show that $16a + 9b = 56$

(2)

Given further that when $f(x)$ is divided by $(x - 2)$ the remainder is b ,

(b) find the value of a and the value of b .

(4)

(c) Hence, using algebra, fully factorise $f(x)$.

(3)

(Total for Question 5 is 9 marks)

Q31**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q4

P2 · TOPIC 2 12 marks

4. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$f(x) = 4x^3 + 13x^2 - 10x + 8$$

- (a) When $f(x)$ is divided by $(x - 2)$ the remainder is R and the quotient is $Q(x)$.

(i) Find $Q(x)$.

(ii) Find R .

(4)

- (b) (i) Use the factor theorem to show that $(x + 4)$ is a factor of $f(x)$.

(ii) Hence prove, using algebra, that the equation $f(x) = 0$ has only one real solution.

(5)

- (c) Find the range of values of x for which $f(x)$ is decreasing.

(3)

(Total for Question 4 is 12 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Division** $0x^k$ for gaps; $f = (\text{divisor})Q + R$.
- 02 Remainder theorem** Root of the divisor into f ; one fact = one equation.
- 03 Factor theorem** $f(a) = 0$; divide; factorise; $b^2 - 4ac < 0 \rightarrow$ no more roots.

BOTTOM LINE Root in $\rightarrow$ remainder $\cdot$ zero $\rightarrow$ factor $\rightarrow$ divide $\rightarrow$ factorise the quadratic.

Answers

Polynomials: Division, Remainder and Factor Theorems · for the teacher.
Every answer checked with a computer algebra system.

IDEA 01 · ALGEBRAIC DIVISION: QUOTIENT AND REMAINDER

- Q1** Dr Eslam · Long division quotient $2x^2 - x - 3$, remainder 12
- Q2** Dr Eslam · Write in the form $(x - 2)(x^2 + 2x - 3) - 3$: $a = 1$, $b = 2$, $c = -3$, $d = -3$
- Q3** Practice · Easy $x^2 + 4x + 3$ (remainder 0)
- Q4** Practice · Medium (i) quotient $4x^2 + 14x + 21$, remainder 44 (ii) $a = 4$, $b = 14$, $c = 21$, $d = 44$
- Q5** Practice · Very Hard $2x^3 + x^2 - 18x - 9$, remainder 0
- Q6** Pearson IAL WMA12 · January 2026 · Q3 (i) 574. (ii) $Q(x) = 3x^2 - 12x + \frac{15}{2}$, $R = -\frac{29}{2}$
- Q7** Pearson IAL WMA12 · June 2026 · Q1 (a) $f(2) = 2$ (b) $2x^2 - 3x - 1$

IDEA 02 · THE REMAINDER THEOREM: FIND THE UNKNOWN

- Q8** Dr Eslam · Put the root in (a) $f(3) = 22$ (b) $f(\frac{1}{2}) = \frac{41}{8}$
- Q9** Dr Eslam · Two facts, two equations $a + b = 5, 2a + b = 8: a = 3, b = 2$
- Q10** Practice · Easy $f(4) = 64 - 16k = 0 \Rightarrow k = 4$
- Q11** Practice · Medium (a) $r = 6, s = -56$ (b) $(x - 2)(x + 4)(x + 7)$
- Q12** Practice · Hard $f(-4) = 0: 8p - 11q = -7; f(-1) = -12: 4p - 7q = -11; p = 6, q = 5$
- Q13** Pearson IAL WMA12 · January 2021 · Q1 (a) Proven: $a + b = -1$. (b) $a = -5, b = 4$
- Q14** Pearson IAL WMA12 · January 2022 · Q5 (a) Proven: $11A + 9B = 83$. (b) $A = -8, B = 19$. (c) $g(x) = x^2 - 2x + 5$
- Q15** Pearson IAL WMA12 · January 2024 · Q1 $a = \frac{5}{8}$
- Q16** Pearson IAL WMA12 · October 2024 · Q3 (a) Proven. (b) $A = -26, B = 40$. (c) $2x^2 + 13x + 65$
- Q17** Pearson IAL WMA12 · June 2025 · Q4 (a) Proven. (b) $a = 2, b = 8$. (c) Quotient $2x^2 + 12x + 29$, remainder 55

IDEA 03 · THE FACTOR THEOREM: FACTORISE FULLY AND COUNT THE ROOTS

- Q18** Dr Eslam · Factor, then factorise
 $f(-3) = 0; (x + 3)(2x + 1)(x - 1)$
- Q19** Dr Eslam · Only one real root
 $(x - 2)(x^2 + 2); x^2 + 2$ has discriminant $-8 < 0$, so $x = 2$ only
 $f(2) = 8 - 4 - 28 + 24 = 0$, so $(x - 2)$ is a factor
- Q20** Practice · Easy
 (a) $f(-2) = 0$ (b) $(x + 2)(2x - 5)(2x + 3)$ (c)
 $x = -2, \frac{5}{2}, -\frac{3}{2}$
- Q21** Practice · Medium
 (a) both give 0 (b) $(x + 1)(x + 2)(x - 7)(2x - 7):$
 $x = -2, -1, \frac{7}{2}, 7$
- Q22** Practice · Very Hard
 (a) -35 . (b) $a = -17$. (c) $f(x) = (3x - 2)(x + 2)(x - 4)$
- Q23** Pearson IAL WMA12 · October 2019 · Q4
 (a) Proven. (b) $f(x) = (2x + 3)(3x - 2)(x + 2)$. (c) $\theta = 2.03, 2.16$ radians
- Q24** Pearson IAL WMA12 · January 2020 · Q3
 (a) -21 . (b) Proven. (c)(i) $f(x) = (x - 3)(2x^2 + 3x + 5)$. (c)(ii) Only real root: $x = 3$
- Q25** Pearson IAL WMA12 · October 2021 · Q4
 (a) Proven. (b) Proven. (c) $p = 6, q = -105$. (d) $g(x) = x^2 + 12x + 35$
- Q26** Pearson IAL WMA12 · January 2023 · Q5
 (a) $a = -3$. (b) $f(x) = 0$ has only one real root, $x = \frac{3}{2}$
- Q27** Pearson IAL WMA12 · June 2023 · Q2
 (a) Proven. (b) $4a + b = 1$. (c)(i) $a = 4, b = -15$. (c)(ii) $f(x) = (2x + 1)(x - 2)(2x + 15)$
- Q28** Pearson IAL WMA12 · October 2023 · Q4
 (a) 21. (b) $k = 11$. (c)(i) $(2x - 1)(x^2 + x + 1)$. (c)(ii) One real solution
- Q29** Pearson IAL WMA12 · June 2024 · Q4
 (a) Proven. (b) $a = -1, b = 8$. (c) $f(x) = (x - 1)(x + 2)(3x - 4)$
- Q30** Pearson IAL WMA12 · January 2025 · Q5
 (a)(i) $Q(x) = 4x^2 + 21x + 32$. (a)(ii) $R = 72$. (b) Proven; only real solution is $x = -4$. (c)
 $-\frac{5}{2} < x < \frac{1}{3}$
- Q31** Pearson IAL WMA12 · October 2025 · Q4

Circles

Complete the square for the centre and radius.
Then find one right angle: tangent, chord bisector
or semicircle.



DR ESLAM AHMED
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THE TOPIC ON ONE MAP

Circles

01 — Centre and radius

complete the square

from a diameter

$$r^2 > 0$$

02 — Chords and diameters

perpendicular bisector

angle in a semicircle

03 — Tangents

radius $\perp$ tangent

$$b^2 - 4ac = 0$$

04 — Two circles, nearest point

$$\text{touch: } d = r_1 \pm r_2$$

nearest point on the line of centres

THE CIRCLE EQUATION

$$(x - a)^2 + (y - b)^2 = r^2 \quad \text{centre } (a, b), \text{ radius } r$$

$$x^2 + y^2 + 2fx + 2gy + c = 0 \Rightarrow \text{centre } (-f, -g), \quad r^2 = f^2 + g^2 - c$$

01 Centre and Radius: Complete the Square

Group the x terms and the y terms and complete the square on each. The numbers inside the brackets change sign for the centre; the number on the right is r^2 — so it must be positive.

WHEN YOU SEE → YOU DO

“find the centre and the exact radius”

→ complete the square in x and in y

“state the possible range of values for k ”

→ $r^2 > 0$ gives an inequality in k

“ $y = k$ is a tangent”

→ $k = b \pm r$ (top and bottom of the circle)

“the x -axis is a tangent”

→ $r = |b|$

“a chord of length 4”

→ half-chord, radius and distance to centre: Pythagoras

WORKED MODEL • JANUARY 2020

$x^2 + y^2 + 6x - 4y - 14 = 0$. Centre and exact radius.

1 $(x + 3)^2 - 9 + (y - 2)^2 - 4 - 14 = 0$

complete both squares

2 $(x + 3)^2 + (y - 2)^2 = 27$

ANSWER centre $(-3, 2)$, $r = \sqrt{27} = 3\sqrt{3}$

IDEA 01

Practise: Centre and Radius: Complete the Square

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Complete the square twice

P2 · TOPIC 3 3 marks

Find the centre and the radius of the circle $x^2 + y^2 - 8x + 2y - 8 = 0$.

Q2 **DR ESLAM** From a diameter

P2 · TOPIC 3 3 marks

The points $(-1, 3)$ and $(5, 11)$ are the ends of a diameter. Find an equation of the circle.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 3**3 marks**

1 Write down the equations of the circles with the following centres and radii

(i) Centre: $(0, 0)$ Radius: $r = 4$,

(ii) Centre: $(3, -4)$ Radius: $r = 2$,

(iii) Centre: $(-5, 0)$ Radius: $r = 5$.

(3 marks)**Q4****PRACTICE**

Practice · Medium

P2 · TOPIC 3**4 marks**

1 A circle has centre $(6, -5)$ and goes through the point $(1, 7)$. Find the equation of the circle.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 3 4 marks

2 Find the centre and radius of the circle with equation $x^2 + y^2 + x - 3y + 2 = 0$.

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · January 2020 · Q6

P2 · TOPIC 3 8 marks

6. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0$$

(a) Find

(i) the coordinates of the centre of C ,(ii) the exact radius of C .

(3)

The line with equation $y = k$, where k is a constant, is a tangent to C .(b) Find the possible values of k .

(2)

The line with equation $y = p$, where p is a negative constant, is a chord of C .

Given that the length of this chord is 4 units,

(c) find the value of p .

(3)

(Total 8 marks)

Q7 **PAST PAPER** Pearson IAL WMA12 · June 2021 · Q6**P2 · TOPIC 3** **7 marks**

6. A circle has equation

$$x^2 - 6x + y^2 + 8y + k = 0$$

where k is a positive constant.

Given that the x -axis is a tangent to this circle,

- (a) find the value of k .

(3)

The circle meets the coordinate axes at the points R , S and T .

- (b) Find the exact area of the triangle RST .

(4)

(Total 7 marks)

Q8 PAST PAPER Pearson IAL WMA12 · October 2021 · Q6

P2 · TOPIC 3 8 marks

6. (i) The circle C_1 has equation

$$x^2 + y^2 + 10x - 12y = k \quad \text{where } k \text{ is a constant}$$

- (a) Find the coordinates of the centre of C_1 (2)

- (b) State the possible range in values for k . (2)

- (ii) The point $P(p, 0)$, the point $Q(-2, 10)$ and the point $R(8, -14)$ lie on a different circle, C_2

Given that

- p is a positive constant
- QR is a diameter of C_2

find the exact value of p . (4)

(Total 8 marks)

Q9 PAST PAPER Pearson IAL WMA12 · January 2024 · Q3

P2 · TOPIC 3 6 marks

3. The circle C

- has centre $A(3, 5)$
- passes through the point $B(8, -7)$

(a) Find an equation for C .

(3)

The points M and N lie on C such that MN is a chord of C .Given that MN

- lies above the x -axis
- is parallel to the x -axis
- has length $4\sqrt{22}$

(b) find an equation for the line passing through points M and N .

(3)

(Total for Question 3 is 6 marks)

02 Chords, Diameters and Right Angles

Two circle facts turn into straight-line work. The perpendicular bisector of **any** chord passes through the centre. The angle in a semicircle is 90° — so if PQR is a right angle, PR is a diameter.

WHEN YOU SEE → YOU DO

“show that angle $PQR = 90^\circ$ ”

→ $m_{PQ} \times m_{QR} = -1$

“hence find the centre”

→ the right angle makes PR a diameter:
centre = midpoint

“the line through M (midpoint of AB) and P ”

→ perpendicular bisector: gradient $-\frac{1}{m_{AB}}$

“ $P(7, k)$ lies on l ”

→ substitute into the bisector

three points on a circle

→ two perpendicular bisectors meet at the centre

ZONE DIAMETER

ends A, B

centre = midpoint

ZONE CHORD

AB

bisector passes through the centre

ZONE SEMICIRCLE

angle at Q

$90^\circ \Leftrightarrow PR$ is a diameter

WORKED MODEL • JANUARY 2025

$A(-1, 2)$ and $B(11, 6)$ lie on a circle, centre P . Find the line l through M (midpoint) and P .

1 $M(5, 4), m_{AB} = \frac{4}{12} = \frac{1}{3}$

midpoint, gradient

2 $m_l = -3: y - 4 = -3(x - 5)$

perpendicular bisector

ANSWER $y = -3x + 19$

IDEA 02

Practise: Chords, Diameters and Right Angles

My questions first, then selected practice, then real past papers.

Q10 **DR ESLAM** The perpendicular bisector

P2 · TOPIC 3 4 marks

$A(1, 2)$ and $B(5, 6)$ lie on a circle whose centre is on the x -axis. Find an equation of the circle.

The perpendicular bisector of any chord passes through the centre.

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Medium

P2 · TOPIC 3 7 marks

6 (a) The points $A(3, 5)$, $B(5, 3)$ and $C(9, 7)$ lie on a circle.

Show that triangle ABC is a right-angle triangle.

(2 marks)

(b) Explain why the line segment AC must be the diameter of the circle.

(1 mark)

(c) Hence find the equation of the circle.

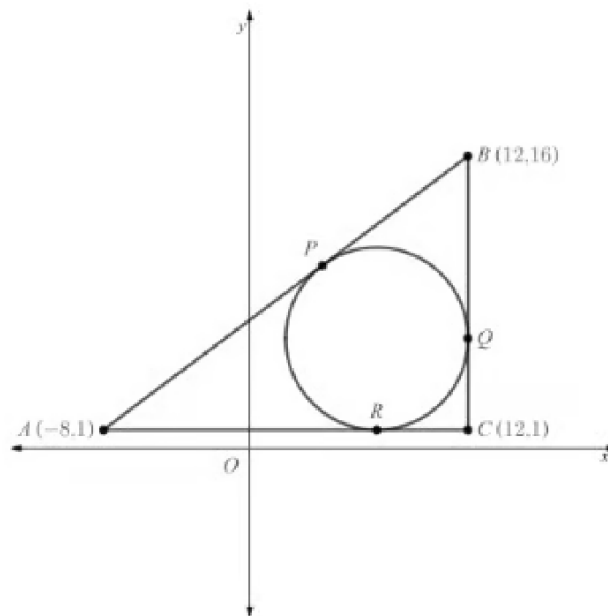
(4 marks)

Q12**PRACTICE**

Practice · Hard

P2 · TOPIC 3 7 marks

- 7 (a)** Triangle ABC has vertices $A(-8, 1)$, $B(12, 16)$ and $C(12, 1)$. A circle with equation $(x - 7)^2 + (y - 6)^2 = 25$ touches Triangle ABC at the three points P , Q and R , as shown in the diagram below:



Write down the coordinates of points R and Q .

(2 marks)

- (b)** Find the coordinates of point P .

(5 marks)

Q13**PRACTICE**

Practice · Very Hard

P2 · TOPIC 3 7 marks

6 The points $A(4, 6)$, $B(7, 2)$ and $C(12, 12)$ lie on a circle.

Find the equation of the circle.

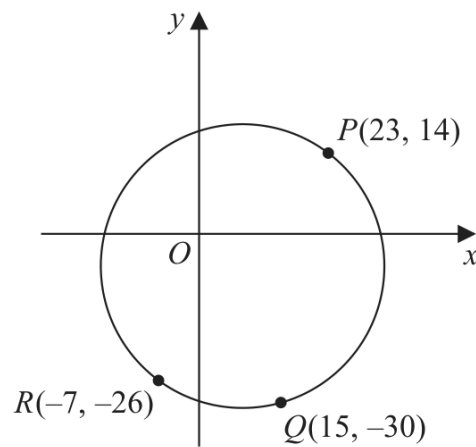
(7 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA12 · January 2022 · Q6

P2 · TOPIC 3 8 marks

6.

**Figure 1**

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

(a) Show that angle $PQR = 90^\circ$ (2)

(b) Hence, or otherwise, find

- (i) the centre of C ,
- (ii) the radius of C . (3)

Given that the point S lies on C such that the distance QS is greatest,

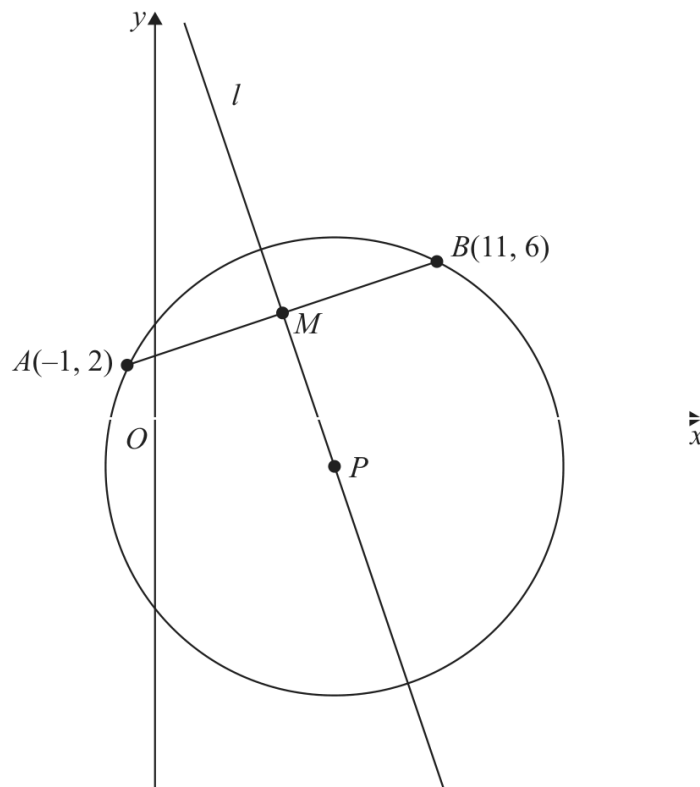
(c) find an equation of the tangent to C at S , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (3)

(Total 8 marks)

Q15 PAST PAPER Pearson IAL WMA12 · January 2025 · Q6

P2 · TOPIC 3 8 marks

6.

**Figure 2**

The point $A(-1, 2)$ and the point $B(11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

- (a) find an equation for l , giving your answer in the form $y = mx + c$, where m and c are constants.

(4)

Given that P has coordinates $(7, k)$, where k is a constant,

- (b) find the value of k ,

(1)

- (c) find an equation for the circle.

(3)

(Total for Question 6 is 8 marks)

03 Tangents and Lines Meeting a Circle

A tangent is perpendicular to the radius at the point of contact. When the point is unknown, substitute the line into the circle: a tangent gives one repeated root, so $b^2 - 4ac = 0$.

WHEN YOU SEE → YOU DO

“the tangent to C at P”

→ radius gradient, then $-\frac{1}{m}$, then $y - y_1 = m(x - x_1)$

“ $y = mx + 1$ is a tangent to C”

→ substitute, then $b^2 - 4ac = 0$

“find the points where the line meets the circle”

→ substitute the line, solve the quadratic

“the tangent ... in the form $ax + by + c = 0$ ”

→ integers: clear the fractions

“the line through the centre and the point of contact”

→ it is the normal: perpendicular to the tangent

Centre and radius

Gradient of the radius

Negative reciprocal

Line through the point

WORKED MODEL • JUNE 2023

Centre (2, 5), $P(8, -3)$ on the circle. Tangent at P .

1 radius gradient $\frac{-3-5}{8-2} = -\frac{4}{3}$

centre to P

2 tangent gradient $\frac{3}{4}$: $y + 3 = \frac{3}{4}(x - 8)$

perpendicular

ANSWER $3x - 4y - 36 = 0$

IDEA 03

Practise: Tangents and Lines Meeting a Circle

My questions first, then selected practice, then real past papers.

Q16 DR ESLAM Tangent at a point

P2 · TOPIC 3 3 marks

Find the tangent to $(x - 2)^2 + (y + 1)^2 = 25$ at $(5, 3)$, in the form $ax + by + c = 0$.

Q17 DR ESLAM Tangent by the discriminant

P2 · TOPIC 3 4 marks

Find the values of k for which $y = x + k$ is a tangent to $x^2 + y^2 = 8$.

SELECTED PRACTICE

Q18 PRACTICE Practice · Easy

P2 · TOPIC 3 2 marks

7 On a sketch show how a circle and a line can either have 0, 1 or 2 intersections.

(2 marks)**Q19** PRACTICE Practice · Hard

P2 · TOPIC 3 4 marks

3 The line $y + 2x = 11$ meets the circle with equation $x^2 + y^2 + 6x - 14y = -38$.

(i) Show that the line and circle meet at one point only.

(ii) Find the coordinates of the point of intersection.

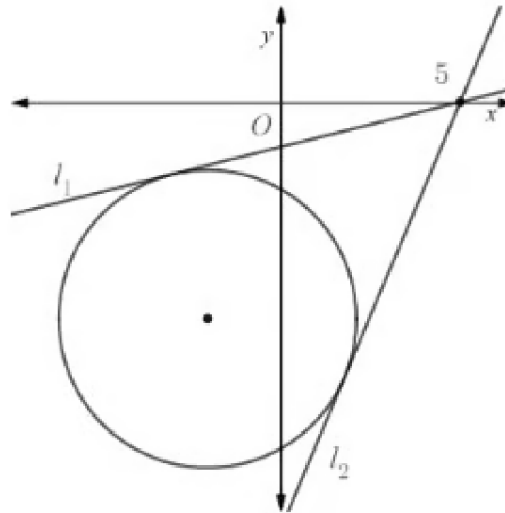
(4 marks)

Q20 PRACTICE Practice · Very Hard

P2 · TOPIC 3 8 marks

7 A circle has equation $x^2 + y^2 + 4x + 12y = -23$.

The lines l_1 and l_2 are both tangents to the circle, and they intersect at the point $(5, 0)$.



Find the equations of l_1 and l_2 , giving your answers in the form $y = mx + c$.

(8 marks)

PAST PAPERS

Q21 PAST PAPER Pearson IAL WMA12 · October 2019 · Q6

P2 · TOPIC 3 7 marks

6.

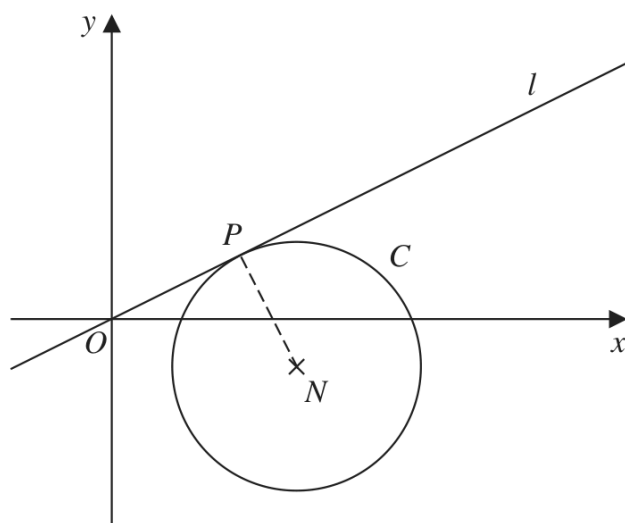


Figure 1

Figure 1 shows a sketch of a circle C with centre $N(4, -1)$.

The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P .

Find

(a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (2)

(b) the equation of C . (5)

(Total 7 marks)

Q22 PAST PAPER Pearson IAL WMA12 · January 2021 · Q9

P2 · TOPIC 3 10 marks

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

(a) Write down, in terms of k ,

(i) the coordinates of the centre of C ,

(ii) the radius of C .

(2)

Given that the point $P(2, 3)$ lies on C

(b) (i) show that $5k^2 - 17k + 6 = 0$

(ii) hence find the possible values of k .

(3)

The tangent to the circle at P intersects the x -axis at point T .

Given that $k < 2$

(c) calculate the exact area of triangle OPT .

(5)

(Total 10 marks)

Q23 PAST PAPER Pearson IAL WMA12 · June 2022 · Q10

P2 · TOPIC 3 12 marks

10. The circle C has centre $X(3, 5)$ and radius r

The line l has equation $y = 2x + k$, where k is a constant.

(a) Show that l and C intersect when

$$5x^2 + (4k - 26)x + k^2 - 10k + 34 - r^2 = 0 \quad (3)$$

Given that l is a tangent to C ,

(b) show that $5r^2 = (k + p)^2$, where p is a constant to be found. (3)

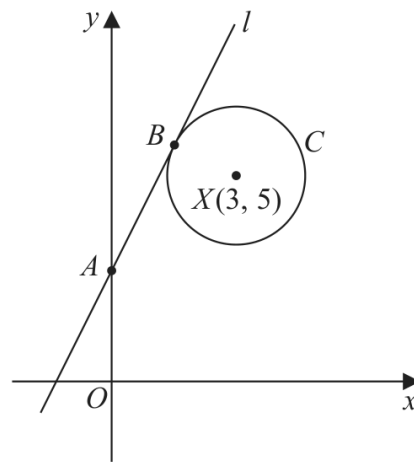


Figure 2

The line l

- cuts the y -axis at the point A
- touches the circle C at the point B

as shown in Figure 2.

Given that $AB = 2r$

(c) find the value of k (6)

9. In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

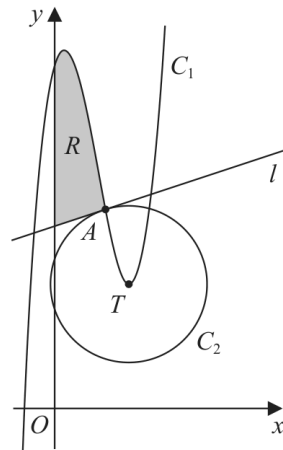


Figure 3

Figure 3 shows

- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
- the circle C_2 with centre T

The point T is the minimum turning point of C_1

Using Figure 3 and calculus,

(a) find the coordinates of T

(3)

The curve C_1 intersects the circle C_2 at the point A with x coordinate 2

(b) Find an equation of the circle C_2

(3)

The line l shown in Figure 3, is the tangent to circle C_2 at A

(c) Show that an equation of l is

$$y = \frac{1}{3}x + \frac{22}{3}$$

(3)

The region R , shown shaded in Figure 3, is bounded by C_1 , l and the y -axis.

(d) Find the exact area of R .

(3)

(Total for Question 9 is 12 marks)

Q25 **PAST PAPER** Pearson IAL WMA12 · January 2023 · Q6**P2 · TOPIC 3** **10 marks**

6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact radius of C .

(3)

The point P lies on C .

Given that the tangent to C at P has equation $x + 2y + 10 = 0$

(b) find the coordinates of P

(4)

(c) Find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found.

(3)

(Total for Question 6 is 10 marks)

Q26 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q3**P2 · TOPIC 3** **7 marks**

3. A circle C has centre $(2, 5)$

Given that the point $P(8, -3)$ lies on C

- (a) (i) find the radius of C

- (ii) find an equation for C

(3)

- (b) Find the equation of the tangent to C at P giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

(Total for Question 3 is 7 marks)

7.

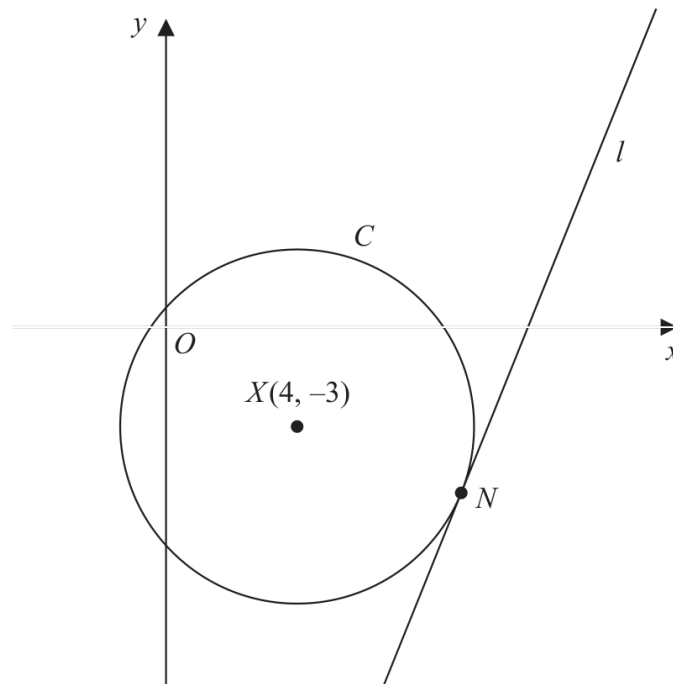


Figure 2

Figure 2 shows a sketch of

- the circle C with centre $X(4, -3)$
- the line l with equation $y = \frac{5}{2}x - \frac{55}{2}$

Given that l is the tangent to C at the point N ,

(a) show that an equation for the straight line passing through X and N is

$$2x + 5y + 7 = 0 \quad (3)$$

(b) Hence find

- the coordinates of N ,
- an equation for C .

(5)

(Total for Question 7 is 8 marks)

Q28 **PAST PAPER** Pearson IAL WMA12 · October 2024 · Q10**P2 · TOPIC 3** **10 marks**

10. The circle C has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact value of the radius of C .

(3)

The line L has equation $y = mx + 1$, where m is a constant.

Given that L is the tangent to C at the point P ,

(b) show that

$$2m^2 - 7m - 22 = 0$$

(3)

(c) Hence find the possible pairs of coordinates of P .

(4)

(Total for Question 10 is 10 marks)

04 Two Circles and the Nearest Point

Everything happens on the line joining the centres. Two circles touch when the distance between the centres is $r_1 + r_2$ (outside) or $|r_1 - r_2|$ (inside). The nearest point of a circle to a point O is on the line from O to the centre, at distance $d - r$.

WHEN YOU SEE → YOU DO

“the circles touch”

$$\rightarrow d = r_1 + r_2 \text{ or } d = |r_1 - r_2|$$

“neither touch nor intersect”

$$\rightarrow d > r_1 + r_2 \text{ (or } d < |r_1 - r_2| \text{)}$$

“the point P on the circle nearest the origin”

$$\rightarrow OP = d - r \text{ along the line to the centre}$$

“nearest the x -axis”

$$\rightarrow \text{straight below the centre: } (a, b - r)$$

WORKED MODEL • JUNE 2024

C_1 : centre $(-4, 5)$, $r = \sqrt{70}$. C_2 : centre $(5, -8)$, $r = \sqrt{52}$. Do they meet?

1 $d = \sqrt{81 + 169} = \sqrt{250} \approx 15.8$

distance between centres

2 $r_1 + r_2 \approx 8.37 + 7.21 = 15.6$

sum of radii

ANSWER $d > r_1 + r_2$: they neither touch nor intersect

× **TRAP** Comparing d^2 with $r_1^2 + r_2^2$. Compare d with $r_1 + r_2$.

IDEA 04

Practise: Two Circles and the Nearest Point

My questions first, then selected practice, then real past papers.

Q29 DR ESLAM Do they touch?

P2 · TOPIC 3 3 marks

Show that the circles $x^2 + y^2 = 4$ and $(x - 6)^2 + (y - 8)^2 = 25$ do not meet, and find the shortest distance between them.

SELECTED PRACTICE**Q30** PRACTICE Practice · Easy

P2 · TOPIC 3 3 marks

6 Determine if the circles with equations

$$(x + 4)^2 + y^2 = 9 \text{ and } (x - 2)^2 + y^2 = 9$$

intersect once, twice or not at all. Fully explain your answer.

(3 marks)

Q31 PRACTICE Practice · Medium

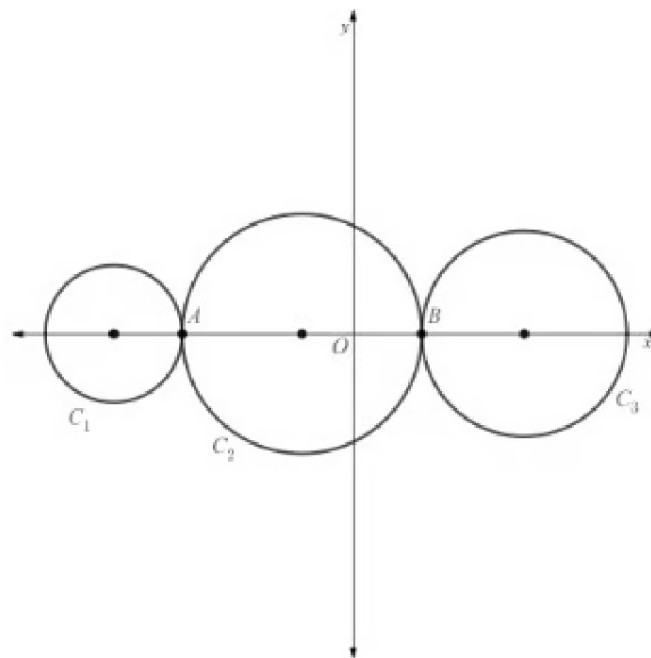
P2 · TOPIC 3 6 marks

7 C_2 Circles C_1 , C_2 and C_3 all have their centres on the x -axis.

Circle C_1 has equation $(x + 7)^2 + y^2 = 4$.

Circle C_3 has equation $x^2 + y^2 - 10x + 16 = 0$.

Circles C_1 and C_2 touch at point A , and circles C_2 and C_3 touch at point B .



Find the coordinates of the centre of circle C_2 .

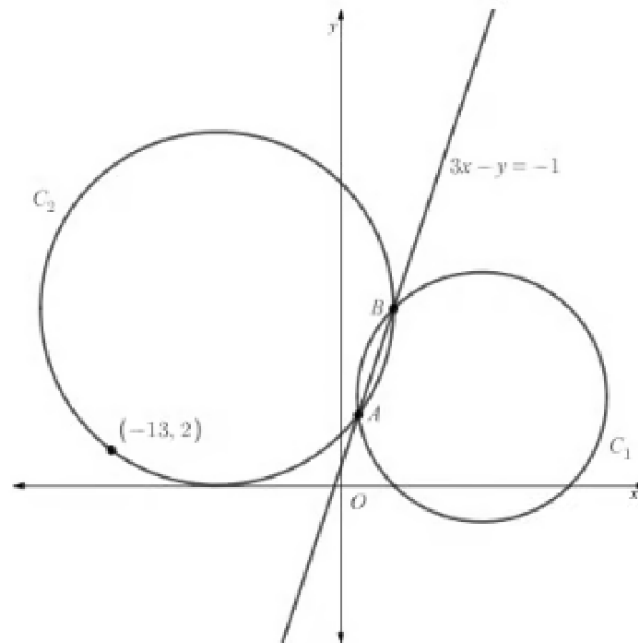
(6 marks)

Q32**PRACTICE**

Practice · Very Hard

P2 · TOPIC 3**11 marks**

- 8 The diagram below shows circles C_1 and C_2 which intersect at the two points A and B . Circle C_1 has equation $x^2 + y^2 - 16x - 10y + 39 = 0$, and points A and B lie along the line with equation $3x - y = -1$. Circle C_2 also passes through the point $(-13, 2)$.



Find an equation of circle C_2 .

(11 marks)

PAST PAPERS

Q33 PAST PAPER Pearson IAL WMA12 · June 2024 · Q7

P2 · TOPIC 3 6 marks

7. The circle C_1 has equation

$$x^2 + y^2 + 8x - 10y = 29$$

(a) (i) Find the coordinates of the centre of C_1 (ii) Find the exact value of the radius of C_1

(3)

In part (b) you must show detailed reasoning.The circle C_2 has equation

$$(x - 5)^2 + (y + 8)^2 = 52$$

(b) Prove that the circles C_1 and C_2 neither touch nor intersect.

(3)

(Total for Question 7 is 6 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June (R paper) 2024 · Q3

P2 · TOPIC 3 5 marks

3. A circle has equation

$$x^2 + y^2 + 8x - 14y - 79 = 0$$

(a) Find

- (i) the coordinates of the centre of the circle,
- (ii) the radius of the circle.

(3)

Given that P is the point on the circle that is nearest the origin O ,

- (b) find the exact length of OP

(2)

(Total for Question 3 is 5 marks)

Q35 **PAST PAPER** Pearson IAL WMA12 · June 2025 · Q2**P2 · TOPIC 3** **7 marks**

2: The line joining the points $(-2, 5)$ and $(4, 15)$ is the diameter of a circle C .

(a) Find an equation for C .

(5)

(b) Hence find the exact coordinates of the point on C that is nearest the x -axis.

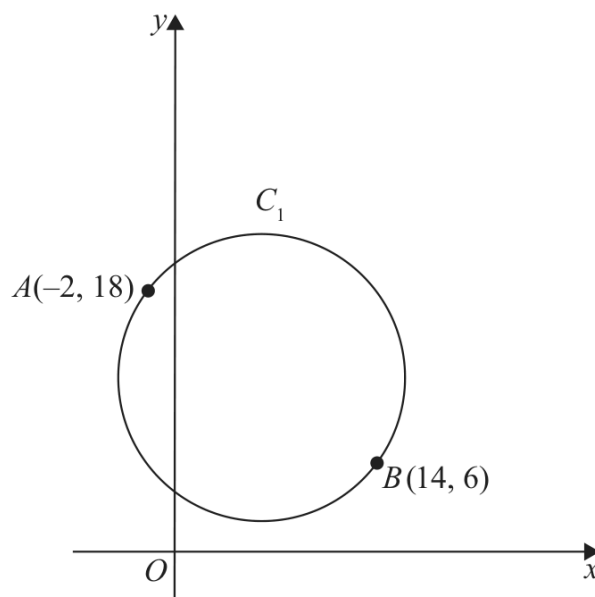
(2)

(Total for Question 2 is 7 marks)

Q36 PAST PAPER Pearson IAL WMA12 · October 2025 · Q6

P2 · TOPIC 3 9 marks

6.

**Figure 2**

The points $A(-2, 18)$ and $B(14, 6)$ lie on a circle C_1 as shown in Figure 2.

Given that AB is a diameter of the circle C_1

(a) find an equation for C_1 making your method clear.

(5)

A circle C_2 has its centre at the origin.

Given that circles C_1 and C_2 touch,

(b) find possible equations for C_2

(4)

(Total for Question 6 is 9 marks)

Q37 PAST PAPER Pearson IAL WMA12 · January 2026 · Q5

P2 · TOPIC 3 6 marks

5. The circle C_1 has equation

$$x^2 + y^2 - 6x + 5y - 41 = 0$$

(a) Find

- (i) the coordinates of the centre of C_1
- (ii) the radius of C_1

(3)

The circle C_2 has

- centre $(-k, 0)$ where k is a positive constant
- radius 5

Given that circles C_1 and C_2 touch

(b) find the exact value of k .

(3)

(Total for Question 5 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|---|
| 01 | Centre, radius | Complete the square; $r^2 > 0$. |
| 02 | Chords, diameters | Bisector through the centre; semicircle 90° . |
| 03 | Tangents | Radius $\perp$ tangent; unknown point $\rightarrow b^2 - 4ac = 0$. |
| 04 | Two circles | $d = r_1 \pm r_2$; nearest point at $d - r$. |

BOTTOM LINE Centre and radius first • then one right angle: tangent, semicircle or chord bisector.

Answers

Circles · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · CENTRE AND RADIUS: COMPLETE THE SQUARE

- Q1** Dr Eslam · Complete the square twice $(x - 4)^2 + (y + 1)^2 = 25$: centre $(4, -1)$, radius 5
- Q2** Dr Eslam · From a diameter centre $(2, 7)$, $r = 5$: $(x - 2)^2 + (y - 7)^2 = 25$
- Q3** Practice · Easy (i) $x^2 + y^2 = 16$ (ii) $(x - 3)^2 + (y + 4)^2 = 4$ (iii) $(x + 5)^2 + y^2 = 25$
- Q4** Practice · Medium $r^2 = 5^2 + 12^2 = 169$: $(x - 6)^2 + (y + 5)^2 = 169$
- Q5** Practice · Very Hard $(x + \frac{1}{2})^2 + (y - \frac{3}{2})^2 = \frac{1}{2}$: centre $(-\frac{1}{2}, \frac{3}{2})$, radius $\frac{\sqrt{2}}{2}$
- Q6** Pearson IAL WMA12 · January 2020 · Q6 (a)(i) $(-3, 2)$. (a)(ii) $3\sqrt{3}$. (b) $k = 2 \pm 3\sqrt{3}$. (c) $p = 2 - \sqrt{23}$
- Q7** Pearson IAL WMA12 · June 2021 · Q6 (a) $k = 9$. (b) $3\sqrt{7}$
- Q8** Pearson IAL WMA12 · October 2021 · Q6 (i)(a) $(-5, 6)$. (i)(b) $k > -61$. (ii) $p = 3 + \sqrt{165}$
- Q9** Pearson IAL WMA12 · January 2024 · Q3 (a) $(x - 3)^2 + (y - 5)^2 = 169$. (b) $y = 14$

IDEA 02 · CHORDS, DIAMETERS AND RIGHT ANGLES

- Q10** Dr Eslam · The perpendicular bisector
 bisector $y = 7 - x$; centre $(7, 0)$: $(x - 7)^2 + y^2 = 40$
- Q11** Practice · Medium
 (a) $m_{AB} = -1$, $m_{BC} = 1$: perpendicular, right angle at B (b) the angle in a semicircle is 90°
 (c) $(x - 6)^2 + (y - 6)^2 = 10$
- Q12** Practice · Hard
 (a) $R(7, 1)$, $Q(12, 6)$ (b) $P(4, 10)$
- Q13** Practice · Very Hard
 $x^2 + y^2 - 19x - 14y + 108 = 0$, i.e. $(x - \frac{19}{2})^2 + (y - 7)^2 = \frac{125}{4}$
- Q14** Pearson IAL WMA12 · January 2022 · Q6
 (a) Proven: $\angle PQR = 90^\circ$.
 (b)(i) $(8, -6)$. (b)(ii) 25. (c) $7x - 24y + 425 = 0$
- Q15** Pearson IAL WMA12 · January 2025 · Q6
 (a) $y = -3x + 19$. (b) $k = -2$. (c) $(x - 7)^2 + (y + 2)^2 = 80$

IDEA 03 · TANGENTS AND LINES MEETING A CIRCLE

- Q16** Dr Eslam · Tangent at a point
radius gradient $\frac{4}{3}$, tangent $-\frac{3}{4}$: $3x+4y-27=0$
- Q17** Dr Eslam · Tangent by the discriminant
 $2x^2 + 2kx + k^2 - 8 = 0$; $b^2 - 4ac = 0 \Rightarrow k = \pm 4$
- Q18** Practice · Easy
misses (0), tangent (1), chord (2)
- Q19** Practice · Hard
(i) the quadratic has a repeated root (ii) (1, 9)
- Q20** Practice · Very Hard
 $m = \frac{1}{4}$ or $\frac{19}{8}$: $y = \frac{1}{4}x - \frac{5}{4}$, $y = \frac{19}{8}x - \frac{95}{8}$
- Q21** Pearson IAL WMA12 · October 2019 · Q6
(a) $y = -2x + 7$. (b) $(x-4)^2 + (y+1)^2 = \frac{36}{5}$
- Q22** Pearson IAL WMA12 · January 2021 · Q9
(a)(i) $(k, 2k)$. (a)(ii) $\sqrt{k+7}$. (b)(ii) $k = 3$ or $\frac{2}{5}$. (c) $\frac{147}{16}$
- Q23** Pearson IAL WMA12 · June 2022 · Q10
(a) Proven. (b) $5r^2 = (k+1)^2$, so $p = 1$.
(c) $k = \frac{11}{4}$
- Q24** Pearson IAL WMA12 · October 2022 · Q9
(a) $T = (3, 5)$. (b) $(x-3)^2 + (y-5)^2 = 10$. (c) $y = \frac{1}{3}x + \frac{22}{3}$. (d) $\frac{28}{3}$
- Q25** Pearson IAL WMA12 · January 2023 · Q6
(a)(i) $(-4, 2)$. (a)(ii) $2\sqrt{5}$. (b) $P = (-6, -2)$.
(c) $y = 2x + 10$
- Q26** Pearson IAL WMA12 · June 2023 · Q3
(a)(i) 10. (a)(ii) $(x-2)^2 + (y-5)^2 = 100$. (b) $3x - 4y - 36 = 0$
- Q27** Pearson IAL WMA12 · October 2023 · Q7
(a) Proven. (b)(i) $N = (9, -5)$. (b)(ii) $(x-4)^2 + (y+3)^2 = 29$
- Q28** Pearson IAL WMA12 · October 2024 · Q10
(a)(i) $(-2, 15)$. (a)(ii) $2\sqrt{5}$. (b) Proven.
(c) $(-6, 13)$ and $(\frac{12}{5}, \frac{71}{5})$

IDEA 04 · TWO CIRCLES AND THE NEAREST POINT

- Q29** Dr Eslam · Do they touch? $d = 10 > 2 + 5$; shortest distance $10 - 7 = 3$
- Q30** Practice · Easy centres $(-4, 0)$, $(2, 0)$: distance $6 = 3 + 3$, so they touch once
- Q31** Practice · Medium $A(-5, 0)$, $B(2, 0)$: centre of C_2 is $(-\frac{3}{2}, 0)$
- Q32** Practice · Very Hard $A(1, 4)$, $B(3, 10)$; $C_2: (x + 7)^2 + (y - 10)^2 = 100$
- Q33** Pearson IAL WMA12 · June 2024 · Q7 $(a)(i) (-4, 5)$. $(a)(ii) \sqrt{70}$. (b) The circles neither touch nor intersect
- Q34** Pearson IAL WMA12 · June (R paper) 2024 · Q3 $(a)(i) (-4, 7)$. $(a)(ii) 12$. $(b) 12 - \sqrt{65}$
- Q35** Pearson IAL WMA12 · June 2025 · Q2 $(a) (x - 1)^2 + (y - 10)^2 = 34$. $(b) (1, 10 - \sqrt{34})$
- Q36** Pearson IAL WMA12 · October 2025 · Q6 $(a) (x - 6)^2 + (y - 12)^2 = 100$. $(b) x^2 + y^2 = 280 - 120\sqrt{5}$ or $x^2 + y^2 = 280 + 120\sqrt{5}$
- Q37** Pearson IAL WMA12 · January 2026 · Q5 $(a)(i) (3, -\frac{5}{2})$. $(a)(ii) \frac{15}{2}$. $(b) k = 5\sqrt{6} - 3$

DR ESLAM AHMED
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Binomial Expansion

Every term is $\binom{n}{r} a^{n-r} (bx)^r$. Keep the sign and the number inside the bracket with x , and every question becomes one term.

THE TOPIC ON ONE MAP

Binomial Expansion

01 Expand $(a + bx)^n$

first 4 terms

one term: $\binom{n}{r}$

simplest form

02 Find the constant

coefficient given

ratio of two coefficients

03 Multiply by a bracket

pick the pairs

term independent of x

04 Approximations

choose x

over or under?

THE GENERAL TERM

$$(a + bx)^n: \text{ term in } x^r = \binom{n}{r} a^{n-r} (bx)^r$$

powers of a go down, powers of x go up, they always add to n · $\binom{n}{r}$ on the calculator: nCr

01 Expand $(a + bx)^n$: One Term at a Time

Write each term as $\binom{n}{r} a^{n-r} (bx)^r$ and simplify it on its own line. The bracket round (bx) matters: the power applies to the number **and** the sign.

WHEN YOU SEE → YOU DO

“the first 4 terms, in ascending powers of x ”

→ $r = 0, 1, 2, 3$

“giving each term in simplest form”

→ work out every coefficient to a number (or in k)

“find the coefficient of x^7 ”

→ only the term with $r = 7$

$(2 - \frac{kx}{4})^8$

→ $b = -\frac{k}{4}$: odd powers are negative

“the 2nd, 3rd and 5th terms”

→ $r = 1, 2, 4$

WORKED MODEL • JUNE 2022

First four terms of $(2 + \frac{3}{8}x)^{10}$.

$$1 \quad 2^{10} + \binom{10}{1} 2^9 \left(\frac{3}{8}x\right) + \binom{10}{2} 2^8 \left(\frac{3}{8}x\right)^2 + \binom{10}{3} 2^7 \left(\frac{3}{8}x\right)^3$$

general term

ANSWER $1024 + 1920x + 1620x^2 + 810x^3$

IDEA 01

Practise: Expand $(a + bx)^n$: One Term at a Time

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM First four terms

P2 · TOPIC 4 3 marks

Find the first four terms, in ascending powers of x , of $(1 + 2x)^6$.

Q2 DR ESLAM One term only

P2 · TOPIC 4 2 marks

Find the coefficient of x^3 in $(2 - 3x)^5$.

$\binom{5}{3} 2^2 (-3x)^3$ — the sign goes with the x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 4 3 marks

1 Evaluate

(i) $4!$

(ii) 5C_2

(iii) 6C_3

(3 marks)

Q4**PRACTICE**

Practice · Medium

P2 · TOPIC 4 3 marks

1 Expand $(2 + x)^4$.

(3 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 4**3 marks**

3 Given that ${}^nC_3 = 35$ find the value of n .

(3 marks)**PAST PAPERS****Q6****PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q1

P2 · TOPIC 4**4 marks**

1. Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(2 + \frac{3}{8}x\right)^{10}$$

Give each coefficient as an integer.

(4)**(Total 4 marks)**

Q7 PAST PAPER Pearson IAL WMA12 · October 2023 · Q10

P2 · TOPIC 4 12 marks

10. (i) (a) Find, in ascending powers of x , the 2nd, 3rd and 5th terms of the binomial expansion of

$$(3 + 2x)^6 \quad (3)$$

For a particular value of x , these three terms form consecutive terms in a geometric series.

- (b) Find this value of x . (3)

- (ii) In a **different** geometric series,

- the first term is $\sin^2 \theta$
- the common ratio is $2 \cos \theta$
- the sum to infinity is $\frac{8}{5}$

- (a) Show that

$$5 \cos^2 \theta - 16 \cos \theta + 3 = 0 \quad (3)$$

- (b) Hence find the exact value of the 2nd term in the series. (3)

(Total for Question 10 is 12 marks)

Q8 **PAST PAPER** Pearson IAL WMA12 · January 2024 · Q2**P2 · TOPIC 4** **3 marks**

2. Find the coefficient of the term in x^7 of the binomial expansion of

$$\left(\frac{3}{8} + 4x\right)^{12}$$

giving your answer in simplest form.

(3)

(Total for Question 2 is 3 marks)

02 Find the Unknown Constant

Write the terms with the constant still in them, then turn the sentence into an equation: “the coefficient is 540”, “ $B = 18D$ ”, “the coefficient of x^2 is 3 times the coefficient of x^3 ”.

WHEN YOU SEE → YOU DO

“the first three terms are $1, -4x, px^2$ ”

→ match term by term: first find k , then p

“one of the terms is $540x^4$ ”

→ $\binom{6}{4}3^2a^4 = 540$

“coefficient of x^2 is 3 times coefficient of x^3 ”

→ one equation, cancel common powers of k

“the coefficients form an arithmetic progression”

→ $2 \times \text{middle} = \text{first} + \text{last}$

“when $f(x)$ is divided by $(4x - 5)$ the remainder is ...”

→ remainder theorem: $f(\frac{5}{4})$

WORKED MODEL • JANUARY 2026

$(3 + kx)^7$: the coefficient of x^2 is 3 times the coefficient of x^3 . Find k .

1 x^2 : $5103k^2$ x^3 : $2835k^3$

the two terms

2 $5103k^2 = 3 \times 2835k^3$

the sentence as an equation

ANSWER $k = \frac{5103}{8505} = \frac{3}{5}$

IDEA 02

Practise: Find the Unknown Constant

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Coefficient given

P2 · TOPIC 4 3 marks

In the expansion of $(1 + kx)^8$, $k > 0$, the coefficient of x^2 is 112. Find k .

SELECTED PRACTICE**Q10****PRACTICE**

Practice · Easy

P2 · TOPIC 4 3 marks

7 In the expansion of $(p + x)^{12}$, the coefficient of the x^5 term is 12 976 128. Find the value of p .

(3 marks)

Q11**PRACTICE**

Practice · Hard

P2 · TOPIC 4**3 marks**

- 6 In the expansion of $(3a - 2x)^6$, the coefficient of the x^3 term is equal to the coefficient of the x^4 term. Find the value of a .

(3 marks)**Q12****PRACTICE**

Practice · Very Hard

P2 · TOPIC 4**5 marks**

- 10 In the expansion of $(a + bx)^8$, the coefficient of the x^5 term is -870 912.
In the expansion of $(a + bx)^{12}$, the coefficient of the x^3 term is -1 557 135 360.

Find the possible values of a and b .

(5 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2020 · Q2

P2 · TOPIC 4 7 marks

2. One of the terms in the binomial expansion of $(3 + ax)^6$, where a is a constant, is $540x^4$

(a) Find the possible values of a . (4)

(b) Hence find the term independent of x in the expansion of

$$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3 + ax)^6 \quad (3)$$

(Total 7 marks)

Q14 PAST PAPER Pearson IAL WMA12 · October 2021 · Q1

P2 · TOPIC 4 6 marks

1. The first three terms, in ascending powers of x , of the binomial expansion of $(1 + kx)^{16}$ are

$$1, -4x \text{ and } px^2$$

where k and p are constants.

- (a) Find, in simplest form,

(i) the value of k

(ii) the value of p

(3)

$$g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$$

Using the value of k found in part (a),

- (b) find the term in x^2 in the expansion of $g(x)$.

(3)

(Total 6 marks)

Q15 PAST PAPER Pearson IAL WMA12 · October 2022 · Q2

P2 · TOPIC 4 7 marks

2. A curve C has equation $y = f(x)$ where

$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

- (a) show that $k = \frac{2}{5}$

(2)

- (b) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{2}{5}x\right)^5$$

giving each term in simplest form.

(3)

Using the solution to part (b) and making your method clear,

- (c) find the gradient of C at the point where $x = 0$

(2)

(Total for Question 2 is 7 marks)

Q16**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q3

P2 · TOPIC 4 7 marks

3.
$$f(x) = \left(2 + \frac{kx}{8}\right)^7$$
 where k is a non-zero constant

- (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of $f(x)$.
Give each term in simplest form.

(4)

Given that, in the binomial expansion of $f(x)$, the coefficients of x , x^2 and x^3 are the first 3 terms of an arithmetic progression,

- (b) find, using algebra, the possible values of k .

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

(Total for Question 3 is 7 marks)

Q17 PAST PAPER Pearson IAL WMA12 · June 2023 · Q4

P2 · TOPIC 4 7 marks

4. The binomial expansion, in ascending powers of x , of

$$(3 + px)^5$$

where p is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \dots$$

where A , B , C and D are constants.

- (a) Find the value of A

(1)

Given that

- $B = 18D$
- $p < 0$

- (b) find

- (i) the value of p
- (ii) the value of C

(6)

(Total for Question 4 is 7 marks)

Q18 **PAST PAPER** Pearson IAL WMA12 · October 2025 · Q2**P2 · TOPIC 4** **6 marks**

2. The first 4 terms, in ascending powers of x , in the binomial expansion of

$$(1 + px)^{10}$$

are

$$1 + 15x + qx^2 + rx^3$$

where p , q and r are constants.

Find the value of p , the value of q and the value of r .

(6)

(Total for Question 2 is 6 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q1

P2 · TOPIC 4 6 marks

1. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + kx)^7$$

where k is a non-zero constant. Give each term in simplest form.

(4)

Given that, in this expansion, the coefficient of x^2 is 3 times the coefficient of x^3

- (b) find, in simplest form, the value of k .

(2)

(Total for Question 1 is 6 marks)

03 Multiply by a Bracket: Pick the Pairs

For $(p + qx)(\text{expansion})$ you only need a few products: to get x^2 , pair p with the x^2 term and qx with the x term. For a term in $\frac{1}{x}$ or $\frac{1}{x^6}$, pair it with the term that cancels the power.

WHEN YOU SEE → YOU DO

“the coefficient of x^2 in $(3 - \frac{1}{2}x)(2 + px)^6$ ”

→ $3 \times (x^2 \text{ term}) - \frac{1}{2} \times (x \text{ term})$

“the term independent of x ”

→ powers must add to 0

$(\frac{x^2+8}{x^5})(1 + \frac{x}{4})^{12}$

→ split: x^{-3} and $8x^{-5}$, pair with x^3 and x^5

“the constant term is 3 times the coefficient of x ”

→ two pair-sums, one equation

WORKED MODEL • JUNE 2024

Coefficient of x^3 in $(10x + 3)(1 - \frac{1}{6}x)^9$, given $1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3$.

1 $10x \times x^2$ and $3 \times (-\frac{7}{18}x^3)$

pick the pairs

ANSWER $10 - \frac{7}{6} = \frac{53}{6}$

× **TRAP** Multiplying out every term. Only the pairs that make the power you want matter.

IDEA 03

Practise: Multiply by a Bracket: Pick the Pairs

My questions first, then selected practice, then real past papers.

Q20 DR ESLAM Pick the pairs

P2 · TOPIC 4 3 marks

Find the coefficient of x^2 in $(2 + x)(1 - x)^6$.

SELECTED PRACTICE**Q21** PRACTICE Practice · Medium

P2 · TOPIC 4 6 marks

8 (a) Find the first three terms in the expansion of $(3 + 2x)^8$.

(3 marks)

(b) Given that x is small such that x^3 and higher powers of x can be ignored show that

$$(1 + x)(3 + 2x)^8 \approx 6561 + 41553x + 116640x^2$$

(3 marks)

Q22**PRACTICE**

Practice · Hard

P2 · TOPIC 4 6 marks

7 (a) Find the first three terms in the expansion of $(2 - 3x)^7$.

(3 marks)

(b) Given that x is small such that x^3 and higher powers of x can be ignored show that
 $(1 - 2x)(2 - 3x)^7 \approx 128 - 1600x + 8736x^2$

(3 marks)

Q23**PRACTICE**

Practice · Very Hard

P2 · TOPIC 4 6 marks

7 (a) Find the first three terms in the expansion of $(4 - 3x)^9$.

(3 marks)

(b) Given that x is small such that x^3 and higher powers of x can be ignored show that

$$(3 - 2x^2)(4 - 3x)^9 \approx 786432 - 5308416x + 15400960x^2$$

(3 marks)

PAST PAPERS

Q24 PAST PAPER Pearson IAL WMA12 · October 2019 · Q3

P2 · TOPIC 4 6 marks

3. (a) Find the first 4 terms, in ascending powers of x , in the binomial expansion of

$$\left(1 + \frac{x}{4}\right)^{12}$$

giving each coefficient in its simplest form.

(3)

- (b) Find the term independent of x in the expansion of

$$\left(\frac{x^2 + 8}{x^5}\right)\left(1 + \frac{x}{4}\right)^{12}$$

(3)

(Total 6 marks)

Q25 PAST PAPER Pearson IAL WMA12 · January 2021 · Q4

P2 · TOPIC 4 8 marks

4. (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$(2 + px)^6$$

where p is a constant. Give each term in simplest form.

(4)

Given that in the expansion of

$$\left(3 - \frac{1}{2}x\right)(2 + px)^6$$

the coefficient of x^2 is $-\frac{3}{4}$

- (b) find the possible values of p .

(4)

(Total 8 marks)

Q26 **PAST PAPER** Pearson IAL WMA12 · January 2022 · Q3**P2 · TOPIC 4** **7 marks**

3. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{kx}{4}\right)^8$$

where k is a non-zero constant. Give each term in simplest form.

(4)

$$f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$$

In the expansion of $f(x)$, the constant term is 3 times the coefficient of x .

- (b) Find the value of k .

(3)

(Total 7 marks)

Q27 PAST PAPER Pearson IAL WMA12 · June 2024 · Q1

P2 · TOPIC 4 5 marks

1. (a) Find the first four terms, in ascending powers of x , of the binomial expansion of

$$\left(1 - \frac{1}{6}x\right)^9$$

giving each term in simplest form.

(3)

- (b) Hence find the coefficient of x^3 in the expansion of

$$(10x + 3)\left(1 - \frac{1}{6}x\right)^9$$

giving the answer in simplest form.

(2)

(Total for Question 1 is 5 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q4

P2 · TOPIC 4 7 marks

4. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + 2x)^6$$

giving each coefficient in simplest form.

(4)

- (b) Hence find the coefficient of x^2 in the expansion of

$$\left(2x^2 - \frac{1}{6x}\right)(3 + 2x)^6$$

(3)

(Total for Question 4 is 7 marks)

Q29**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q5

P2 · TOPIC 4 7 marks

5. (a) Find, in terms of a , the first 3 terms, in ascending powers of x , of the binomial expansion of

$$(2 + ax)^6$$

where a is a non-zero constant. Give each term in simplest form.

(3)

$$f(x) = \left(3 + \frac{1}{x}\right)^2 (2 + ax)^6$$

Given that the constant term in the expansion of $f(x)$ is 576

- (b) find the value of a .

(4)

(Total for Question 5 is 7 marks)

Q30 **PAST PAPER** Pearson IAL WMA12 · June 2025 · Q1**P2 · TOPIC 4** **6 marks**

1: (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(1 - 4x)^7$$

giving each term in simplest form.

(4)

In the series expansion of

$$(5 + kx)(1 - 4x)^7 \quad \text{where } k \text{ is a constant}$$

the coefficient of the term in x^2 is 1316

(b) Use the answer to part (a) to find the value of k .

(2)

(Total for Question 1 is 6 marks)

04 Approximations: Choose the Right x

Match the bracket to the number you want: $(2 + \frac{x}{8})^{13}$ with $2 + \frac{x}{8} = 2.0125$ gives $x = 0.1$. Substitute into the expansion. Small x keeps the ignored terms tiny; state over- or under-estimate from the signs of those terms.

WHEN YOU SEE → YOU DO

“use your expansion to find an approximation for 2.0125^{13} ”

→ solve $2 + \frac{x}{8} = 2.0125$ for x

“state the value of x that should be used”

→ just that equation — no calculation

“overestimate or underestimate?”

→ the missing terms are positive → underestimate

“show that an estimate for $(\frac{33}{32})^{10}$ is $\frac{1389}{k}$ ”

→ $1 + \frac{5}{16}x = \frac{33}{32} \Rightarrow x = \frac{1}{10}$

WORKED MODEL • JUNE 2026

$(1 + \frac{5}{16}x)^{10} \approx 1 + \frac{25}{8}x + \frac{1125}{256}x^2$. Estimate $(\frac{33}{32})^{10}$.

1 $x = \frac{1}{10}$: $1 + \frac{5}{16} + \frac{45}{1024}$

substitute

ANSWER $\frac{1389}{1024}$, $k = 1024$

IDEA 04

Practise: Approximations: Choose the Right x

My questions first, then selected practice, then real past papers.

Q31 DR ESLAM Choose x

P2 · TOPIC 4 3 marks

Expand $(1 + 2x)^5$ up to x^2 and use it to estimate 1.02^5 .

SELECTED PRACTICE**Q32** PRACTICE Practice · Easy

P2 · TOPIC 4 5 marks

8 (a) Find the first three terms in the expansion of $(5 + 2x)^5$.

(3 marks)

(b) Use your answer to part (a) to estimate the value of $(5.04)^5$.

(2 marks)

Q33**PRACTICE**

Practice · Medium

P2 · TOPIC 4 5 marks

5 (a) Find the first three terms in the expansion of $(9 - 2x)^5$.

(3 marks)

(b) Use your answer to part (a) to estimate $(8.9)^5$.

(2 marks)

Q34**PRACTICE**

Practice · Very Hard

P2 · TOPIC 4 7 marks

4 (a) Use the first three terms, in ascending powers of x , in the expansion of $(3 - 5x)^4$ to find an approximation for $(2.6)^4$.

(5 marks)

(b) Using your calculator, find the percentage error in the approximation from part (a) to the exact value of $(2.6)^4$.

(2 marks)

PAST PAPERS

Q35 PAST PAPER Pearson IAL WMA12 · June 2021 · Q4

P2 · TOPIC 4 8 marks

4. (a) Find, in ascending powers of x , up to and including the term in x^3 , the binomial expansion of

$$\left(2 + \frac{x}{8}\right)^{13}$$

fully simplifying each coefficient.

(4)

- (b) Use the answer to part (a) to find an approximation for 2.0125^{13}

Give your answer to 3 decimal places.

(3)

Without calculating 2.0125^{13}

- (c) state, with a reason, whether the answer to part (b) is an overestimate or an underestimate.

(1)

(Total 8 marks)

Q36 **PAST PAPER** Pearson IAL WMA12 · January 2025 · Q2**P2 · TOPIC 4** **5 marks**

2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(2 - 5x)^8$$

giving each term in simplest form.

(4)

This expansion is to be used to find an approximation for 2.05^8

- (b) State the value of x that should be used.

(There is no need to carry out this calculation.)

(1)

(Total for Question 2 is 5 marks)

Q37 PAST PAPER Pearson IAL WMA12 · June 2026 · Q7

P2 · TOPIC 4 9 marks

7. (a) Find, in ascending powers of x , up to and including the term in x^2 , the binomial expansion of

$$\left(1 + \frac{5}{16}x\right)^{10}$$

fully simplifying each coefficient.

(4)

- (b) Use $x = \frac{1}{10}$ and the answer to part (a) to show that an estimate for $\left(\frac{33}{32}\right)^{10}$ is

$$\frac{1389}{k}$$

where k is an integer to be found.

(3)

- (c) Hence find a rational estimate for $\sqrt[10]{1389}$

(2)

(Total for Question 7 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|--|
| 01 | Expand | $\binom{n}{r}a^{n-r}(bx)^r$; brackets round (bx) . |
| 02 | Constants | Terms with k in; the sentence becomes an equation. |
| 03 | Times a bracket | Pick only the pairs that make the power. |
| 04 | Approximations | Bracket = the number $\rightarrow x$; sign of the next terms. |

BOTTOM LINE One general term • keep the sign with x • pick the pairs • choose x from the bracket.

Answers

Binomial Expansion · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · EXPAND $(a + bx)^n$: ONE TERM AT A TIME

Q1	Dr Eslam · First four terms	$1 + 12x + 60x^2 + 160x^3$
Q2	Dr Eslam · One term only	$10 \times 4 \times (-27) = -1080$
Q3	Practice · Easy	(i) 24 (ii) 10 (iii) 20
Q4	Practice · Medium	$16 + 32x + 24x^2 + 8x^3 + x^4$
Q5	Practice · Very Hard	$\frac{n(n-1)(n-2)}{6} = 35 \Rightarrow n = 7$
Q6	Pearson IAL WMA12 · June 2022 · Q1	$1024 + 1920x + 1620x^2 + 810x^3$
Q7	Pearson IAL WMA12 · October 2023 · Q10	(i)(a) $2916x, 4860x^2, 2160x^4$. (i)(b) $x = \frac{15}{4}$. (ii)(a) Proven. (ii)(b) $\frac{48}{125}$
Q8	Pearson IAL WMA12 · January 2024 · Q2	96228

IDEA 02 · FIND THE UNKNOWN CONSTANT

- Q9** Dr Eslam · Coefficient given $28k^2 = 112 \Rightarrow k = 2$
- Q10** Practice · Easy $\binom{12}{5}p^7 = 12\,976\,128 \Rightarrow p^7 = 16384: p = 4$
- Q11** Practice · Hard $a = -\frac{1}{2}$
- Q12** Practice · Very Hard $a = 4, b = -3$ or $a = -4, b = 3$
- Q13** Pearson IAL WMA12 · January 2020 · Q2 $(a) a = \pm\sqrt{2}. (b) 17$
- Q14** Pearson IAL WMA12 · October 2021 · Q1 $(a)(i) k = -\frac{1}{4}. (a)(ii) p = \frac{15}{2}. (b) -125x^2$
- Q15** Pearson IAL WMA12 · October 2022 · Q2 $(a) k = \frac{2}{5}. (b) 32 - 32x + \frac{64}{5}x^2. (c) -32$
- Q16** Pearson IAL WMA12 · January 2023 · Q3 $(a) 128 + 56kx + \frac{21}{2}k^2x^2 + \frac{35}{32}k^3x^3. (b) k = \frac{16}{5}$
or $k = 16$
- Q17** Pearson IAL WMA12 · June 2023 · Q4 $(a) A = 243. (b)(i) p = -\frac{1}{2}. (b)(ii) C = \frac{135}{2}$
- Q18** Pearson IAL WMA12 · October 2025 · Q2 $p = \frac{3}{2}, q = \frac{405}{4}, r = 405$
- Q19** Pearson IAL WMA12 · January 2026 · Q1 $(a) 2187 + 5103kx + 5103k^2x^2 + 2835k^3x^3. (b) k = \frac{3}{5}$

IDEA 03 · MULTIPLY BY A BRACKET: PICK THE PAIRS

- Q20** Dr Eslam · Pick the pairs $(1 - x)^6 = 1 - 6x + 15x^2 \dots: 2(15) + 1(-6) = 24$
- Q21** Practice · Medium (a) $6561 + 34992x + 81648x^2$ (b) multiply by $(1 + x)$, keep up to x^2
- Q22** Practice · Hard (a) $128 - 1344x + 6048x^2$ (b) multiply by $(1 - 2x)$, keep up to x^2
- Q23** Practice · Very Hard (a) $262144 - 1769472x + 5308416x^2$ (b) multiply by $(3 - 2x^2)$, keep up to x^2
- Q24** Pearson IAL WMA12 · October 2019 · Q3 (a) $1 + 3x + \frac{33}{8}x^2 + \frac{55}{16}x^3$. (b) $\frac{77}{8}$
- Q25** Pearson IAL WMA12 · January 2021 · Q4 (a) $64 + 192px + 240p^2x^2$. (b) $p = \frac{1}{8}$ or $p = \frac{1}{120}$
- Q26** Pearson IAL WMA12 · January 2022 · Q3 (a) $256 - 256kx + 112k^2x^2 - 28k^3x^3$. (b) $k = -\frac{14}{15}$
- Q27** Pearson IAL WMA12 · June 2024 · Q1 (a) $1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3$. (b) $\frac{53}{6}$
- Q28** Pearson IAL WMA12 · June (R paper) 2024 · Q4 (a) $729 + 2916x + 4860x^2 + 4320x^3$. (b) 738
- Q29** Pearson IAL WMA12 · October 2024 · Q5 (a) $64 + 192ax + 240a^2x^2$. (b) $a = -\frac{24}{5}$
- Q30** Pearson IAL WMA12 · June 2025 · Q1 (a) $1 - 28x + 336x^2 - 2240x^3$. (b) $k = 13$

IDEA 04 · APPROXIMATIONS: CHOOSE THE RIGHT x

- Q31** Dr Eslam · Choose x $1 + 10x + 40x^2$; $x = 0.01$: 1.104
- Q32** Practice · Easy (a) $3125 + 6250x + 5000x^2$ (b) $x = 0.02$: 3252
- Q33** Practice · Medium (a) $59049 - 65610x + 29160x^2$ (b) $x = 0.05$: 55841.4
- Q34** Practice · Very Hard (a) $81 - 540x + 1350x^2$, $x = 0.08$: 46.44 (b) exact 45.6976: error $\approx 1.62\%$
- Q35** Pearson IAL WMA12 · June 2021 · Q4 (a) $8192 + 6656x + 2496x^2 + 572x^3$. (b) 8883.132. (c) Underestimate
- Q36** Pearson IAL WMA12 · January 2025 · Q2 (a) $256 - 5120x + 44800x^2 - 224000x^3$. (b) $x = -0.01$
- Q37** Pearson IAL WMA12 · June 2026 · Q7 (a) $1 + \frac{25}{8}x + \frac{1125}{256}x^2$ (b) $1 + \frac{5}{16} + \frac{45}{1024} = \frac{1389}{1024}$. $k = 1024$ (c) $\frac{33}{16}$

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Arithmetic Sequences and Series

Add the same d every time. Two formulas do everything: the n th term and the sum. Two facts give two equations in a and d .

THE TOPIC ON ONE MAP

Arithmetic Sequences and Series

01 — The n th term

$$a + (n - 1)d$$

two terms $\rightarrow a, d$

algebraic terms

02 — The sum

$$\frac{n}{2}(2a + (n - 1)d)$$

$$\frac{n}{2}(a + l)$$

quadratic in n

03 — In context

year n total after N years

rows of seats

THE TWO FORMULAS

$$u_n = a + (n - 1)d \quad S_n = \frac{n}{2}\{2a + (n - 1)d\} = \frac{n}{2}(a + l)$$

a first term d common difference l last term n number of terms (a whole number)

01 The n th Term: $u_n = a + (n - 1)d$

Any term is the first term plus $(n - 1)$ steps of d . Two terms given means two equations: subtract them to find d . Three consecutive terms in algebra: the middle one is the average of its neighbours.

WHEN YOU SEE → YOU DO

“the 4th and 8th terms are 20 and 64”

→ $a + 3d = 20, a + 7d = 64$: subtract

“ $(p - 9), -7, (9 - 3p)$ are the first three terms”

→ $2(\text{middle}) = \text{first} + \text{third}$

“the number of terms in $2 + 5 + 8 + \dots + 254$ ”

→ $254 = 2 + (n - 1)3$

“find the 18th term”

→ $a + 17d$

WORKED MODEL · JANUARY 2025

$2 + 5 + 8 + 11 + \dots + 254$. How many terms?

1 $254 = 2 + (n - 1) \times 3$

last term

ANSWER $n - 1 = 84 \Rightarrow n = 85$

IDEA 01

Practise: The n th Term: $u_n = a + (n - 1)d$

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Two terms given

P2 · TOPIC 5 3 marks

The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find a and d .

Subtract the two equations: $5d = 25$.

Q2 DR ESLAM Algebraic terms

P2 · TOPIC 5 3 marks

x , $2x + 1$, $5x - 2$ are consecutive terms of an arithmetic sequence. Find x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 5

2 marks

1 Write down the next three terms in these arithmetic sequences

(i) 30, 18, 6, ...

(ii) $\frac{1}{4}, \frac{5}{12}, \frac{7}{12}, \frac{3}{4}, \dots$

(2 marks)

Q4**PRACTICE**

Practice · Medium

P2 · TOPIC 5

4 marks

3 An arithmetic sequence has first term r^2 and common difference $3r$, where $r > 0$. The fifth term of the sequence is 85.

Find:

(i) the value of r

(ii) the ninth term in the sequence.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 5 4 marks

2 The first three terms in an arithmetic sequence are $2, 3q^2, q, \dots$

Given that the first three terms in the sequence are all positive, find the fortieth term in the sequence.

(4 marks)

PAST PAPERS

Q6 **PAST PAPER** Pearson IAL WMA12 · January 2025 · Q1**P2 · TOPIC 5** **4 marks**

1. The arithmetic series S is given by

$$S = 2 + 5 + 8 + 11 + \dots + 254$$

Find

- (a) the number of terms in the series, (2)
- (b) the sum of the series. (2)

(Total for Question 1 is 4 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q1

P2 · TOPIC 5 4 marks

1. An arithmetic series starts

$$78 + 75 + 72 + \dots$$

Find

- (a) the 18th term in the series, (2)

- (b) the sum of the first 100 terms of the series. (2)

(Total for Question 1 is 4 marks)

02 The Sum: $S_n = \frac{n}{2}\{2a + (n - 1)d\}$

Use $\frac{n}{2}(a + l)$ when you know the last term, the long form otherwise. A sum that is given as a number, with n unknown, becomes a quadratic in n : keep the positive whole-number root.

WHEN YOU SEE → YOU DO

“the sum of the first ten terms is -80”

→ $5(2a + 9d) = -80$

“the sixth term is 2” and a sum

→ two equations in a, d

“find the smallest n for which $S_n > 8000$ ”

→ solve $S_n = 8000$, then round **up**

“show that $S_n = \frac{n}{2}\{2a + (n - 1)d\}$ ”

→ write the series forwards and backwards, add

“900 + 892 + 884 + ... + 500”

→ n from the last term, then $\frac{n}{2}(a + l)$

Write the facts

Two equations

Solve for a and d

Answer the question

WORKED MODEL · JUNE 2024

$u_6 = 2$ and $S_{10} = -80$. Find a and d .

1 $a + 5d = 2$ and $5(2a + 9d) = -80 \Rightarrow 2a + 9d = -16$

two equations

ANSWER $d = 20, a = -98$

× **TRAP** Giving $n = 34.6$ as the answer. n counts terms: round up for “exceeds”, and check it.

IDEA 02

Practise: The Sum: $S_n = \frac{n}{2}\{2a + (n - 1)d\}$

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM A plain sum

P2 · TOPIC 5 2 marks

Find the sum of the first 20 terms of $5 + 8 + 11 + \dots$

Q9 DR ESLAM Smallest n

P2 · TOPIC 5 4 marks

Find the smallest n for which $3 + 7 + 11 + \dots$ (to n terms) exceeds 1000.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 5 2 marks

- 2 Find the sum of the first four terms in the sequence defined by $u_n = 2n + 3$.

Justify why this sequence is an arithmetic sequence.

(2 marks)

Q11**PRACTICE**

Practice · Medium

P2 · TOPIC 5 9 marks

8 (a) Calculate the sum of all the odd numbers between 0 and 150,

$$1 + 3 + 5 + \dots + 149$$

(3 marks)

(b) An arithmetic series is defined by

$$k + 2k + 3k + \dots + 360$$

where k is an integer and a positive factor of 360.

(i) In terms of k , find an expression for the number of terms in this series.

(ii) Show that the sum of this series is $180 + \frac{64800}{k}$.

(4 marks)

(c) In terms of q , find the 100th term of the arithmetic sequence defined by

$$(3q - 7), (5q - 4), (7q - 1), \dots$$

Give your answer in simplest form.

(2 marks)

Q12**PRACTICE**

Practice · Very Hard

P2 · TOPIC 5 4 marks

- 5 The sum of the first 24 terms in an arithmetic series is nine times the sum of the first two terms in the series.

Find the sum of the first 90 terms in the series.

(4 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · June 2024 · Q2

P2 · TOPIC 5 7 marks

2. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

In an arithmetic series,

- the sixth term is 2
- the sum of the first ten terms is -80

For this series,

- (a) find the value of the first term and the value of the common difference. (4)

- (b) Hence find the smallest value of n for which

$$S_n > 8000 \quad (3)$$

(Total for Question 2 is 7 marks)

Q14 PAST PAPER Pearson IAL WMA12 · June (R paper) 2024 · Q8

P2 · TOPIC 5 13 marks

8. (i) (a) In an **arithmetic** series the first term is a and the common difference is d .

Show that

$$S_n = \frac{n}{2} \{2a + (n-1)d\} \quad (3)$$

- (b) Hence find

$$900 + 892 + 884 + \dots + 500 \quad (3)$$

- (ii) Given that the first three terms of a **geometric** series are

$$k+4 \quad k-2 \quad 11-k$$

where k is a constant,

- (a) show that

$$2k^2 - 11k - 40 = 0 \quad (3)$$

Given also that this series is convergent,

- (b) find the value of S_∞ (4)

(Total for Question 8 is 13 marks)

03 Arithmetic Series in Context

Year 1, row 1, month 1 is the first term. “Each year ...less than the previous” is $d < 0$. “Mass extracted in year 14” is a term; “total extracted” is a sum.

WHEN YOU SEE → YOU DO

“480 tonnes in year 1, 465 in year 2 ...”

→ $a = 480, d = -15$

“the mass extracted in year 14”

→ u_{14}

“after a total of 7770 tonnes ...at the end of year N ”

→ $S_N = 7770$: quadratic in N

“each row has 4 more seats”

→ $d = 4$

“hence state the value of N ”

→ the root that makes sense

WORKED MODEL · OCTOBER 2021

480, 465, 450, ... tonnes. Total 7770 tonnes after N years. Show $N^2 - 65N + 1036 = 0$.

$$1 \quad \frac{N}{2}\{960 - 15(N - 1)\} = 7770$$

sum

$$2 \quad N(975 - 15N) = 15540 \Rightarrow 15N^2 - 975N + 15540 = 0$$

expand

$$\text{ANSWER} \quad N^2 - 65N + 1036 = 0 \Rightarrow N = 28 \text{ (37 gives negative years)}$$

IDEA 03

Practise: Arithmetic Series in Context

My questions first, then real past papers.

Q15 **DR ESLAM** Savings plan

P2 · TOPIC 5 3 marks

Sam saves £50 in month 1 and £10 more each month than the month before. How much has Sam saved after 2 years?

PAST PAPERS

Q16 PAST PAPER Pearson IAL WMA12 · October 2021 · Q5

P2 · TOPIC 5 6 marks

5. A company that owned a silver mine

- extracted 480 tonnes of silver from the mine in year 1
- extracted 465 tonnes of silver from the mine in year 2
- extracted 450 tonnes of silver from the mine in year 3

and so on, forming an arithmetic sequence.

(a) Find the mass of silver extracted in year 14

(2)

After a total of 7770 tonnes of silver was extracted, the company stopped mining.

Given that this occurred at the end of year N ,

(b) show that

$$N^2 - 65N + 1036 = 0$$

(3)

(c) Hence, state the value of N .

(1)

(Total 6 marks)

Q17 PAST PAPER Pearson IAL WMA12 · October 2023 · Q8

P2 · TOPIC 5 7 marks

8. In a large theatre there are n rows of seats, where n is a constant.

The number of seats in the first row is a , where a is a constant.

In each subsequent row there are 4 more seats than in the previous row so that

- in the 2nd row there are $(a + 4)$ seats
- in the 3rd row there are $(a + 8)$ seats
- the number of seats in each row form an **arithmetic** sequence

Given that the **total** number of seats in the first 10 rows is 360

(a) find the value of a . (2)

Given also that the total number of seats in the n rows is 2146

(b) show that (2)

$$n^2 + 8n - 1073 = 0$$

(c) Hence (3)

- (i) state the number of rows of seats in the theatre,
- (ii) find the maximum number of seats in any one row.

(Total for Question 8 is 7 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|---|
| 01 | <i>n</i>th term | $a + (n - 1)d$; two terms, subtract; middle = average. |
| 02 | Sum | $\frac{n}{2}\{2a + (n - 1)d\}$ or $\frac{n}{2}(a + l)$; n is a whole number. |
| 03 | Context | Year 1 = first term; term vs total. |

BOTTOM LINE Name a and d · term or sum? · two facts → two equations
· n must be a whole number.

Answers

Arithmetic Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE n TH TERM: $u_n = a + (n - 1)d$

- Q1** Dr Eslam · Two terms given $d = 5, a = 2$
- Q2** Dr Eslam · Algebraic terms $2(2x + 1) = x + (5x - 2) \Rightarrow x = 2$
- Q3** Practice · Easy (i) $-6, -18, -30$ (ii) $d = \frac{1}{6}; \frac{11}{12}, \frac{13}{12}, \frac{5}{4}$
- Q4** Practice · Medium (i) $r^2 + 12r = 85 \Rightarrow r = 5$ (ii) $u_9 = 25 + 8(15) = 145$
- Q5** Practice · Very Hard $6q^2 = 2 + q \Rightarrow q = \frac{2}{3}$ (positive); $d = -\frac{2}{3}; u_{40} = 2 + 39(-\frac{2}{3}) = -24$
- Q6** Pearson IAL WMA12 · January 2025 · Q1 (a) 85. (b) 10880
- Q7** Pearson IAL WMA12 · October 2025 · Q1 (a) 27. (b) -7050

IDEA 02 · THE SUM: $S_n = \frac{n}{2}\{2a + (n - 1)d\}$

- Q8** Dr Eslam · A plain sum $10(10 + 57) = 670$
- Q9** Dr Eslam · Smallest n $S_n = 2n^2 + n > 1000$: $n = 22$ gives 990, $n = 23$ gives 1081: $n = 23$
- Q10** Practice · Easy $5 + 7 + 9 + 11 = 32$; the difference $u_{n+1} - u_n = 2$ is constant
- Q11** Practice · Medium (a) 75 terms: 5625 (b)(i) $\frac{360}{k}$ (ii) $\frac{360}{2k}(k + 360) = 180 + \frac{64800}{k}$ (c) $d = 2q + 3$: $201q + 290$
- Q12** Practice · Very Hard $12(2a + 23d) = 9(2a + d) \Rightarrow a = -44.5d$; $S_{90} = 45(2a + 89d) = 0$
- Q13** Pearson IAL WMA12 · June 2024 · Q2 (a) First term = -98, common difference = 20. (b) $n = 35$
- Q14** Pearson IAL WMA12 · June (R paper) 2024 · Q8 (i)(a) Proven. (i)(b) 35700. (ii)(a) Proven. (ii)(b) $S_\infty = 24$

IDEA 03 · ARITHMETIC SERIES IN CONTEXT

- Q15** Dr Eslam · Savings plan $a = 50, d = 10, n = 24$: $12(100 + 230) = £3960$
- Q16** Pearson IAL WMA12 · October 2021 · Q5 (a) 285 tonnes. (b) Proven: $N^2 - 65N + 1036 = 0$. (c) $N = 28$
- Q17** Pearson IAL WMA12 · October 2023 · Q8 (a) $a = 18$. (b) Proven. (c)(i) 29 rows. (c)(ii) 130 seats

Geometric Sequences and Series

Multiply by the same r every time. Divide terms to find r , use logs when n is in the power, and only add to infinity when $|r| < 1$.



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THE TOPIC ON ONE MAP

Geometric Sequences and Series

01 - The n th term

$$ar^{n-1}$$

divide two terms

middle² = product

02 - The sum S_n

$$\frac{a(1-r^n)}{1-r}$$

smallest n : logs

prove the formula

03 - Sum to infinity

$$\frac{a}{1-r}$$

$$|r| < 1$$

choose the right r

THE THREE FORMULAS

$$u_n = ar^{n-1} \quad S_n = \frac{a(1-r^n)}{1-r} \quad S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

n in the power $\Rightarrow$ take logs $\cdot$ log of a number less than 1 is negative: the inequality flips

01 The n th Term: $u_n = ar^{n-1}$

Divide one term by another to cancel a and leave a power of r . Three consecutive terms in algebra: (middle)² = first $\times$ third.

WHEN YOU SEE $\rightarrow$ YOU DO

“the 3rd term is 20 and the 5th term is 12.8”

$$\rightarrow \frac{ar^4}{ar^2} = r^2$$

“ $2k-15$, k , $k+4$ are consecutive terms”

$$\rightarrow k^2 = (2k-15)(k+4)$$

“the n th term is greater than 10^{100} ”

$$\rightarrow 4 \times 6^{n-1} > 10^{100}: \text{logs}$$

“is 8192 a term?”

$$\rightarrow \text{solve } ar^{n-1} = 8192: \text{ is } n \text{ a whole number?}$$

WORKED MODEL • JUNE 2023

Downloads $2k-15$, k , $k+4$ (thousands) are geometric. Show $k^2 - 7k - 60 = 0$.

$$1 \quad \frac{k}{2k-15} = \frac{k+4}{k}$$

same ratio

$$2 \quad k^2 = 2k^2 - 7k - 60$$

cross-multiply

ANSWER $k^2 - 7k - 60 = 0 \Rightarrow k = 12 \ (k > 7.5)$

IDEA 01

Practise: The n th Term: $u_n = ar^{n-1}$

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Divide the terms

P2 · TOPIC 6 3 marks

The 3rd term of a geometric sequence is 12 and the 6th term is 96. Find a and r .

$$\frac{ar^5}{ar^2} = r^3.$$

Q2 DR ESLAM Algebraic terms

P2 · TOPIC 6 3 marks

x , $x + 4$, $4x$ are consecutive terms of a geometric sequence, $x > 0$. Find x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 6**4 marks**

1 Identify which of the following are geometric sequences. For those that are, write down the first term and the common ratio.

(i) 3, 8, 13, 18, ...

(ii) 5, 15, 45, 135, ...

(iii) 5, -10, 20, -40, ...

(iv) $\frac{1}{3}, \frac{1}{6}, \frac{1}{12}, \dots$

(4 marks)

Q4**PRACTICE**

Practice · Medium

P2 · TOPIC 6 6 marks

- 1 (a) The first three terms of a geometric sequence are given by x^2 , $4x$, and $x + 14$ respectively, where $x > 0$.

Show that $x^3 - 2x^2 = 0$.

(2 marks)

- (b) Find the value of the 15th term of the sequence.

(3 marks)

- (c) State, with a reason, whether 8192 is a term in the sequence.

(1 mark)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 6**5 marks**

- 1 The first three terms of a geometric sequence are given by $x + 11$, $5x$, and $3x^2$ respectively, where x is a non-zero real number.

Find the value of the sixth term in the sequence, giving your answer as a fraction.

(5 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · June 2023 · Q6

P2 · TOPIC 6 9 marks

6. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \quad k \quad k + 4$$

where k is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that $k^2 - 7k - 60 = 0$ (2)

(b) predict the number of downloads in the 4th month. (4)

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the N th month.

(c) Calculate the value of N according to the model. (3)

(Total for Question 6 is 9 marks)

02 The Sum S_n and the Smallest n

Write S_n with the numbers in, make r^n the subject, then take logs of both sides. Round n **up** for “greater than”. Dividing by $\log r < 0$ flips the inequality.

WHEN YOU SEE → YOU DO

“the sum of the first n terms is greater than 156”

→ $S_n > 156$, isolate r^n , logs

“smallest possible value of n ”

→ solve, then round up; check n and $n - 1$

“prove that $S_n = \frac{a(1-r^n)}{1-r}$ ”

→ write S_n and rS_n , subtract

“8% less each year”, “150 litres at the start”

→ $r = 0.92$, $a = 150$

“solutions based entirely on graphical or numerical methods are not acceptable”

→ show the log step

Write S_n

Isolate r^n

Take logs

Round up

WORKED MODEL • JANUARY 2023

$a = 31.25$, $r = 0.8$. Smallest n with $S_n > 156$.

$$1 \quad \frac{31.25(1-0.8^n)}{0.2} > 156 \Rightarrow 0.8^n < 0.0016$$

isolate r^n

$$2 \quad n > \frac{\log 0.0016}{\log 0.8} = 28.8$$

logs — the sign flips

ANSWER $n = 29$

IDEA 02

Practise: The Sum S_n and the Smallest n

My questions first, then selected practice, then real past papers.

Q7 DR ESLAM A sum

P2 · TOPIC 6 2 marks

Find the sum of the first 10 terms of $2 + 6 + 18 + \dots$

Q8 DR ESLAM Smallest n with logs

P2 · TOPIC 6 4 marks

Find the smallest n for which $3 + 6 + 12 + \dots$ (to n terms) exceeds 1000.

SELECTED PRACTICE

Q9**PRACTICE**

Practice · Easy

P2 · TOPIC 6 4 marks**2 (a)** Evaluate

$$\sum_{r=1}^5 3(2^r)$$

(2 marks)**(b)** Evaluate.

$$\sum_{r=4}^8 (-1)^r(2^r)$$

(2 marks)

Q10**PRACTICE**

Practice · Hard

P2 · TOPIC 6 5 marks

- 2 A geometric series has first term 14 and common ratio $\frac{99}{100}$.

Given that the sum of the first k terms of the series is less than 1000, find the largest possible value of k .

(5 marks)

Q11**PRACTICE**

Practice · Very Hard

P2 · TOPIC 6**4 marks**

- 4 A geometric series has first term a , and its terms are connected by the relationship $u_{n+4} = 9u_n$ for all $n \geq 1$.

Given that all the terms of the series are positive, show that the sum of the first twelve terms of the series may be written in the form

$$S_{12} = ka(\sqrt{n} + 1)$$

where k and n are positive integers and $\sqrt{n}$ is a surd.

(4 marks)

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA12 · January 2023 · Q7

P2 · TOPIC 6 7 marks

7. A geometric sequence has first term a and common ratio r , where $r > 0$

Given that

- the 3rd term is 20
- the 5th term is 12.8

(a) show that $r = 0.8$

(1)

(b) Hence find the value of a .

(2)

Given that the sum of the first n terms of this sequence is greater than 156

(c) find the smallest possible value of n .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

(Total for Question 7 is 7 marks)

Q13**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q8

P2 · TOPIC 6**8 marks**

8. (i) A geometric series has first term a and common ratio r .

Prove that the sum of the first n terms of this series S_n is given by

$$S_n = \frac{a(1-r^n)}{1-r} \quad (3)$$

- (ii) A liquid is to be stored in a barrel.

Due to evaporation, the volume of the liquid in the barrel at the end of each year is 8% less than the volume of the liquid in the barrel at the start of the year.

At the start of the first year, the barrel is filled with 150 litres of the liquid.

- (a) Show that the amount of the liquid in the barrel at the end of 6 years is approximately 91 litres. (2)

At the start of each year a new barrel is filled with 150 litres of the liquid so that, at the end of 40 years, there are 40 barrels containing the liquid.

- (b) Calculate the total amount of the liquid, to the nearest litre, in the 40 barrels at the end of 40 years. (3)

(Total for Question 8 is 8 marks)

03 Sum to Infinity: $|r| < 1$

A sum to infinity exists only when $-1 < r < 1$. When a quadratic gives two values of r , the condition chooses one. $S_{\infty} = \frac{a}{1-r}$ together with a term or a finite sum gives two equations.

WHEN YOU SEE → YOU DO

“given that the series has a sum to infinity”

→ reject $|r| \geq 1$

“ $S_{\infty} = 3a$ ”

→ $\frac{a}{1-r} = 3a \Rightarrow r = \frac{2}{3}$

“the second term is -6 , $S_{\infty} = 25$ ”

→ $ar = -6$, $\frac{a}{1-r} = 25$: eliminate a

“ $S_3 = 70.2$ and $S_{\infty} = 75$ ”

→ $\frac{S_3}{S_{\infty}} = 1 - r^3$

“is convergent”, $-1 + 3x - 9x^2 + \dots$

→ $|r| = |-3x| < 1$

WORKED MODEL · JANUARY 2025

$S_3 = 70.2$ and $S_{\infty} = 75$. Find r and a .

$$1 \quad \frac{a(1-r^3)}{1-r} \div \frac{a}{1-r} = 1 - r^3 = \frac{70.2}{75} = 0.936$$

divide

$$2 \quad r^3 = 0.064 \Rightarrow r = 0.4$$

ANSWER $a = 75(1 - 0.4) = 45$

× **TRAP** Keeping $r = \frac{6}{5}$ when the series has a sum to infinity. $|r|$ must be less than 1.

IDEA 03

Practise: Sum to Infinity: $|r| < 1$

My questions first, then selected practice, then real past papers.

Q14 **DR ESLAM** Sum to infinity

P2 · TOPIC 6 3 marks

(a) Find S_{∞} for $24 + 12 + 6 + \dots$ (b) A series has $a = 5$ and $S_{\infty} = 20$. Find r .

SELECTED PRACTICE

Q15 **PRACTICE** Practice · Easy

P2 · TOPIC 6 3 marks

8 (a) The first term of a geometric sequence is 6.

The sum to infinity is 8.

Show that the common ratio is 0.25.

(2 marks)

(b) Briefly explain why the geometric sequence with first term 6 and common ratio 0.25 has a sum to infinity.

(1 mark)

Q16**PRACTICE**

Practice · Hard

P2 · TOPIC 6**8 marks**

- 6 (a)** The second and fifth terms of a geometric series are 13.44 and 5.67 respectively. The series has first term a and common ratio r .

By first determining the values of a and r , calculate the sum to infinity of the series.

(6 marks)

- (b)** Calculate the difference between the sum to infinity of the series and the sum of the first 20 terms of the series. Give your answer accurate to 2 decimal places.

(2 marks)

Q17**PRACTICE**

Practice · Very Hard

P2 · TOPIC 6**13 marks**

- 7 (a)** The geometric series $S = u_1 + u_2 + u_3 + \dots + u_n + \dots$ is convergent, and the sum to infinity of the series is S_∞ . The first term of the series is a , and the common ratio is r .

A different series $T = u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2 + \dots$ is formed by squaring all the terms of the series S above.

Show that $T = u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2 + \dots$ is also a convergent geometric series.

(4 marks)

- (b)** The sum to infinity of the series T is T_∞ .

Express the ratio $\frac{T_\infty}{S_\infty}$ in terms of a and r , simplifying your answer as far as possible.

(3 marks)

- (c)** Show that if $T_\infty = S_\infty$, then $u_k^2 = u_{2k-1} + u_{2k}$ for all $k \geq 1$. Comment on what this shows about the relationship between the terms of the two series.

(6 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA12 · October 2021 · Q7

P2 · TOPIC 6 10 marks

7. (i) A geometric sequence has first term 4 and common ratio 6
Given that the n^{th} term is greater than 10^{100} , find the minimum possible value of n . (3)

- (ii) A different geometric sequence has first term a and common ratio r .

Given that

- the second term of the sequence is -6
- the sum to infinity of the series is 25

- (a) show that

$$25r^2 - 25r - 6 = 0 \quad (3)$$

- (b) Write down the solutions of

$$25r^2 - 25r - 6 = 0 \quad (1)$$

Hence,

- (c) state the value of r , giving a reason for your answer, (1)

- (d) find the sum of the first 4 terms of the series. (2)

(Total 10 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q6

P2 · TOPIC 6 8 marks

6. In a geometric sequence $u_1, u_2, u_3, \dots$

- the common ratio is r
- $u_2 + u_3 = 6$
- $u_4 = 8$

(a) Show that r satisfies

$$3r^2 - 4r - 4 = 0$$

(3)

Given that the geometric sequence has a sum to infinity,

(b) find u_1

(3)

(c) find S_∞

(2)

(Total 8 marks)

Q20 **PAST PAPER** Pearson IAL WMA12 · October 2022 · Q8**P2 · TOPIC 6** **7 marks**

8. A geometric sequence has first term a and common ratio r

Given that $S_{\infty} = 3a$

- (a) show that $r = \frac{2}{3}$ (2)

Given also that

$$u_2 - u_4 = 16$$

where u_k is the k^{th} term of this sequence,

- (b) find the value of S_{10} giving your answer to one decimal place. (5)

(Total for Question 8 is 7 marks)

Q21**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q4

P2 · TOPIC 6**6 marks**

4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Given that, in a particular geometric series,

- the sum of the first three terms is 70.2
- the sum to infinity is 75

find, for this series,

(a) the common ratio, (4)

(b) the first term. (2)

(Total for Question 4 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------------|---|
| 01 | nth term | Divide terms for r ; middle ² = first × third. |
| 02 | Sum S_n | Isolate r^n , logs, round up (sign flips for $r < 1$). |
| 03 | S_∞ | Only if $ r < 1$; it picks the right root. |

BOTTOM LINE Same ratio · n in the power → logs · infinity only when $|r| < 1$.

Answers

Geometric Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE n TH TERM: $u_n = ar^{n-1}$

- Q1** Dr Eslam · Divide the terms $r^3 = 8 \Rightarrow r = 2, a = 3$
- Q2** Dr Eslam · Algebraic terms $(x + 4)^2 = 4x^2 \Rightarrow x + 4 = 2x: x = 4$ (terms 4, 8, 16)
- Q3** Practice · Easy (i) not geometric (ii) $a = 5, r = 3$ (iii) $a = 5, r = -2$ (iv) $a = \frac{1}{3}, r = \frac{1}{2}$
- Q4** Practice · Medium (a) $16x^2 = x^2(x + 14) \Rightarrow x^3 - 2x^2 = 0$ (b) $x = 2: 4, 8, 16; u_{15} = 4(2^{14}) = 65536$ (c) yes: $8192 = 4(2^{11})$, the 12th term
- Q5** Practice · Very Hard $25x^2 = 3x^2(x + 11) \Rightarrow x = -\frac{8}{3}; a = \frac{25}{3}, r = -\frac{8}{5}; u_6 = -\frac{32768}{375}$
- Q6** Pearson IAL WMA12 · June 2023 · Q6 (a) Proven. (b) $\frac{64}{3}$ thousand downloads, about 21300 downloads. (c) $N = 17$

IDEA 02 · THE SUM S_n AND THE SMALLEST n

- Q7** Dr Eslam · A sum $\frac{2(3^{10}-1)}{2} = 59048$
- Q8** Dr Eslam · Smallest n with logs $3(2^n - 1) > 1000 \Rightarrow 2^n > 334.3 \Rightarrow n > 8.38:$
 $n = 9$
- Q9** Practice · Easy (a) $3(2 + 4 + 8 + 16 + 32) = 186$ (b) $16 - 32 + 64 - 128 + 256 = 176$
- Q10** Practice · Hard $1400(1 - 0.99^k) < 1000 \Rightarrow 0.99^k > \frac{2}{7} \Rightarrow k < 124.6: k = 124$
- Q11** Practice · Very Hard $r^4 = 9 \Rightarrow r = \sqrt{3}; S_{12} = \frac{a(729-1)}{\sqrt{3}-1} = 364a(\sqrt{3} + 1)$
- Q12** Pearson IAL WMA12 · January 2023 · Q7 (a) $r = 0.8.$ (b) $a = 31.25.$ (c) $n = 29$
- Q13** Pearson IAL WMA12 · October 2025 · Q8 (i) Proven. (ii)(a) Approximately 91 litres.
(ii)(b) 1664 litres

IDEA 03 · SUM TO INFINITY: $|r| < 1$

- Q14** Dr Eslam · Sum to infinity (a) $\frac{24}{1-\frac{1}{2}} = 48$ (b) $\frac{5}{1-r} = 20 \Rightarrow r = \frac{3}{4}$
- Q15** Practice · Easy (a) $\frac{6}{1-r} = 8 \Rightarrow r = \frac{1}{4}$ (b) $|r| < 1$
- Q16** Practice · Hard (a) $r^3 = 0.421875 \Rightarrow r = 0.75, a = 17.92:$
 $S_{\infty} = 71.68$ (b) $71.68 \times 0.75^{20} \approx 0.23$
- Q17** Practice · Very Hard (a) T has first term a^2 , ratio r^2 , and $r^2 < 1$ (b) $\frac{a}{1+r}$ (c) $a = 1+r$, so $u_{2k-1} + u_{2k} = ar^{2k-2}(1+r) = a^2 r^{2k-2} = u_k^2$
- Q18** Pearson IAL WMA12 · October 2021 · Q7 (i) $n = 129.$ (ii)(a) Proven. (ii)(b) $r = \frac{6}{5}$ or $-\frac{1}{5}.$ (ii)(c) $r = -\frac{1}{5}.$ (ii)(d) $\frac{624}{25}$
- Q19** Pearson IAL WMA12 · June 2022 · Q6 (a) Proven: $3r^2 - 4r - 4 = 0.$ (b) $u_1 = -27.$ (c) $S_{\infty} = -\frac{81}{5}$
- Q20** Pearson IAL WMA12 · October 2022 · Q8 (a) $r = \frac{2}{3}$ (b) $a = 43.2; S_{10} = \frac{92840}{729} \approx 127.4$
- Q21** Pearson IAL WMA12 · January 2025 · Q4 (a) $r = 0.4.$ (b) 45

DR ESLAM AHMED
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Sequences and Series

Write the first few terms. A recurrence builds each from the last; a sigma sum hides a known series; a periodic one repeats.

THE TOPIC ON ONE MAP

Sequences and Series

01 — Recurrence relations

$u_{n+1} = f(u_n)$ terms in k sum given $\rightarrow k$

02 — Sigma notation

write the first terms arithmetic or geometric? split the sum

03 — Periodic, increasing

order of a cycle sum of many cycles $(-1)^n$

WRITE THE FIRST FEW TERMS

$$\sum_{r=1}^n u_r = u_1 + u_2 + \cdots + u_n$$

number of terms = top – bottom + 1 · the pattern tells you: arithmetic, geometric, or periodic

01 Recurrence Relations: $u_{n+1} = f(u_n)$

Put $n = 1$ to get u_2 from u_1 , then $n = 2$ for u_3 . Keep the unknown constant as a letter; a sum or a given term then gives the equation for it.

WHEN YOU SEE → YOU DO

“find, in terms of k , an expression for u_3 ”

→ u_2 in k , then u_3 in k — expand

“given that $\sum_{n=1}^3 u_n = 117$ ”

→ $u_1 + u_2 + u_3 = 117$: an equation in k

“ $u_{n+2} = 3u_{n+1} - 2u_n$ ”

→ needs two starting terms: use the given ones

“ $u_{n+1} = (-1)^n u_n + k$ ”

→ $(-1)^n$ alternates the sign each step

“find the larger possible value”

→ solve the quadratic, test both

WORKED MODEL • OCTOBER 2021

$u_1 = 6$, $u_{n+1} = ku_n + 3$, $k > 0$, and $u_1 + u_2 + u_3 = 117$. Find k .

1 $u_2 = 6k + 3$, $u_3 = 6k^2 + 3k + 3$

generate

2 $6 + 6k + 3 + 6k^2 + 3k + 3 = 117 \Rightarrow 6k^2 + 9k - 105 = 0$

sum

ANSWER $2k^2 + 3k - 35 = 0 \Rightarrow k = \frac{7}{2} (k > 0)$

IDEA 01

Practise: Recurrence Relations: $u_{n+1} = f(u_n)$

My questions first, then real past papers.

Q1**DR ESLAM**

Generate terms

P2 · TOPIC 7

2 marks

$u_1 = 2$, $u_{n+1} = 3u_n - 1$. Find u_2 , u_3 and u_4 .

Q2**DR ESLAM**

Find the constant

P2 · TOPIC 7

4 marks

$u_1 = 4$, $u_{n+1} = ku_n + 2$, $k > 0$. Given $u_3 = 22$, find k .

Write u_2 and u_3 in k first.

PAST PAPERS

Q3 **PAST PAPER** Pearson IAL WMA12 · January 2021 · Q8**P2 · TOPIC 7** **7 marks**

8. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$
$$a_1 = p - 3$$

where p is a constant.

- (a) Find an expression for a_2 in terms of p , giving your answer in simplest form.

(1)

Given that $\sum_{n=1}^3 a_n = p + 15$

- (b) find the possible values of a_2

(6)

(Total 7 marks)

Q4 **PAST PAPER** Pearson IAL WMA12 · October 2021 · Q2**P2 · TOPIC 7** **5 marks**

2. A sequence is defined by

$$u_1 = 6$$
$$u_{n+1} = ku_n + 3$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for u_3 (2)

Given that $\sum_{n=1}^3 u_n = 117$

- (b) find the value of k . (3)

(Total 5 marks)

Q5 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q11**P2 · TOPIC 7** **8 marks**

11. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where a and b are constants.

(a) Find, in terms of a and b ,

(i) u_2

(ii) u_3

(2)

Given

- $\sum_{n=1}^3 u_n = 153$

- $b = a + 9$

(b) show that

$$a^2 - 5a - 66 = 0$$

(3)

(c) Hence find the larger possible value of u_2

(3)

(Total for Question 11 is 8 marks)

Q6**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q1

P2 · TOPIC 7 5 marks

1. The sequence $u_1, u_2, u_3, \dots$ satisfies

$$u_{n+2} = 3u_{n+1} - 2u_n$$

Given that

- $u_1 = 7$
- $u_3 = 4$

- (a) find the value of u_2

(2)

- (b) find $\sum_{r=1}^4 (u_r + 2r)$

(3)

(Total for Question 1 is 5 marks)

2. A sequence of numbers $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where k is a constant.

- (a) Show that $u_5 = 7$

(3)

Given that $\sum_{r=1}^4 u_r = 30$

- (b) find the value of k .

(2)

- (c) Hence find the value of $\sum_{r=1}^{150} u_r$

(2)

(Total for Question 2 is 7 marks)

02 Sigma Notation: Spot the Series

Write out the first two or three terms and the last one. A constant difference: arithmetic, use $\frac{n}{2}(a+l)$. A constant ratio: geometric. A mixed expression: split the sum into parts you know.

WHEN YOU SEE → YOU DO

$$\sum_{r=1}^k (31 - 6r)$$

→ arithmetic: $a = 25, d = -6$

$$\sum_{r=1}^{\infty} 6 \times 0.25^r$$

→ geometric: $a = 1.5, r = 0.25, S_{\infty}$

$$\sum_{n=1}^{59} \{5n + 3(-1)^n\}$$

→ split: $5 \sum n$ plus $3 \sum (-1)^n$

$$\sum_{r=5}^{k+2}$$

→ number of terms = top - bottom + 1

$$\text{"given } \sum_{r=1}^k \dots = 620004, \text{ show } k = \dots"$$

→ geometric sum = number, then logs

WORKED MODEL · JANUARY 2020

$$u_n = 5n + 3(-1)^n. \text{ Find } \sum_{n=1}^{59} u_n.$$

$$1 \quad 5 \sum_{n=1}^{59} n = 5 \times \frac{59 \times 60}{2} = 8850$$

arithmetic part

$$2 \quad 3(-1 + 1 - 1 + \dots - 1) = 3(-1) = -3$$

59 terms: one -1 left

$$\text{ANSWER} \quad 8850 - 3 = 8847$$

IDEA 02

Practise: Sigma Notation: Spot the Series

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Arithmetic inside

P2 · TOPIC 7 2 marks

Evaluate $\sum_{r=1}^{20} (3r + 2)$.

Q9 DR ESLAM Geometric inside

P2 · TOPIC 7 2 marks

Evaluate $\sum_{r=1}^8 3 \times 2^r$.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 7 4 marks

1 (a) Calculate

$$\sum_{r=1}^5 2r + 1$$

(2 marks)

- (b)** The sum given in part [a] is an arithmetic series.
Write down the first term and the common difference.

(2 marks)

Q11**PRACTICE**

Practice · Hard

P2 · TOPIC 7**4 marks**

1 Given that $\sum_{r=1}^k (31 - 6r) = -943,$

(i) Show that $(3k + 41)(k - 23) = 0$

(ii) Hence, find the value of k .

(4 marks)**Q12****PRACTICE**

Practice · Very Hard

P2 · TOPIC 7**4 marks**

7 Prove that, for all $n \geq 1,$

$$\sum_{r=1}^n (2r)^2 - \sum_{r=1}^n (2r-1)^2 = \sum_{r=1}^{2n} r$$

(4 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2020 · Q8

P2 · TOPIC 7 7 marks

8. (i) An arithmetic series has first term a and common difference d .

Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a + (n-1)d\} \quad (3)$$

- (ii) A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_n = 5n + 3(-1)^n$$

Find the value of

(a) u_5 (1)

(b) $\sum_{n=1}^{59} u_n$ (3)

(Total 7 marks)

Q14 PAST PAPER Pearson IAL WMA12 · June 2025 · Q7

P2 · TOPIC 7 9 marks

7: **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) A **geometric** series begins

$$10 + 8 + 6.4 + \dots$$

- (a) Find the sum to infinity of this series.

(2)

Given that the k th term of this series is less than 0.0005

- (b) use algebra to find the smallest possible value of k .

(3)

- (ii) An **arithmetic** series begins

$$850 + 843 + 836 + \dots$$

Given that the sum of the first n terms of this series is S_n
find the greatest possible value of S_n

(4)

(Total for Question 7 is 9 marks)

03 Periodic, Increasing and Decreasing Sequences

Generate terms until one repeats: the number of terms in one cycle is the order. For a long sum, add one cycle, multiply by the number of full cycles, and add the terms left over.

WHEN YOU SEE → YOU DO

“show that the sequence is periodic”

→ generate until u_1 comes back

“state the order”

→ the length of one cycle

“ $\sum_{n=1}^{70} u_n$ ” for order 3

→ 23 cycles + u_1

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

→ exact values repeat every 3

“periodic of period 4” with a sum given

→ 12 cycles + $u_1 + u_2 =$ the sum

“increasing, decreasing, or neither?”

→ compare u_{n+1} with u_n

WORKED MODEL • JANUARY 2024

$u_1 = 3, u_{n+1} = \frac{u_n - 3}{u_n - 2}$. Show it is periodic and find $\sum_{n=1}^{70} u_n$.

1 $u_2 = 0, u_3 = \frac{3}{2}, u_4 = 3$

repeats: order 3

2 one cycle: $3 + 0 + \frac{3}{2} = \frac{9}{2}$; $70 = 23 \times 3 + 1$

count cycles

ANSWER $23 \times \frac{9}{2} + 3 = \frac{213}{2}$

IDEA 03

Practise: Periodic, Increasing and Decreasing Sequences

My questions first, then selected practice, then real past papers.

Q15 DR ESLAM Periodic

P2 · TOPIC 7 4 marks

$u_1 = 2, u_{n+1} = \frac{1}{1 - u_n}$. Show that the sequence is periodic, state its order, and find $\sum_{r=1}^{30} u_r$.

SELECTED PRACTICE

Q16**PRACTICE**

Practice · Medium

P2 · TOPIC 7 6 marks

- 5 (a) The terms of a sequence are defined by $u_k = k^2$ for all $k \geq 1$.

State, with a reason, whether this sequence is increasing, decreasing, or neither.

(1 mark)

- (b) It can be shown that, for all $n \geq 1$,

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$$

Using that formula,

Calculate $\sum_{r=1}^{50} u_r$

(2 marks)

- (c) Find the value of $51^2 + 52^2 + 53^2 + \dots + 99^2 + 100^2$, i.e. the sum of the squares of all the integers between 51 and 100 inclusive.

(3 marks)

Q17**PRACTICE**

Practice · Hard

P2 · TOPIC 7 7 marks

5 (a) The terms of a sequence are defined, for all $k \geq 1$, by $u_k = (-1)^k \times k^2$.

State, with a reason, whether this sequence is increasing, decreasing, or neither.

(1 mark)

(b) It can be shown that, for all $n \geq 1$,

$$\sum_{r=1}^n (2r)^2 = \frac{2n(n+1)(2n+1)}{3} \text{ and } \sum_{r=1}^n (2r-1)^2 = \frac{n(2n+1)(2n-1)}{3}$$

Using those formulas,

Show that $\sum_{r=1}^{100} u_r = \sum_{r=1}^{100} r$.

(6 marks)

Q18**PRACTICE**

Practice · Very Hard

P2 · TOPIC 7 5 marks

6 (a) A sequence is defined for all $k \geq 1$ by

$$u_k = -2k \times (\cos(k\pi))^{k+1}$$

Determine, giving reasons for your answer, whether the sequence is increasing, decreasing, or neither.

(3 marks)

(b) A different sequence is defined for all $k \geq 1$ by

$$v_k = \sin(kq\pi)$$

where q is a real constant.

Given that the sequence is not periodic,
suggest a possible value for q , giving a reason for your answer.

(2 marks)

PAST PAPERS

Q19 PAST PAPER Pearson IAL WMA12 · October 2022 · Q3

P2 · TOPIC 7 7 marks

3. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(a) (i) a_1

(ii) a_2

(iii) a_3

(3)

(b) Hence find the exact value of

$$\sum_{n=1}^{50} \left\{ n + \cos^2\left(\frac{n\pi}{3}\right) \right\}$$

You must make your method clear.

(4)

(Total for Question 3 is 7 marks)

Q20 **PAST PAPER** Pearson IAL WMA12 · October 2023 · Q2**P2 · TOPIC 7** **5 marks**

2. A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$

$$u_{n+1} = 2 - \frac{4}{u_n}$$

(a) Find the value of u_2 , the value of u_3 and the value of u_4

(3)

(b) Find the value of

$$\sum_{r=1}^{100} u_r$$

(2)

(Total for Question 2 is 5 marks)

Q21 PAST PAPER Pearson IAL WMA12 · January 2024 · Q5

P2 · TOPIC 7 8 marks

5. (i) Find the value of

$$\sum_{r=1}^{\infty} 6 \times (0.25)^r$$

(3)

(ii) A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$

$$u_{n+1} = \frac{u_n - 3}{u_n - 2} \quad n \in \mathbb{N}$$

(a) Show that this sequence is periodic. (2)(b) State the order of this sequence. (1)

(c) Hence find

$$\sum_{n=1}^{70} u_n$$

(2)

(Total for Question 5 is 8 marks)

Q22 PAST PAPER Pearson IAL WMA12 · June 2026 · Q3

P2 · TOPIC 7 9 marks

3. The first four terms of a sequence are

$$3, -6, u_3, -24, \dots$$

(a) Decide, showing detailed reasoning, whether the sequence could be

- (i) an arithmetic sequence,
- (ii) a geometric sequence.

State, for each part, the value of u_3 if such a sequence is possible.

(6)

(b) Given instead that

- the sequence is periodic of period 4
- $\sum_{i=1}^{50} u_i = 141$

find the value of u_3

(3)

(Total for Question 3 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-------------------|---|
| 01 | Recurrence | Generate in k ; the sum gives the equation. |
| 02 | Sigma | Write the terms; arithmetic, geometric or split. |
| 03 | Periodic | One cycle; cycles $\times$ cycle-sum + leftovers. |

BOTTOM LINE Always write the first few terms $\rightarrow$ the pattern tells you which formula to use.

Answers

Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · RECURRENCE RELATIONS: $u_{n+1} = f(u_n)$

- Q1** Dr Eslam · Generate terms 5, 14, 41
- Q2** Dr Eslam · Find the constant $u_3 = 4k^2 + 2k + 2 = 22 \Rightarrow (2k + 5)(k - 2) = 0$:
 $k = 2$
- Q3** Pearson IAL WMA12 · January 2021 · Q8 $(a) a_2 = 2p^2 - 7. \quad (b) a_2 = -7 \text{ or } \frac{1}{2}$
- Q4** Pearson IAL WMA12 · October 2021 · Q2 $(a) u_3 = 6k^2 + 3k + 3. \quad (b) k = \frac{7}{2}$
- Q5** Pearson IAL WMA12 · June 2023 · Q11 $(a)(i) u_2 = b - 3a. \quad (a)(ii) u_3 = b - ab + 3a^2.$
 $(b) \text{ Proven.} \quad (c) 21$
- Q6** Pearson IAL WMA12 · June (R paper) 2024 · Q1 $(a) u_2 = 6. \quad (b) 37$
- Q7** Pearson IAL WMA12 · October 2024 · Q2 $(a) \text{ Proven.} \quad (b) k = 15. \quad (c) 1125$

IDEA 02 · SIGMA NOTATION: SPOT THE SERIES

- Q8** Dr Eslam · Arithmetic inside $a = 5, l = 62: 10(5 + 62) = 670$
- Q9** Dr Eslam · Geometric inside $a = 6, r = 2: \frac{6(2^8 - 1)}{1} = 1530$
- Q10** Practice · Easy (a) $3 + 5 + 7 + 9 + 11 = 35$ (b) $a = 3, d = 2$
- Q11** Practice · Hard (i) $31k - 3k(k + 1) = -943 \Rightarrow 3k^2 - 28k - 943 = 0$
(ii) $k = 23$
- Q12** Practice · Very Hard $\sum\{(2r)^2 - (2r - 1)^2\} = \sum(4r - 1) = 2n^2 + n = \frac{2n(2n+1)}{2} = \sum_{r=1}^{2n} r$
- Q13** Pearson IAL WMA12 · January 2020 · Q8 (i) Proven. (ii)(a) 22. (ii)(b) 8847
- Q14** Pearson IAL WMA12 · June 2025 · Q7 (i)(a) 50. (i)(b) 46. (ii) 52033

IDEA 03 · PERIODIC, INCREASING AND DECREASING SEQUENCES

- Q15** Dr Eslam · Periodic $2, -1, \frac{1}{2}, 2, \dots$: order 3; $10 \times 1.5 = 15$
- Q16** Practice · Medium (a) increasing: k^2 grows for $k \geq 1$ (b) 42925
(c) $338350 - 42925 = 295425$
- Q17** Practice · Hard (a) neither: the sign alternates (b) $171700 - 166650 = 5050 = \sum_{r=1}^{100} r$
- Q18** Practice · Very Hard (a) $(\cos k\pi)^{k+1} = (-1)^{k(k+1)} = 1$, so $u_k = -2k$: decreasing (b) any irrational q , e.g. $\sqrt{2}$: kq is never a whole number of periods
- Q19** Pearson IAL WMA12 · October 2022 · Q3 (a) $a_1 = \frac{1}{4}, a_2 = \frac{1}{4}, a_3 = 1$. (b) $\frac{2599}{2}$
- Q20** Pearson IAL WMA12 · October 2023 · Q2 (a) $u_2 = \frac{2}{3}, u_3 = -4, u_4 = 3$. (b) -8
- Q21** Pearson IAL WMA12 · January 2024 · Q5 (i) 2. (ii)(a) Periodic. (ii)(b) Order 3.
(ii)(c) $\frac{213}{2}$
- Q22** Pearson IAL WMA12 · June 2026 · Q3 (a)(i) $d = -9$ and $3 + 3(-9) = -24$: yes, $u_3 = -15$
(ii) $r = -2$ and $3(-2)^3 = -24$: yes, $u_3 = 12$ (b)
 $12(u_3 - 27) + 3 - 6 = 141$: $u_3 = 39$

DR ESLAM AHMED
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Modelling with Sequences and Series

Read the story for one word: “the same amount” means arithmetic, “the same percentage” means geometric. Then it is Topic 5 or Topic 6.

THE TOPIC ON ONE MAP

Modelling with Sequences and Series

01 – Arithmetic models

same amount year n total to date

02 – Geometric models

same % $r = 1 \pm \frac{p}{100}$ logs for n

03 – Two models, limits

model A vs B limit = S_{∞} $N = ka^t$

TRANSLATE THE STORY

“increases by 5 each year” $\rightarrow d = 5$ “increases by 2% each year”
 $\rightarrow r = 1.02$ “decreases by 8%” $\rightarrow r = 0.92$

year 1 is the first term · “in year n ” = a term · “total” = a sum · “limit to the total”
 $= S_{\infty}$

01 Arithmetic Models: the Same Amount Each Time

The first year (week, row, hit) is a ; the fixed change is d . “How much in week 10” is u_{10} ; “how much altogether” is S_n . If the question gives a total, solve the quadratic for n .

WHEN YOU SEE $\rightarrow$ YOU DO

“£5 in week 1, £5.25 in week 2, £5.50 in week 3”

$\rightarrow$ arithmetic: $a = 5, d = 0.25$

“state with a reason whether arithmetic or geometric”

$\rightarrow$ constant *difference* $\rightarrow$ arithmetic

“the number of weeks to save £350”

$\rightarrow S_n \geq 350$: solve, round up

“the additional distance each hit”

$\rightarrow$ each hit is a term; total depth is the sum

WORKED MODEL · JUNE 2021

Adina saves £5, £5.25, £5.50, How many weeks to save £350?

1 $\frac{n}{2}\{10 + 0.25(n - 1)\} \geq 350$

sum

2 $n^2 + 39n - 2800 \geq 0 \Rightarrow n \geq 36.9$

quadratic

ANSWER 37 weeks

IDEA 01

Practise: Arithmetic Models: the Same Amount Each Time

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Salary rises

P2 · TOPIC 8 3 marks

A salary is £24 000 in year 1 and rises by £1500 each year. Find the salary in year 10 and the total earned over 10 years.

SELECTED PRACTICE

Q2**PRACTICE**

Practice · Easy

P2 · TOPIC 8**8 marks**

- 1 (a)** Lauren is to start training in order to run a marathon.

Each week she will run a number of miles according to the formula

$$u_n = 4n - 1$$

where u_n is the number of miles to be run in week n .

Work out how far Lauren will run in weeks 1, 2 and 3.

(2 marks)

- (b)** Work out how far Lauren will run in her 10th week of training.

(2 marks)

- (c)** Lauren tends to train for 10 weeks.

Find the total number of miles Lauren will run across all 10 weeks of her training schedule.

(3 marks)

- (d)** Explain why the model would become unrealistic for large values of n , for example for the training schedule of an elite athlete.

(1 mark)

Q3**PRACTICE** Practice · Medium**P2 · TOPIC 8 7 marks**

- 1 (a)** Lloyd is to start training in order to run a marathon.

For the first week of training he will run a total of 2 miles.

Each subsequent week he'll increase the total number of miles run by 3 miles.

He intends training for 15 weeks.

Calculate how far Lloyd will run during his eighth week of training

(2 marks)

- (b)** Work out how much further Lloyd will run in his last week of training compared to his first.

(2 marks)

- (c)** Find the total number of miles Lloyd will run across all 15 weeks of his training schedule.

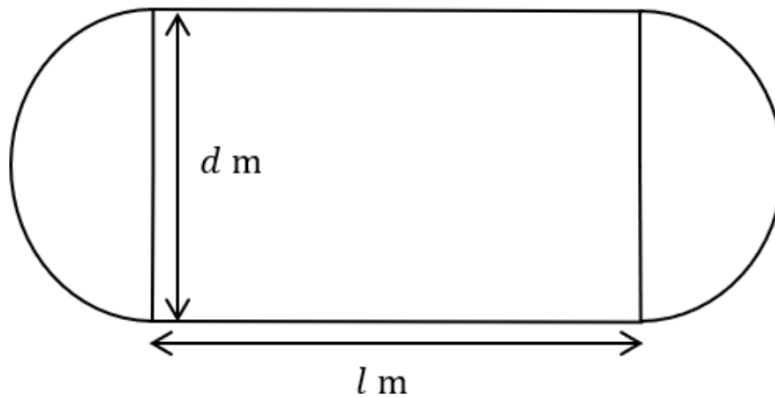
(3 marks)

Q4**PRACTICE**

Practice · Very Hard

P2 · TOPIC 8 7 marks

- 4 (a) A training track for cyclists is in the shape of a rectangle and two semi-circles as shown below. The track is also made up of several lanes.



Not
to scale

The shortest, inner lane, as shown in the diagram, has straight runs of l m, with the semi-circles at each end having a diameter of d m.

Each lane moving outwards increases the diameter by e m compared to the previous lane.

Show that the distances of each lane form an arithmetic sequence with first term $\pi d + 2l$ m and common difference πe m.

(2 marks)

- (b) There is a total of 12 lanes on the training track.

Given that l, d and e are integers and that the total distance for all 12 laps is $96(4\pi + 5)$ m, find the value of l and show that $2d + 11e = 64$.

(3 marks)

- (c) It is recommended that lanes are at least 2 m wide to allow sufficient space between cyclists in different lanes.

Find the least value of e and the associated value of d .

(2 marks)

PAST PAPERS

Q5 PAST PAPER Pearson IAL WMA12 · June 2021 · Q1

P2 · TOPIC 8 7 marks

1. Adina is saving money to buy a new computer. She saves £5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she has enough money, in total, to buy the computer.

She decides to model her savings using either an arithmetic series or a geometric series.

Using the information given,

- (a) (i) state with a reason whether an arithmetic series or a geometric series should be used,
- (ii) write down an expression, in terms of n , for the amount, in pounds (£), saved in week n .

(3)

Given that the computer Adina wants to buy costs £350

- (b) find the number of weeks it will take for Adina to save enough money to buy the computer.

(4)

(Total 7 marks)

02 Geometric Models: the Same Percentage Each Time

Growth of $p\%$ means multiply by $1 + \frac{p}{100}$; decay means $1 - \frac{p}{100}$. The value after n steps is ar^n if you start counting at 0, ar^{n-1} if year 1 is the first term — read carefully. n in the power: logs.

WHEN YOU SEE → YOU DO

“increase by 2% each year”

→ $r = 1.02$

“the population at the start of 2032”
(2019 = first)

→ 2032 is the 14th term: ar^{13}

“three years later there were 34000”

→ $30000r^3 = 34000$

“the smallest integer N for which it exceeds 75000”

→ logs, then round up

“ $N = k\lambda^t$ ”, “half its initial value”

→ $\lambda^{5700} = \frac{1}{2}$

“in the form $N = ab^t$ ”

→ two data points: divide to find b

Find r

Write the term or sum

Take logs

Round up

WORKED MODEL • JANUARY 2020

30000 bees; 34000 three years later; $p\%$ growth. Find p .

1 $30000r^3 = 34000 \Rightarrow r = \sqrt[3]{\frac{34}{30}} = 1.0426$

three steps

ANSWER $p = 4.26$

× **TRAP** Using $r = 0.02$ for “increases by 2%”. The multiplier is 1.02.

IDEA 02

Practise: Geometric Models: the Same Percentage Each Time

My questions first, then selected practice, then real past papers.

Q6 DR ESLAM Percentage growth

P2 · TOPIC 8 3 marks

A population of 8000 grows by 5% a year. After how many complete years does it first exceed 20 000?

$r = 1.05$; take logs.

Q7 DR ESLAM Depreciation

P2 · TOPIC 8 2 marks

A car worth £20 000 loses 15% of its value each year. Find its value after 6 years.

SELECTED PRACTICE

Q8**PRACTICE** Practice · Easy**P2 · TOPIC 8 8 marks**

- 3 (a)** A ball is dropped from the top of a building and is allowed to bounce until it comes to rest. The height the ball reaches after each bounce is modelled by the geometric sequence

$$u_n = 2 \times (0.8)^{n-1}$$

[i] Write down the height the ball reaches after its first bounce and the common ratio between subsequent bounces.

[ii] Find the height the ball bounces to after its fifth bounce.

(3 marks)

- (b)** [i] Find the sum of the heights the ball reaches after 10 bounces.

[ii] Explain why the total distance travelled by the ball from the moment it first hits the ground to the moment it returns to the ground after the 10th bounce is twice your answer to part (i).

(4 marks)

- (c)** State whether the sequence $u_1, u_2, u_3, \dots$ is increasing or decreasing.

(1 mark)

Q9**PRACTICE**

Practice · Hard

P2 · TOPIC 8 9 marks

2 (a) Stephen opens a savings account with £600.

Compound interest is paid annually at a rate of 1.2%.

At the start of each new year Stephen pays another £600 into his account.

Show that at the end of two years Stephen has £1221.69 in the account

(2 marks)

(b) Show that at the end of year n , the amount of money, in pounds, Stephen will have in his account is given by

$$600(1.012 + 1.012^2 + 1.012^3 + \dots + 1.012^n)$$

(2 marks)

(c) Hence show that the total amount, in pounds, in Stephen's account after n years is

$$50\,600(1.012^n - 1)$$

(2 marks)

(d) Find the year in which the amount in Stephen's account first exceeds £10 000.

(2 marks)

(e) State one assumption that has been made about this scenario.

(1 mark)

Q10**PRACTICE**

Practice · Very Hard

P2 · TOPIC 8**5 marks**

- 3 (a)** A ball is dropped from a height of A m.
It bounces to a height of x m.

Each subsequent bounce height is 25% shorter than the previous bounce.

Show that no matter how many times the ball bounces, it will not travel further than a total distance of $(A + 8x)$ m.

(4 marks)

- (b)** Write down one assumption that has been made using this model when calculating the distance.

(1 mark)

PAST PAPERS

Q11 PAST PAPER Pearson IAL WMA12 · October 2019 · Q2

P2 · TOPIC 8 6 marks

2. The adult population of a town at the start of 2019 is 25 000

A model predicts that the adult population will increase by 2% each year, so that the number of adults in the population at the start of each year following 2019 will form a geometric sequence.

- (a) Find, according to the model, the adult population of the town at the start of 2032
(3)

It is also modelled that every member of the adult population gives £5 to local charity at the start of each year.

- (b) Find, according to these models, the total amount of money that would be given to local charity by the adult population of the town from 2019 to 2032 inclusive. Give your answer to the nearest £1 000
(3)

(Total 6 marks)

Q12**PAST PAPER**

Pearson IAL WMA12 · January 2020 · Q5

P2 · TOPIC 8**8 marks**

5. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A colony of bees is being studied.

The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by $p\%$ each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

(a) find the value of p , giving your answer to 2 decimal places.

(3)

According to the model, at the end of N years of study the number of bees in the colony exceeds 75 000

(b) Find, showing all steps in your working, the smallest integer value of N .

(5)

(Total 8 marks)

Q13 PAST PAPER Pearson IAL WMA12 · June 2022 · Q9

P2 · TOPIC 8 9 marks

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

- (a) Sketch the graph of N against t for $k = 1$ (2)

Given that it takes 5700 years for the amount of carbon-14 to reduce to half its initial value,

- (b) show that the value of the constant λ is 0.999878 to 6 decimal places. (2)

Given that Item A

- is known to have had 15 grams of carbon-14 present initially
 - is thought to be 3250 years old
- (c) calculate, to 3 significant figures, how much carbon-14 the equation predicts is currently in Item A . (2)

Item B is known to have initially had 25 grams of carbon-14 present, but only 18 grams now remain.

- (d) Use algebra to calculate the age of Item B to the nearest 100 years. (3)

(Total 9 marks)

Q14 PAST PAPER Pearson IAL WMA12 · June 2024 · Q10

P2 · TOPIC 8 8 marks

10. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The number of dormice and the number of voles on an island are being monitored.

Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

- (a) Find, according to the model, the number of dormice on the island 6 years after monitoring began. Give your answer to 3 significant figures.

(2)

The number of voles on the island is being monitored over the same period of time.

Given that

- 4 years after monitoring began there were 3690 voles on the island
- 7 years after monitoring began there were 3470 voles on the island
- the number of voles on the island at the end of each year is modelled as a geometric sequence

- (b) find the equation of this model in the form

$$N = ab^t$$

where N is the number of voles, t years after monitoring began and a and b are constants. Give the value of a and the value of b to 2 significant figures.

(3)

When $t = T$, the number of dormice on the island is equal to the number of voles on the island.

- (c) Find, according to the models, the value of T , giving your answer to one decimal place.

(3)

(Total for Question 10 is 8 marks)

03 Two Models, Limits and Comparisons

Many questions compare Model A (arithmetic) with Model B (geometric) fitted to the same two facts. Find d for A and r for B from the same data, then answer each part with the right formula. A geometric model with $r < 1$ has a limit: S_{∞} .

WHEN YOU SEE → YOU DO

“model A: same amount; model B: same percentage”

→ d from $a + (n - 1)d$; r from ar^{n-1}

“the difference between the totals”

→ S_n for each model, subtract

“there is a limit to the total”

→ geometric with $r < 1$: $S_{\infty} = \frac{a}{1-r}$

“after the 20th hit the distances form a geometric sequence”

→ switch formula halfway

“state one reason why the model may not be realistic”

→ growth cannot go on for ever; values must be whole numbers

WORKED MODEL · JANUARY 2024

300 tonnes in year 1, 4000 in year 12. Model A: same amount; model B: same percentage.

1 A: $300 + 11d = 4000 \Rightarrow d = 336.4$: year 4 ≈ 1310

arithmetic

2 B: $300r^{11} = 4000 \Rightarrow r = 1.2656$: year 2 ≈ 380

geometric

ANSWER A: 1310 t, B: 380 t

IDEA 03

Practise: Two Models, Limits and Comparisons

My questions first, then selected practice, then real past papers.

Q15 **DR ESLAM** Arithmetic or geometric?

P2 · TOPIC 8 4 marks

A company sells 500 units in year 1 and 620 in year 3. Model A: arithmetic. Model B: geometric. Find, for each model, the sales in year 6.

SELECTED PRACTICE

Q16**PRACTICE**

Practice · Easy

P2 · TOPIC 8 10 marks

6 (a) Two sequences are being used to model the value of a car, n years after it was new.

At new, the car's value is £30 000.

Model 1 is an arithmetic sequence where the value of the car, u_n , at n years old, is given by the formula $u_n = 30\,000 - 5000n$.

Model 2 is a geometric sequence where the value of the car, u_n , at n years old, is given by the formula $u_n = 30\,000(0.6)^n$.

[i] Find the age of the car when Model 1 predicts its value has halved.

[ii] Find the age of the car when Model 2 predicts its value has halved.

(4 marks)

(b) Which model predicts the greater value for the car when it is 5 years old?

(3 marks)

(c) [i] State a problem with Model 1 for higher values of n .

[ii] Justify why Model 2 never predicts the value of the car to be £0.

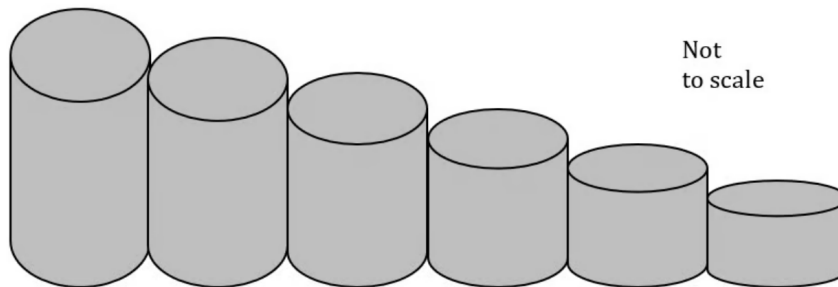
(3 marks)

Q17**PRACTICE**

Practice · Hard

P2 · TOPIC 8 5 marks

- 6 (a)** A model maker is constructing part of a model building using a series of hollow tubes, stacked next to each other as illustrated below.



Each tube is one-tenth shorter than the one to its left.

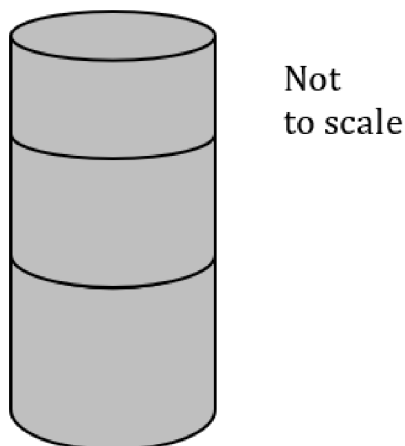
All tubes are the same width as they are all cut from one longer tube.

Show that no matter how many tubes the model maker uses, the longer tube they are cut from need not be any longer than ten times the height of the tallest tube.

(2 marks)

- (b)** In a different part of the model building, tubes are required to be stacked on top of each other, as shown below.

The tallest tube is the same length as the tallest tube above.



All the other tubes are 1 cm shorter than the tube immediately below it.

The longer tube all others are cut from is ten times the height of the tallest tube in the stack.

All of the longer tube shall be used in the stack.

If the length of the tallest tube is and there are n tubes in the stack, show that

$$n^2 - (2a + 1)n + 20a = 0.$$

(3 marks)

Q18**PRACTICE**

Practice · Very Hard

P2 · TOPIC 8 10 marks

- 5 (a)** A student is investigating a sequence of values returned from a computer program that has been reported as having a bug in it. The sequence of values is

$$3, 1, 6, \frac{1}{2}, 9, \frac{1}{4}, 12, \frac{1}{8}, 15, \frac{1}{16}, \dots$$

Explain why this sequence is neither an increasing nor a decreasing sequence.

(1 mark)

- (b)** The bug in the computer program appears to have made it output two different sequences in the same list, rather than outputting them separately.

Suggest what the two sequences could be.

(2 marks)

- (c)** To fix the bug without rewriting or searching the computer program for errors the student decides it would be easier to separate the output into one sequence using the odd numbered terms and another sequence using the even numbered terms.

Show that the odd numbered terms in the sequence form an arithmetic sequence with first term 3 and common difference 3

Find an expression for the sum of the first m terms of this sequence.

(2 marks)

- (d)** Find an expression for the sum of the first n even numbered terms of the sequence.

(2 marks)

- (e)** The computer program also outputs the sum of the first 20 terms of the sequence. (Instead of the sum of the first 20 terms for each sequence).

Find the value the computer would output, giving your answer to 4 decimal places.

(2 marks)

- (f)** Is the computer-generated sequence convergent or divergent? Give a reason for your answer.

(1 mark)

PAST PAPERS

Q19 PAST PAPER Pearson IAL WMA12 · January 2021 · Q10

P2 · TOPIC 8 11 marks

10. In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan *A*, he will increase the distance he runs each Sunday by the same amount.

- (a) Calculate the distance he will run on Sunday in week 5 under training plan *A*. (3)

In training plan *B*, he will increase the distance he runs each Sunday by the same percentage.

- (b) Calculate the distance he will run on Sunday in week 5 under training plan *B*.
Give your answer in km to one decimal place. (3)

Owen will also run a fixed distance, x km, each Wednesday over the 12-week period.

Given that

- x is an integer
 - the total distance that Owen will run on Sundays and Wednesdays over the 12 weeks will not exceed 360 km
- (c) (i) find the maximum value of x , if he uses training plan *A*,
(ii) find the maximum value of x , if he uses training plan *B*. (5)

Q20 PAST PAPER Pearson IAL WMA12 · January 2022 · Q8

P2 · TOPIC 8 9 marks

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

(a) show that the post is driven an **additional** 62 mm into the ground with the 20th hit. (1)

(b) Find the **total distance** that the post has been driven into the ground after 20 hits. (2)

Given that for each subsequent hit after the 20th hit

- the **additional** distances the post travels form a geometric sequence with common ratio r
- on the 22nd hit, the post is driven an **additional** 60 mm into the ground

(c) find the value of r , giving your answer to 3 decimal places. (2)

After a total of N hits, the post will have been driven more than 3 m into the ground.

(d) Find, showing all steps in your working, the smallest possible value of N . (4)

(Total 9 marks)

Q21 PAST PAPER Pearson IAL WMA12 · January 2024 · Q7

P2 · TOPIC 8 9 marks

7. Wheat is grown on a farm.

- In year 1, the farm produced 300 tonnes of wheat.
- In year 12, the farm is predicted to produce 4 000 tonnes of wheat.

Model *A* assumes that the amount of wheat produced on the farm will increase by the same amount each year.

- (a) Using model *A*, find the amount of wheat produced on the farm in year 4.
Give your answer to the nearest 10 tonnes.

(3)

Model *B* assumes that the amount of wheat produced on the farm will increase by the same percentage each year.

- (b) Using model *B*, find the amount of wheat produced on the farm in year 2.
Give your answer to the nearest 10 tonnes.

(3)

- (c) Calculate, according to the two models, the difference between the total amounts of wheat predicted to be produced on the farm from year 1 to year 12 inclusive.
Give your answer to the nearest 10 tonnes.

(3)

(Total for Question 7 is 9 marks)

Q22**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q7

P2 · TOPIC 8**6 marks**

7. Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

(a) Show that Jem pays £69.50 into scheme A in month 100

(1)

(b) Find the **total** amount that Jem pays into scheme A over the period of 300 months.

(2)

Kim pays money into a different savings scheme, B , over the same period of 300 months.

In a model, the amounts Kim pays into scheme B increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme B in month 1
- £250 into scheme B in month 300

(c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme A and the total amount paid into scheme B over the period of 300 months.

(3)

(Total for Question 7 is 6 marks)

Q23**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q9

P2 · TOPIC 8 10 marks

9. A company mines lithium.

- in year 1, the company mined 12 400 tonnes of lithium
- in year 15, the company is predicted to mine 7 500 tonnes of lithium

Model *A* assumes that the mass of lithium the company mines will decrease by the same amount each year.

According to model *A*,

- (a) find the mass of lithium the company will mine in year 5 (3)
- (b) calculate the **total** mass of lithium the company will mine from year 1 to year 15 inclusive. (2)

Model *B* assumes that the mass of lithium the company mines will decrease by the same percentage each year.

- (c) Find, according to model *B*, the mass of lithium the company will mine in year 10. Give the answer to the nearest 10 tonnes. (3)

According to model *B*, there is a limit to the **total** mass of lithium that can be mined.

- (d) Calculate the value of this limit. Give the answer to the nearest 100 tonnes. (2)

(Total for Question 9 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Arithmetic model** Same amount = d ; term for “in year n ”, sum for “total”.
- 02 Geometric model** Same % $\rightarrow r = 1 \pm \frac{p}{100}$; logs for n .
- 03 Two models** Fit both to the same data; limit = S_{∞} .

BOTTOM LINE Same amount $\rightarrow$ arithmetic · same percentage $\rightarrow$ geometric · which year is the first term?

Answers

Modelling with Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · ARITHMETIC MODELS: THE SAME AMOUNT EACH TIME

- Q1** Dr Eslam · Salary rises $24000 + 9(1500) = £37\,500$; $5(24000 + 37500) = £307\,500$
- Q2** Practice · Easy (a) 3, 7, 11 miles (b) 39 (c) $5(3 + 39) = 210$
(d) the distance keeps growing without limit
- Q3** Practice · Medium (a) $2 + 7(3) = 23$ (b) $44 - 2 = 42$ miles more
(c) $7.5(2 + 44) = 345$
- Q4** Practice · Very Hard (a) each lane: $2l + \pi d$, next lane $+\pi e$ (b)
 $6\{2(\pi d + 2l) + 11\pi e\} = 384\pi + 480 \Rightarrow l = 20$,
 $2d + 11e = 64$ (c) $e = 4$, $d = 10$
- Q5** Pearson IAL WMA12 · June 2021 · Q1 (a)(i) Arithmetic series. (a)(ii) $5 + 0.25(n-1)$.
(b) 37 weeks

IDEA 02 · GEOMETRIC MODELS: THE SAME PERCENTAGE EACH TIME

- Q6** Dr Eslam · Percentage growth $8000(1.05)^n > 20000 \Rightarrow n > 18.8$: 19 years
- Q7** Dr Eslam · Depreciation $20000(0.85)^6 \approx \text{£}7\,543$
- Q8** Practice · Easy (a)(i) 2 m, $r = 0.8$ (ii) $2(0.8)^4 \approx 0.819$ m (b)(i) $\frac{2(1-0.8^{10})}{0.2} \approx 8.93$ m (ii) each bounce goes up and comes down (c) decreasing
- Q9** Practice · Hard (a) $(607.2 + 600) \times 1.012 \approx 1221.69$ (c) $600 \times 1.012 \frac{1.012^n - 1}{0.012}$ (d) $1.012^n > 1.198 \Rightarrow n > 15.1$: year 16 (e) e.g. the rate stays at 1.2%
- Q10** Practice · Very Hard (a) $A + 2(x + 0.75x + \dots) = A + \frac{2x}{0.25} = A + 8x$ (b) e.g. it bounces for ever / no energy lost to air
- Q11** Pearson IAL WMA12 · October 2019 · Q2 (a) 32341 adults, approximately. (b) £1,997,000
- Q12** Pearson IAL WMA12 · January 2020 · Q5 (a) $p = 4.26$. (b) $N = 22$
- Q13** Pearson IAL WMA12 · June 2022 · Q9 (a) Decreasing exponential through (0, 1). (b) $\lambda = 0.999878$. (c) 10.1 g. (d) 2700 years
- Q14** Pearson IAL WMA12 · June 2024 · Q10 (a) 2390 dormice. (b) $N = 4000(0.98)^t$ to 2 s.f. constants. (c) $T = 13.9$

IDEA 03 · TWO MODELS, LIMITS AND COMPARISONS

- Q15** Dr Eslam · Arithmetic or geometric? A: $d = 60$, $500 + 5(60) = 800$; B: $r^2 = 1.24$, $500(1.24)^{2.5} \approx 856$
- Q16** Practice · Easy (a)(i) $n = 3$ (ii) $0.6^n = 0.5 \Rightarrow n \approx 1.36$ (b) Model 1: £5000, Model 2: £2333 — Model 1 (c)(i) negative after 6 years (ii) $0.6^n > 0$ for all n
- Q17** Practice · Hard (a) $h + 0.9h + 0.81h + \dots = \frac{h}{0.1} = 10h$ (b) $\frac{n}{2}\{2a - (n - 1)\} = 10a \Rightarrow n^2 - (2a + 1)n + 20a = 0$
- Q18** Practice · Very Hard (a) it goes up and down in turn (b) 3, 6, 9, ... and $1, \frac{1}{2}, \frac{1}{4}, \dots$ (c) $\frac{3m(m+1)}{2}$ (d) $2(1 - \frac{1}{2^n})$ (e) 166.9980 (f) divergent: the arithmetic terms grow without limit
- Q19** Pearson IAL WMA12 · January 2021 · Q10 (a) 23 km. (b) 20.8 km. (c)(i) $x = 4$. (c)(ii) $x = 5$
- Q20** Pearson IAL WMA12 · January 2022 · Q8 (a) Proven: 62 mm. (b) 1620 mm. (c) $r = 0.984$. (d) $N = 48$
- Q21** Pearson IAL WMA12 · January 2024 · Q7 (a) 1310 tonnes. (b) 380 tonnes. (c) 7860 tonnes
- Q22** Pearson IAL WMA12 · October 2024 · Q7 (a) Proven. (b) £28,425. (c) £1,060
- Q23** Pearson IAL WMA12 · January 2026 · Q9 (a) 11 000 tonnes (b) 149 250 tonnes (c) about 8980 tonnes (d) about 351 500 tonnes

Exponentials and Logarithms

A log answers one question: what power? Unknown in a power: take logs. Unknown inside logs: combine, then remove them.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Exponentials and Logarithms

01 The laws

$\log ab, \log \frac{a}{b}$ $\log a^n$ in terms of k

02 Exponential equations

$a^x = b$ hidden quadratic unify the base

03 Log equations

one log each side number = log reject negatives

04 Graphs and models

$y = a^x$ asymptote $N = ka^t$

THE DEFINITION AND THE THREE LAWS

$$\log_a b = c \iff a^c = b \quad \log ab = \log a + \log b$$

$$\log \frac{a}{b} = \log a - \log b \quad \log a^n = n \log a$$

$$\log_a a = 1 \quad \log_a 1 = 0 \quad \log_a \sqrt{a} = \frac{1}{2} \quad \text{a number can be written as a log: } 3 = \log_2 8$$

01 The Laws of Logarithms: Simplify and Express in k

Break the expression into the pieces you are given: powers come out as multipliers, products become sums, quotients become differences. Replace each given log by its letter at the end.

WHEN YOU SEE → YOU DO

“given $\log_a b = k$, find $\log_a \left(\frac{\sqrt{a}}{b}\right)$ ”

$$\longrightarrow \frac{1}{2} \log_a a - \log_a b = \frac{1}{2} - k$$

“given $a = \log_2 3$, write $\log_2 9$ ”

$$\longrightarrow 9 = 3^2: 2a$$

“evaluate $\log_5 625$ ”

$$\longrightarrow \text{what power of 5 gives 625? } 4$$

“ $\log_{\sqrt{a}} a^6$ ”

$$\longrightarrow a^6 = (\sqrt{a})^{12}: 12$$

a Σ of logs

$$\longrightarrow \text{simplify one term, then sum (arithmetic)}$$

WORKED MODEL • OCTOBER 2022

$a = \log_2 3$. Write $\log_2 \left(\frac{\sqrt{3}}{16}\right)$ in terms of a .

1 $\log_2 \sqrt{3} - \log_2 16$

quotient law

2 $\frac{1}{2} \log_2 3 - 4$

power law; $16 = 2^4$

ANSWER $\frac{a}{2} - 4$

IDEA 01

Practise: The Laws of Logarithms: Simplify and Express in k

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Three laws

P2 · TOPIC 9 3 marks

Given $\log_a 2 = p$ and $\log_a 3 = q$, write in terms of p and q : (a) $\log_a 12$ (b) $\log_a \frac{9}{2}$ (c) $\log_a \sqrt{6a}$.

SELECTED PRACTICE

Q2**PRACTICE**

Practice · Easy

P2 · TOPIC 9 4 marks**1** Evaluate

(i) $\log_3 27$

(ii) $\log_5 625$

(iii) $\log_2 \frac{1}{4}$

(iv) $\log_a a$

(4 marks)

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 9**2 marks****12**

Use a calculator to find the value of

(i) $5 \log_3 7$

(ii) $2 \log_2 3 + 3 \log_3 2$

giving your answers to four significant figures.

(2 marks)**Q4****PRACTICE**

Practice · Very Hard

P2 · TOPIC 9**3 marks****4**

Without using a calculator, show that

$$\log_4 8 = \log_9 27.$$

(3 marks)

PAST PAPERS

Q5 **PAST PAPER** Pearson IAL WMA12 · October 2019 · Q7**P2 · TOPIC 9** **7 marks**

7. Given $\log_a b = k$, find, in simplest form in terms of k ,

(i) $\log_a \left(\frac{\sqrt{a}}{b} \right)$ (2)

(ii) $\frac{\log_a a^2 b}{\log_a b^3}$ (2)

(iii) $\sum_{n=1}^{50} (k + \log_a b^n)$ (3)

(Total 7 marks)

Q6 **PAST PAPER** Pearson IAL WMA12 · October 2022 · Q10**P2 · TOPIC 9** **7 marks****10.** Given $a = \log_2 3$ (i) write, in simplest form, in terms of a ,

(a) $\log_2 9$

(b) $\log_2 \left(\frac{\sqrt{3}}{16} \right)$

(3)

(ii) Solve

$$3^x \times 2^{x+4} = 6$$

giving your answer, in simplest form, in terms of a .**(4)****(Total for Question 10 is 7 marks)**

02 Exponential Equations: Take Logs

Get the power on its own, then take logs of both sides: the power comes down in front. If the same base appears twice, as a^{2x} and a^x , it is a quadratic in disguise — substitute $y = a^x$.

WHEN YOU SEE → YOU DO

“solve $3^a = 70$ ”

$$\rightarrow a = \frac{\log 70}{\log 3}$$

“in the form $x = \log_7 a$ ”

$$\rightarrow 7^{x+2} = 3 \Rightarrow x + 2 = \log_7 3$$

$$5^{2x} - 8(5^x) + 12 = 0$$

$$\rightarrow y = 5^x: \text{quadratic, then } x = \log_5 y$$

“the curves $y = 3^{2x+1}$ and $y = 6 \times 3^x$ meet at P ”

$$\rightarrow \text{set equal, divide by } 3^x$$

$$8^{2x-5} = 20 \text{ in the form } a + b \log_2 5$$

$$\rightarrow \text{write } 8 = 2^3, \text{ then } \log_2 \text{ both sides}$$

$$y = a^x \text{ negative}$$

$$\rightarrow \text{no solution: } a^x > 0$$

Isolate the power

Take logs

Power comes down

Solve

WORKED MODEL · JUNE 2025

$3^{2x+1} = 6 \times 3^x$. Find x exactly.

$$1 \quad 3 \times (3^x)^2 = 6 \times 3^x$$

write 3^{2x+1} as $3 \cdot 3^{2x}$

$$2 \quad 3^x = 2$$

divide by $3 \cdot 3^x$

ANSWER $x = \log_3 2, \quad P = (\log_3 2, 12)$

IDEA 02

Practise: Exponential Equations: Take Logs

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM**

Take logs

P2 · TOPIC 9**2 marks**

Solve $5^{2x-1} = 40$, giving x to 3 significant figures.

Q8**DR ESLAM**

A hidden quadratic

P2 · TOPIC 9**4 marks**

Solve $9^x - 10(3^x) + 21 = 0$.

Let $y = 3^x$: $9^x = y^2$.

SELECTED PRACTICE

Q9**PRACTICE** Practice · Easy

P2 · TOPIC 9 2 marks

4

Solve the equation

$$7^{2x-1} = 343.$$

(2 marks)

Q10**PRACTICE** Practice · Medium

P2 · TOPIC 9 3 marks

6

Solve $2^{2x} - 24(2^x) + 128 = 0$.

(3 marks)

Q11**PRACTICE**

Practice · Very Hard

P2 · TOPIC 9 4 marks**6**Solve $3^{2(x+1)} + 3 = 28(3^x)$.**(4 marks)**

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA12 · January 2021 · Q3

P2 · TOPIC 9 8 marks

3. (i) Solve

$$7^{x+2} = 3$$

giving your answer in the form $x = \log_7 a$ where a is a rational number in its simplest form.

(3)

(ii) Using the laws of logarithms, solve

$$1 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$$

(5)

(Total 8 marks)

Q13**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q5

P2 · TOPIC 9 6 marks

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve

$$3^a = 70$$

giving the answer to 3 decimal places.

(2)

(ii) Find the exact value of b such that

$$4 + 3 \log_3 b = \log_3 5b$$

(4)

(Total for Question 5 is 6 marks)

Q14**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q6

P2 · TOPIC 9 10 marks**6:**

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(i) Solve

$$2\log_2(4-x) = 3 + \log_2\left(\frac{x+11}{2}\right) \quad (5)$$

(ii) The curves C_1 and C_2 with equations

$$y = 3^{2x+1} \quad \text{and} \quad y = 6 \times 3^x$$

meet at the point P .

Find the exact coordinates of P , writing your answer in the form $(\log_3 a, b)$
where a and b are integers.

(5)**(Total for Question 6 is 10 marks)**

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q7

P2 · TOPIC 9 9 marks

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Find the exact solution of the equation

$$8^{2x-5} = 20$$

giving the answer in the form $a + b \log_2 5$ where a and b are rational constants.

(4)

- (ii) Using the laws of logarithms, solve the equation

$$\log_3(13 + 2y) + 3 = 2\log_3(4 - y)$$

(5)

(Total for Question 7 is 9 marks)

03 Log Equations: One Log Each Side, Then Remove It

Use the laws to collect every log into a single log on each side. Write any plain number as a log of the same base. When it reads $\log_a(\dots) = \log_a(\dots)$, drop the logs. Finally, reject any answer that makes a log of a negative number.

WHEN YOU SEE → YOU DO

“ $\log_3(32 - 12x) = 2 \log_3(1 - x) + 3$ ” → $3 = \log_3 27$; $2 \log_3(1 - x) = \log_3(1 - x)^2$

a number alone on one side → $\log_a X = n \Rightarrow X = a^n$

different bases $\log_4$ and $\log_{16}$ → $\log_{16} X = \frac{1}{2} \log_4 X$

“show that $2x^3 - 3x^2 - 30x + 56 = 0$ ” → remove the logs, expand, tidy

“hence solve” → check each root: every log argument > 0

WORKED MODEL • OCTOBER 2024

Solve $\log_4(12 - 2x) = 2 + 2 \log_4(x + 1)$.

1 $\log_4(12 - 2x) = \log_4\{16(x + 1)^2\}$ $2 = \log_4 16$

2 $16x^2 + 34x + 4 = 0 \Rightarrow (8x + 1)(x + 2) = 0$ remove the logs

ANSWER $x = -\frac{1}{8}$ ($x = -2$ gives $\log_4(-1)$)

× **TRAP** Keeping a root that makes a log argument negative or zero. Always check.

IDEA 03

Practise: Log Equations: One Log Each Side, Then Remove It

My questions first, then selected practice, then real past papers.

Q16 **DR ESLAM** Combine, then remove

P2 · TOPIC 9 4 marks

Solve $\log_2(x + 6) - \log_2 x = 3$.

One log on the left, then 2^3 on the right.

Q17 **DR ESLAM** Numbers become logs

P2 · TOPIC 9 4 marks

Solve $\log_3 x + \log_3(x - 8) = 2$.

SELECTED PRACTICE

Q18 **PRACTICE** Practice · Easy

P2 · TOPIC 9 2 marks

3 Solve the equation

$$\log_x 16 = 2$$

(2 marks)

Q19 **PRACTICE** Practice · Hard

P2 · TOPIC 9 3 marks

4 Solve the equation

$$2 \log_x (x + 2) = 3$$

giving your answer correct to 3 significant figures.

(3 marks)

Q20**PRACTICE**

Practice · Very Hard

P2 · TOPIC 9 7 marks**8**Solve the equation $\log_4(2 - x) = \log_{16}(13 - 4x)$.**(7 marks)**

PAST PAPERS

Q21 PAST PAPER Pearson IAL WMA12 · June 2021 · Q7

P2 · TOPIC 9 10 marks

7. (a) Given that

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

show that

$$2x^3 - 3x^2 - 30x + 56 = 0 \quad (4)$$

(b) Show that -4 is a root of this cubic equation. (2)

(c) Hence, using algebra and showing each step of your working, solve

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25) \quad (4)$$

(Total 10 marks)

Q22**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q4

P2 · TOPIC 9 5 marks

4. Using the laws of logarithms, solve

$$\log_3(32 - 12x) = 2 \log_3(1 - x) + 3 \quad (5)$$

(Total 5 marks)

Q23**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q4

P2 · TOPIC 9 6 marks

4. In this question you must show all stages of your working.

Give your answers in fully simplified surd form.

Given that a and b are positive constants, solve the simultaneous equations

$$\begin{aligned}a - b &= 8 \\ \log_4 a + \log_4 b &= 3\end{aligned}$$

(6)

(Total 6 marks)

Q24 **PAST PAPER** Pearson IAL WMA12 · January 2023 · Q4**P2 · TOPIC 9** **6 marks**

4. (i) Using the laws of logarithms, solve

$$\log_3(4x) + 2 = \log_3(5x + 7) \quad (3)$$

- (ii) Given that

$$\sum_{r=1}^2 \log_a(y^r) = \sum_{r=1}^2 (\log_a y)^r \quad y > 1, a > 1, y \neq a$$

find y in terms of a , giving your answer in simplest form.

(3)

(Total for Question 4 is 6 marks)

Q25 **PAST PAPER** Pearson IAL WMA12 · June 2023 · Q5**P2 · TOPIC 9** **5 marks**

5. Use the laws of logarithms to solve

$$\log_2(16x) + \log_2(x + 1) = 3 + \log_2(x + 6) \quad (5)$$

(Total for Question 5 is 5 marks)

Q26**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q6

P2 · TOPIC 9 8 marks

6. (a) Given that

$$2 \log_4 (x + 3) + \log_4 x = \log_4 (4x + 2) + \frac{1}{2}$$

show that

$$x^3 + 6x^2 + x - 4 = 0 \quad (4)$$

(b) Given also that -1 is a root of the equation

$$x^3 + 6x^2 + x - 4 = 0$$

(i) use algebra to find the other two roots of the equation. (3)

(ii) Hence solve

$$2 \log_4 (x + 3) + \log_4 x = \log_4 (4x + 2) + \frac{1}{2} \quad (1)$$

(Total for Question 6 is 8 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q3

P2 · TOPIC 9 6 marks

3. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Using the laws of logarithms, solve

$$2\log_2(2-x) = 4 + \log_2(x+10) \quad (5)$$

- (ii) Find the value of

$$\log_{\sqrt{a}} a^6$$

where a is a positive constant greater than 1

(1)

(Total for Question 3 is 6 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q9

P2 · TOPIC 9 9 marks

9. Given that

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = 7$$

(a) verify that $t = 4$ is a solution of the above equation,**(2)**

(b) show that

$$t^3 - 116t^2 + 560t - 448 = 0$$

(3)

(c) Hence, using algebra and showing your working, solve

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = 7$$

giving each answer in simplest form.

*(Solutions based entirely on calculator technology are not acceptable.)***(4)****(Total for Question 9 is 9 marks)**

Q29**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q6

P2 · TOPIC 9 5 marks

6. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

Using the laws of logarithms, solve

$$\log_4(12 - 2x) = 2 + 2\log_4(x + 1)$$

(5)

(Total for Question 6 is 5 marks)

Q30**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q7

P2 · TOPIC 9 8 marks

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) The table below shows values of x and y , where $y = \log_{10}(x + 5)$, for x values between -1 and 4

x	-1	0	1	2	3	4
$y = \log_{10}(x + 5)$	$\log_{10} 4$	$\log_{10} 5$	$\log_{10} 6$	$\log_{10} 7$	$\log_{10} 8$	$\log_{10} 9$

Using the trapezium rule with all the y values in the given table, show that

$$\int_{-1}^4 \log_{10}(x + 5) \, dx \approx \log_{10} k$$

where k is an integer to be found.

(3)

- (ii) Find the value of a such that

$$2 \log_5(5 - a) - \log_5(a + 25) = 1$$

(5)

(Total for Question 7 is 8 marks)

Q31**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q9

P2 · TOPIC 9 10 marks

9.

**In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.**

(i) Solve

$$2\log_3(4x + 5) - \log_3(x + 3) = 2 \quad (5)$$

(ii) Given that $a > 0$, $b > 0$ and

$$\log_{10} a + \log_{10} b = \log_{10}(a + b)$$

(a) prove that $a = \frac{b}{b-1}$ (3)

(b) Hence write down the full restriction on the value of b , giving a reason for your answer. (2)

(Total for Question 9 is 10 marks)

04 Exponential Graphs and Models

$y = a^x$ ($a > 1$) rises through $(0, 1)$ with the asymptote $y = 0$; $y = a^{-x}$ is its reflection in the y -axis. A “ $+k$ ” moves the curve and its asymptote up k . In a model $N = ka^t$, k is the start value and a the growth factor.

WHEN YOU SEE → YOU DO

“sketch $y = 3 \times 4^x$ ”

→ through $(0, 3)$, asymptote $y = 0$

“sketch $y = \frac{1}{5^x} + 3$ ”

→ decreasing, through $(0, 4)$, asymptote $y = 3$

“ $y = 4^x - 6$ meets the x -axis”

→ $4^x = 6 \Rightarrow x = \log_4 6$

“ $w = \log_a(t + 5) - \log_a 4$, $w = 10$ when $t = 3$ ”

→ substitute, then solve for a

“find t in terms of w ”

→ undo the log: $a^w = \frac{t+5}{4}$

WORKED MODEL • JUNE 2026

Solve $\frac{1}{5^x} + 3 = 23$, to 3 significant figures.

1 $5^{-x} = 20$

isolate the power

2 $-x \log 5 = \log 20$

take logs

ANSWER $x = -\frac{\log 20}{\log 5} = -1.86$

IDEA 04

Practise: Exponential Graphs and Models

My questions first, then selected practice, then real past papers.

Q32 DR ESLAM Asymptote and intercept

P2 · TOPIC 9 3 marks

Sketch $y = 2^x - 8$. Give the intercepts and the asymptote.

SELECTED PRACTICE**Q33** PRACTICE Practice · Medium

P2 · TOPIC 9 4 marks

4

On the same axes, sketch the graphs of $y = 2^x$ and $y = 3^x$, labelling any points where the graphs cross the coordinate axes and writing down the equation of any asymptotes.

(4 marks)

Q34**PRACTICE**

Practice · Hard

P2 · TOPIC 9 5 marks**5 (a)**

- (a) On the same axes, sketch the graphs of $y = 4^x$ and $y = 5^x$.
Label any points of intersection with the coordinate axes.
Write down the equations of any asymptotes.

(4 marks)**(b)**

- (b) Write down an equation for the graph that is a reflection of $y = 4^x$ in the y -axis.

(1 mark)

Q35**PRACTICE**

Practice · Very Hard

P2 · TOPIC 9 5 marks**5 (a)**(a) On the same axes, sketch the graphs of $y = 0.3^x$ and $y = 0.5^x$.

Label any points of intersection with the coordinate axes.

Write down the equations of any asymptotes.

(4 marks)**(b)**(b) Write down an equation for the graph that is a reflection of $y = 0.5^x$ in the y -axis.**(1 mark)**

PAST PAPERS

Q36 PAST PAPER Pearson IAL WMA12 · January 2020 · Q9

P2 · TOPIC 9 7 marks

9. (a) Sketch the curve with equation

$$y = 3 \times 4^x$$

showing the coordinates of any points of intersection with the coordinate axes.

(2)

The curve with equation $y = 6^{1-x}$ meets the curve with equation $y = 3 \times 4^x$ at the point P .

- (b) Show that the x coordinate of P is $\frac{\log_{10} 2}{\log_{10} 24}$
- (5)**

(Total 7 marks)

Q37 PAST PAPER Pearson IAL WMA12 · June 2021 · Q2

P2 · TOPIC 9 8 marks

2.

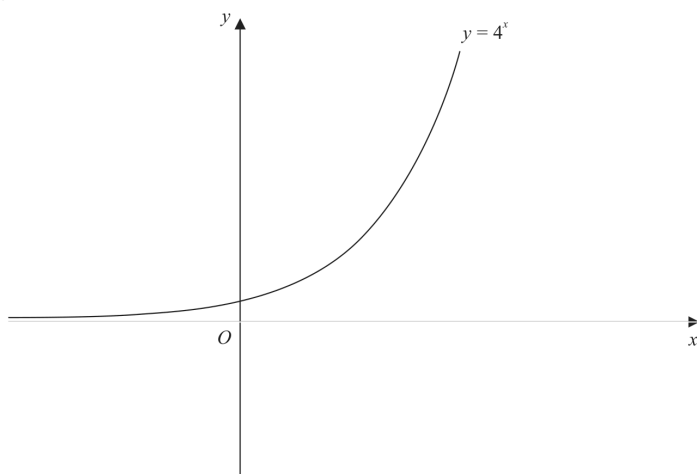
**Figure 1**

Figure 1 shows a sketch of the curve with equation $y = 4^x$

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(a) On Diagram 1, sketch the curve with equation

(i) $y = 2^x$

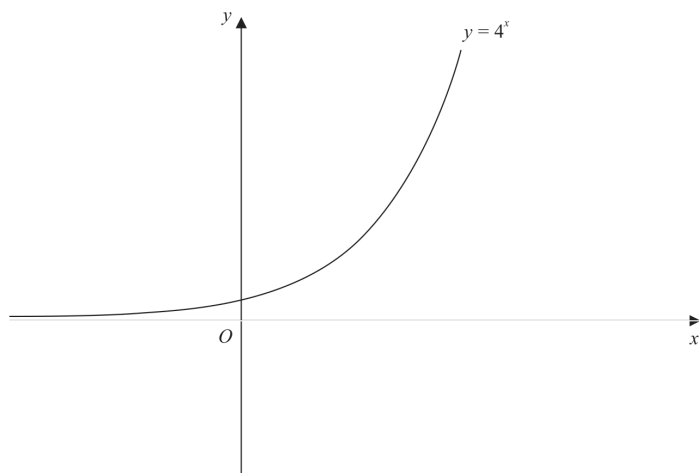
(ii) $y = 4^x - 6$

Label clearly the coordinates of any points of intersection with the coordinate axes.

(4)

The curve with equation $y = 2^x$ meets the curve with equation $y = 4^x - 6$ at the point P .

(b) Using algebra, find the exact coordinates of P .

(4)**2****Diagram 1****(Total 8 marks)**

Q38**PAST PAPER**

Pearson IAL WMA12 · October 2022 · Q4

P2 · TOPIC 9**8 marks**

4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \quad 2 \leq t \leq 18$$

where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

- (a) show that $a = 1.072$ correct to 3 decimal places.

(3)

Using $a = 1.072$

- (b) find an equation for t in terms of w

(3)

- (c) find the value of t when $w = 15$, giving your answer to 3 significant figures.

(2)

(Total for Question 4 is 8 marks)

Q39 **PAST PAPER** Pearson IAL WMA12 · June 2026 · Q6**P2 · TOPIC 9** **10 marks**

6. (i) (a) Sketch a graph of the curve with equation

$$y = \frac{1}{5^x} + 3 \quad x \in \mathbb{R}$$

Show on your sketch

- the coordinates of any points at which the curve crosses the axes
- the equation of any asymptotes

(3)

(b) Using logarithms and making your method clear, solve the equation

$$\frac{1}{5^x} + 3 = 23$$

giving the answer to 3 significant figures.

(2)(ii) Given that $\log_4 x = a$, find in simplest form in terms of a

(a) $\log_4(64x)$

(2)

(b) $\log_4\left(\frac{2}{x^2}\right)$

(3)**(Total for Question 6 is 10 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|---|
| 01 | The laws | Powers out front; products add; quotients subtract. |
| 02 | Unknown in a power | Isolate, take logs; a^{2x} and $a^x \rightarrow$ quadratic. |
| 03 | Unknown in a log | One log each side, numbers as logs, remove, check. |
| 04 | Graphs, models | $(0, 1)$ and $y = 0$ moved by the transformation. |

BOTTOM LINE Power unknown $\rightarrow$ take logs · log unknown $\rightarrow$ remove the logs · always reject log of a negative.

Answers

Exponentials and Logarithms · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE LAWS OF LOGARITHMS: SIMPLIFY AND EXPRESS IN k

- Q1** Dr Eslam · Three laws (a) $2p + q$ (b) $2q - p$ (c) $\frac{1}{2}(p + q + 1)$
- Q2** Practice · Easy (i) 3 (ii) 4 (iii) -2 (iv) 1
- Q3** Practice · Easy (i) 8.856 (ii) 5.063
- Q4** Practice · Very Hard $4^{\frac{3}{2}} = 8$ and $9^{\frac{3}{2}} = 27$, so both equal $\frac{3}{2}$
- Q5** Pearson IAL WMA12 · October 2019 · Q7 (i) $\frac{1}{2} - k$. (ii) $\frac{k+2}{3k}$. (iii) $1325k$
- Q6** Pearson IAL WMA12 · October 2022 · Q10 (i)(a) $2a$. (i)(b) $\frac{a}{2} - 4$. (ii) $x = \frac{a-3}{a+1}$

IDEA 02 · EXPONENTIAL EQUATIONS: TAKE LOGS

- Q7** Dr Eslam · Take logs $2x - 1 = \frac{\log 40}{\log 5} = 2.292: x = 1.65$
- Q8** Dr Eslam · A hidden quadratic $y^2 - 10y + 21 = 0: y = 3 \text{ or } 7; x = 1 \text{ or } x =$
 $\log_3 7 \approx 1.77$
 $7^{2x-1} = 7^3 \Rightarrow x = 2$
- Q9** Practice · Easy
- Q10** Practice · Medium $y = 2^x: y^2 - 24y + 128 = 0, y = 8 \text{ or } 16: x = 3$
or 4
- Q11** Practice · Very Hard $y = 3^x: 9y^2 - 28y + 3 = 0, y = 3 \text{ or } \frac{1}{9}: x = 1 \text{ or}$
 $x = -2$
- Q12** Pearson IAL WMA12 · January 2021 · Q3 $(i) x = \log_7 \left(\frac{3}{49} \right). (ii) y = \frac{1}{2}$
- Q13** Pearson IAL WMA12 · October 2023 · Q5 $(i) a = 3.867. (ii) b = \frac{\sqrt{5}}{9}$
- Q14** Pearson IAL WMA12 · June 2025 · Q6 $(i) x = -2. (ii) P = (\log_3 2, 12)$
- Q15** Pearson IAL WMA12 · January 2026 · Q7 $(i) x = \frac{17}{6} + \frac{1}{6} \log_2 5. (ii) y = -5$

IDEA 03 · LOG EQUATIONS: ONE LOG EACH SIDE, THEN REMOVE IT

- Q16** Dr Eslam · Combine, then remove

$$\frac{x+6}{x} = 8 \Rightarrow x = \frac{6}{7}$$
- Q17** Dr Eslam · Numbers become logs

$$x(x-8) = 9 \Rightarrow x = 9 \text{ (} x = -1 \text{ rejected: log of a negative)}$$
- Q18** Practice · Easy

$$x^2 = 16 \Rightarrow x = 4 \text{ (a base must be positive)}$$
- Q19** Practice · Hard

$$(x+2)^2 = x^3 \Rightarrow x \approx 2.88$$
- Q20** Practice · Very Hard

$$\log_{16}(2-x)^2 = \log_{16}(13-4x) \Rightarrow x^2 = 9; x = -3$$

$$(x=3 \text{ makes } 2-x < 0)$$
- Q21** Pearson IAL WMA12 · June 2021 · Q7
 (a) Proven. (b) -4 is a root of the cubic.
 (c) $x = 2, \frac{7}{2}$
- Q22** Pearson IAL WMA12 · January 2022 · Q4

$$x = -\frac{1}{9}$$
- Q23** Pearson IAL WMA12 · June 2022 · Q4

$$a = 4 + 4\sqrt{5}, \quad b = 4\sqrt{5} - 4$$
- Q24** Pearson IAL WMA12 · January 2023 · Q4
 (i) $x = \frac{7}{31}$. (ii) $y = a^2$
- Q25** Pearson IAL WMA12 · June 2023 · Q5

$$x = \frac{3}{2}$$
- Q26** Pearson IAL WMA12 · January 2024 · Q6
 (a) Proven. (b)(i) $x = \frac{-5 + \sqrt{41}}{2}$ and $x = \frac{-5 - \sqrt{41}}{2}$.
 (b)(ii) $x = \frac{-5 + \sqrt{41}}{2}$
- Q27** Pearson IAL WMA12 · June 2024 · Q3
 (i) $x = -6$. (ii) 12
- Q28** Pearson IAL WMA12 · June (R paper) 2024 · Q9
 (a) Verified. (b) Proven. (c) $t = 4$ or $t = 56 + 12\sqrt{21}$
- Q29** Pearson IAL WMA12 · October 2024 · Q6

$$x = -\frac{1}{8}$$
- Q30** Pearson IAL WMA12 · January 2025 · Q7
 (i) $k = 10080$. (ii) $a = -5$
- Q31** Pearson IAL WMA12 · October
 (i) $x = \frac{1}{16}$. (ii)(a) Proven. (ii)(b) $b > 1$

IDEA 04 · EXPONENTIAL GRAPHS AND MODELS

- Q32** Dr Eslam · Asymptote and intercept
y-intercept -7; x-intercept 3; asymptote $y = -8$
- Q33** Practice · Medium
both through $(0, 1)$, asymptote $y = 0$; 3^x above 2^x for $x > 0$, below for $x < 0$
- Q34** Practice · Hard
(a) both through $(0, 1)$, asymptote $y = 0$; 5^x steeper for $x > 0$ (b) $y = 4^{-x}$
- Q35** Practice · Very Hard
(a) both decreasing through $(0, 1)$, asymptote $y = 0$; 0.3^x below 0.5^x for $x > 0$ (b) $y = 0.5^{-x} = 2^x$
- Q36** Pearson IAL WMA12 · January 2020 · Q9
(a) Increasing exponential curve through $(0, 3)$, with no x-axis intersection. (b) $x = \frac{\log_{10} 2}{\log_{10} 24}$
- Q37** Pearson IAL WMA12 · June 2021 · Q2
(a)(i) Sketch $y = 2^x$ through $(0, 1)$ with no x-axis intercept. (a)(ii) Sketch $y = 4^x - 6$ through $(0, -5)$ and $(\log_4 6, 0)$. (b) $P = (\log_2 3, 3)$
- Q38** Pearson IAL WMA12 · October 2022 · Q4
(a) $a = 1.072$ to 3 d.p. (b) $t = 4(1.072)^w - 5$. (c) $t = 6.35$ months to 3 s.f
- Q39** Pearson IAL WMA12 · June 2026 · Q6
(i)(a) decreasing curve through $(0, 4)$, asymptote $y = 3$ (b) $5^{-x} = 20 \Rightarrow x = -1.86$ (ii)(a) $3 + a$ (b) $\frac{1}{2} - 2a$

DR ESLAM AHMED
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Trigonometric Identities and Equations

Get one trig function equal to a number. Change the range first when the angle is $ax + b$, find every solution with CAST, then undo the change.

THE TOPIC ON ONE MAP

Trigonometric Identities and Equations

01 — Transformed angle

new range CAST undo $ax + b$

02 — The two identities

$\tan = \frac{\sin}{\cos}$ $\sin^2 + \cos^2 = 1$ factorise, never divide

03 — Models and graphs

max and min period first time after ...

THE TWO IDENTITIES

$$\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$$

$$\sin^2 \theta + \cos^2 \theta \equiv 1$$

$$\sin \theta = k \cos \theta \Rightarrow \tan \theta = k \quad \cdot \quad \sin^2 \text{ with } \cos \Rightarrow 1 - \cos^2 \quad \cdot \quad \tan \times \cos = \sin$$

01 Solve $\sin(ax + b) = k$: New Range, Then CAST

Call the whole angle $X = ax + b$ and change the range for X first. Use the calculator for the principal value, find every other value in the new range with CAST (or the graph), then solve back for x .

WHEN YOU SEE → YOU DO

$$3 \sin(2\theta - 10^\circ) = 1 \text{ for } 0 \leq \theta < 180^\circ$$

$$\rightarrow X = 2\theta - 10: -10 \leq X < 350$$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3$$

$$\rightarrow \tan = \pm\sqrt{3}: \text{both signs}$$

$$3 \sin(\dots) = 7 \cos(\dots)$$

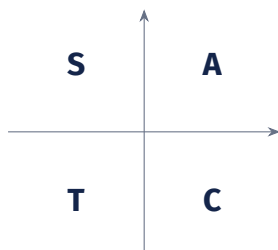
$$\rightarrow \text{divide by } \cos: \tan(\dots) = \frac{7}{3}$$

“give the answers in the form $k\pi$ ”

$$\rightarrow \text{exact values: } \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} \dots$$

“in radians”

$$\rightarrow \text{calculator in radians; the range uses } \pi$$



sin: α and $180^\circ - \alpha$

cos: α and $360^\circ - \alpha$

tan: α and $180^\circ + \alpha$

then add or subtract 360° to fill the range

WORKED MODEL • OCTOBER 2019

Solve $3 \sin(2\theta - 10^\circ) = 1$ for $0 \leq \theta < 180^\circ$.

1 $\sin X = \frac{1}{3}, -10 \leq X < 350: X = 19.47, 160.53$

new range, CAST

ANSWER $\theta = 14.7^\circ, 85.3^\circ$

IDEA 01

Practise: Solve $\sin(ax + b) = k$: New Range, Then CAST

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Transformed angle

P2 · TOPIC 10 4 marks

Solve $2 \cos(2x - 30^\circ) = 1$ for $0 \leq x < 360^\circ$.

Let $X = 2x - 30$: new range $-30 \leq X < 690$.

Q2 DR ESLAM Sine equals k cosine

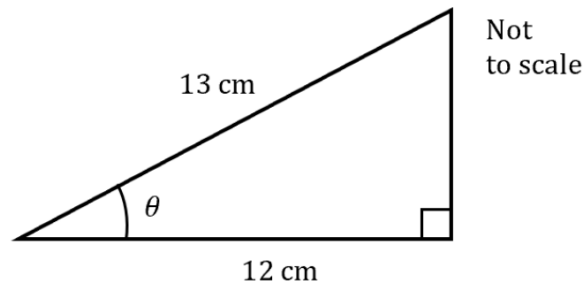
P2 · TOPIC 10 3 marks

Solve $\sin \theta = 3 \cos \theta$ for $0 \leq \theta < 360^\circ$.

SELECTED PRACTICE

Q3**PRACTICE** Practice · Easy**P2 · TOPIC 10** 5 marks**1 (a)**

(a) Work out the length of the missing side in the following right-angled triangle.

**(2 marks)****(b)**

(b) Using your answer from part (a) to help, write down the values of the following:

(i) $\sin \theta$

(ii) $\cos \theta$

(iii) $\tan \theta$

(3 marks)

Q4**PRACTICE**

Practice · Medium

P2 · TOPIC 10**6 marks**

- 6 A right-angled triangle has hypotenuse 8 cm. One of its other sides is 5 cm.

Find exact values for $\sin \theta$, $\cos \theta$ and $\tan \theta$, where θ is the smallest angle in the triangle.

(6 marks)

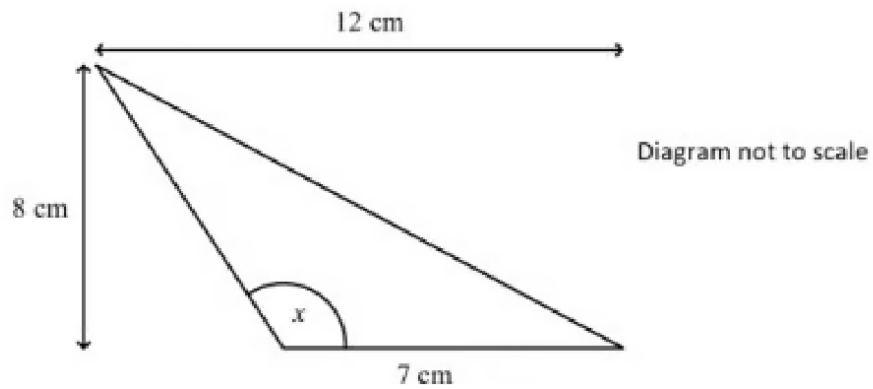
Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 10

6 marks

- 6 For the triangle in the diagram, find exact values for $\sin x$, $\cos x$ and $\tan x$.

**(6 marks)**

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · October 2019 · Q9

P2 · TOPIC 10 12 marks

9. Solutions based entirely on graphical or numerical methods are not acceptable in this question.

- (i) Solve, for $0 \leq \theta < 180^\circ$, the equation

$$3 \sin(2\theta - 10^\circ) = 1$$

giving your answers to one decimal place.

(4)

- (ii) The first three terms of an arithmetic sequence are

$$\sin \alpha, \frac{1}{\tan \alpha} \text{ and } 2 \sin \alpha$$

where α is a constant.

- (a) Show that $2 \cos \alpha = 3 \sin^2 \alpha$

(3)

Given that $\pi < \alpha < 2\pi$,

- (b) find, showing all working, the value of α to 3 decimal places.

(5)

(Total 12 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · October 2021 · Q10

P2 · TOPIC 10 10 marks

10. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3 \quad (5)$$

- (ii) Solve, for $0 < \theta < 360^\circ$

$$(2 \sin \theta - \cos \theta)^2 = 1$$

giving your answers, as appropriate, to one decimal place. (5)

Q8 **PAST PAPER** Pearson IAL WMA12 · January 2022 · Q7**P2 · TOPIC 10** **8 marks**

7. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-90^\circ < x < 90^\circ$, the equation

$$3 \sin(2x - 15^\circ) = \cos(2x - 15^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) Solve, for $0 < \theta < 2\pi$, the equation

$$4 \sin^2 \theta + 8 \cos \theta = 3$$

giving your answers to 3 significant figures.

(4)

(Total 8 marks)

Q9**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q8

P2 · TOPIC 10 9 marks

8. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \pi$, the equation

$$5 \sin(3x + 0.1) + 2 = 0$$

giving your answers, **in radians**, to 2 decimal places.

(4)

- (ii) Solve, for $0 < \theta < 360^\circ$, the equation

$$2 \tan \theta \sin \theta = 5 + \cos \theta$$

giving your answers, **in degrees**, to one decimal place.

(5)

(Total for Question 8 is 9 marks)

Q10**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q8

P2 · TOPIC 10**8 marks****8:**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < \pi$, the equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3}$$

giving the answers in radians in the form $k\pi$, where k is a rational constant to be found.

(3)

- (ii) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$5 \sin \theta \tan \theta = \cos \theta + 4$$

giving your answers, in degrees, to one decimal place.

(5)

(Total for Question 8 is 8 marks)

Q11**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q8

P2 · TOPIC 10 10 marks**8.**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq \theta < 180$

$$4 \tan^2(2\theta - 30)^\circ + 1 = 49$$

giving the answers to one decimal place.

(5)

- (ii) Solve, for $0 \leq x < 2\pi$

$$2 \tan x \sin x + 3 = 0$$

giving the answers, in radians, in the form $k\pi$ where k is a rational constant.

(5)

(Total for Question 8 is 10 marks)

Q12**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q2

P2 · TOPIC 10**6 marks****2.**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < x \leq 2\pi$, the equation

$$1 + \sin^2 x + \cos x = 0 \quad (3)$$

- (ii) Solve, for $0 < \theta < 180^\circ$, the equation

$$\sin(\theta + 60^\circ) = 4 \cos(\theta + 60^\circ)$$

giving the answer to one decimal place.

(3)

(Total for Question 2 is 6 marks)

02 The Two Identities: Make One Trig Function

If the equation mixes $\sin$, $\cos$ and $\tan$, use the identities until only one function is left. Usually you get a quadratic (or cubic) in $\cos x$ or $\sin x$: factorise, and reject values outside $-1 \leq \text{value} \leq 1$.

WHEN YOU SEE → YOU DO

$\tan \theta$ **times something**

→ $\tan = \frac{\sin}{\cos}$, multiply through by $\cos$

$\sin^2 x$ **and** $\cos x$ **together**

→ $\sin^2 = 1 - \cos^2$: quadratic in $\cos$

“show that it can be written as
 $3 \sin^2 \theta + \sin \theta - 1 = 0$ ”

→ then “hence solve” with $2x$ in place of θ

$\sin x$ **on both sides**

→ bring to one side and factor — never divide by $\sin x$

$\cos x = 2$ **or** $\sin x = -5$

→ no solutions: reject

“given $\cos \theta = \frac{1}{3}$, **find** $\tan \theta$ **exactly”**

→ right triangle 1, 3, $\sqrt{8}$; sign from the quadrant

One function only

Quadratic

Factorise

CAST each root

WORKED MODEL · JUNE 2026

Solve $1 + \sin^2 x + \cos x = 0$ for $0 < x \leq 2\pi$.

1 $1 + (1 - \cos^2 x) + \cos x = 0 \Rightarrow \cos^2 x - \cos x - 2 = 0$

one function

2 $(\cos x - 2)(\cos x + 1) = 0$: $\cos x = -1$ only

reject $\cos x = 2$

ANSWER $x = \pi$

× **TRAP** Dividing both sides by $\sin x$ and losing the solutions of $\sin x = 0$.

IDEA 02

Practise: The Two Identities: Make One Trig Function

My questions first, then selected practice, then real past papers.

Q13 **DR ESLAM** One function only

P2 · TOPIC 10 4 marks

Solve $2 \sin^2 x = 3 \cos x$ for $0 \leq x < 360^\circ$.

$\sin^2 x = 1 - \cos^2 x$, then a quadratic in $\cos x$.

Q14 **DR ESLAM** Never divide by $\sin$

P2 · TOPIC 10 4 marks

Solve $\sin x \cos x = \frac{1}{2} \sin x$ for $0 \leq x \leq 180^\circ$.

SELECTED PRACTICE

Q15

PRACTICE

Practice · Easy

P2 · TOPIC 10

2 marks

2

Show that

$$\frac{1 - \cos^2 x}{\tan^2 x} \equiv \cos^2 x$$

(2 marks)

Q16

PRACTICE

Practice · Hard

P2 · TOPIC 10

5 marks

2

Solve the equation $2 \sin^2 \theta = \cos \theta + 1$ for $-180^\circ \leq \theta \leq 180^\circ$.

(5 marks)

Q17**PRACTICE**

Practice · Very Hard

P2 · TOPIC 10 12 marks**8 (a)** Find all the solutions to the equation

$2 \cos 2\theta = 4 \sin 2\theta \cos 2\theta$ in the interval $0 \leq \theta \leq 2\pi$, giving your answers in radians as multiples of π .

(6 marks)

(b) Find all the solutions to the equation $3\cos^2 4x + 13\cos 4x - 10 = 0$ in the interval $0 \leq x \leq \pi$, giving your answers in radians to three significant figures.

(6 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA12 · January 2020 · Q7

P2 · TOPIC 10 7 marks

7. (a) Show that the equation

$$8 \tan \theta = 3 \cos \theta$$

may be rewritten in the form

$$3 \sin^2 \theta + 8 \sin \theta - 3 = 0 \quad (3)$$

- (b) Hence solve, for $0 \leq x \leq 90^\circ$, the equation

$$8 \tan 2x = 3 \cos 2x$$

giving your answers to 2 decimal places. (4)

(Total 7 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2021 · Q6

P2 · TOPIC 10 8 marks

6. (a) Show that the equation

$$\frac{3 \sin \theta \cos \theta}{2 \sin \theta - 1} = 5 \tan \theta \quad \sin \theta \neq \frac{1}{2}$$

can be written in the form

$$3 \sin^3 \theta + 10 \sin^2 \theta - 8 \sin \theta = 0 \quad (4)$$

- (b) Hence solve, for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\frac{3 \sin 2x \cos 2x}{2 \sin 2x - 1} = 5 \tan 2x$$

giving your answers to 3 decimal places where appropriate.

(4)

(Total 8 marks)

Q20**PAST PAPER**

Pearson IAL WMA12 · June 2021 · Q8

P2 · TOPIC 10**10 marks**

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta < 360^\circ$, the equation

$$3 \sin(\theta + 30^\circ) = 7 \cos(\theta + 30^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) (a) Show that the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

can be written in the form

$$\sin x (a \cos^2 x + b \cos x + c) = 0$$

where a , b and c are constants to be found.

- (b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

(6)

(Total 10 marks)

Q21**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q5

P2 · TOPIC 10**6 marks**

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Solve, for $-180^\circ < \theta \leq 180^\circ$, the equation

$$3 \tan(\theta + 43^\circ) = 2 \cos(\theta + 43^\circ)$$

(6)

(Total 6 marks)

Q22**PAST PAPER**

Pearson IAL WMA12 · October 2022 · Q5

P2 · TOPIC 10**8 marks**

5. **In this question you must show detailed reasoning.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$(3 \cos \theta - \tan \theta) \cos \theta = 2$$

can be written as

$$3 \sin^2 \theta + \sin \theta - 1 = 0 \quad (3)$$

(b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$$(3 \cos 2x - \tan 2x) \cos 2x = 2 \quad (5)$$

(Total for Question 5 is 8 marks)

Q23**PAST PAPER**

Pearson IAL WMA12 · June 2023 · Q9

P2 · TOPIC 10 7 marks

9. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 11 - 5 \cos \theta$$

may be written as

$$3 \cos^2 \theta - 14 \cos \theta + 8 = 0 \quad (3)$$

(b) Hence solve, for $0 < x < 360^\circ$

$$3 \cos 2x (\tan 2x \sin 2x + 3) = 11 - 5 \cos 2x$$

giving your answers to one decimal place.

(4)

(Total for Question 9 is 7 marks)

Q24**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q3

P2 · TOPIC 10**7 marks**

3. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

- (a) Solve, for $0 < \theta \leq 360^\circ$ the equation

$$2 \tan \theta + 3 \sin \theta = 0$$

giving your answers, as appropriate, to one decimal place.

(5)

- (b) Hence, or otherwise, find the smallest positive solution of

$$2 \tan(2x + 40^\circ) + 3 \sin(2x + 40^\circ) = 0$$

giving your answer to one decimal place.

(2)

(Total for Question 3 is 7 marks)

Q25**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q7

P2 · TOPIC 10 7 marks

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < 2\pi$, the equation

$$3 \sin x \tan x = 11 + \cos x$$

giving the answers in radians to 3 decimal places.

(5)

- (ii) Given that

- $0 < \theta < 90^\circ$
- $\cos \theta = \frac{1}{3}$

find, in simplest form, the exact value of $\tan \theta$

(2)

(Total for Question 7 is 7 marks)

Q26**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q9

P2 · TOPIC 10**7 marks**

9. **In this question you must show detailed reasoning.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$2 \tan \theta = 3 \cos \theta$$

can be written as

$$3 \sin^2 \theta + 2 \sin \theta - 3 = 0$$

(3)

(b) Hence solve, for $-\pi < x < \pi$, the equation

$$2 \tan \left(2x + \frac{\pi}{3} \right) = 3 \cos \left(2x + \frac{\pi}{3} \right)$$

giving your answers to 3 significant figures.

(4)

(Total for Question 9 is 7 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q10

P2 · TOPIC 10**8 marks****10.**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) \equiv \sin \theta + \frac{2}{\sin \theta} \quad \theta \neq \frac{n\pi}{2} \quad (4)$$

(b) Hence solve, for $0 < x < 2\pi$, the equation

$$\cos x \left(3 \tan x + \frac{2}{\tan x} \right) = 4 \sin x - 5$$

giving your answers to 3 significant figures.

(4)**(Total for Question 10 is 8 marks)**

Q28**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q5

P2 · TOPIC 10**9 marks**

5. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta \leq 360^\circ$, the equation

$$4 \tan \theta + 5 \sin \theta = 0$$

giving any non-exact answers to one decimal place.

(5)

- (ii) Solve, for $0 < x < \pi$, the equation

$$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{5}{\cos x}$$

giving the answers, in radians, to 3 significant figures.

(4)

(Total for Question 5 is 9 marks)

03 Trig Models and Graphs

In $y = A \sin(kt + c) + D$: the maximum is $D + A$, the minimum is $D - A$, and the period is $\frac{360^\circ}{k}$ (or $\frac{2\pi}{k}$). To find “when”, set y equal to the value and solve like Idea 1 — then pick the time the question asks for.

WHEN YOU SEE → YOU DO

“the y coordinate of P (a maximum) is 7”

$$\longrightarrow D + A = 7$$

“the maximum temperature”

$$\longrightarrow \sin = 1: D + A$$

“the time of day when this occurs”

$$\longrightarrow kt + c = 90^\circ, \text{ then convert hours to a clock time}$$

“the first time after midday”

$$\longrightarrow \text{solve, then choose the first } t > 12$$

“how long above 0.5 m?”

$$\longrightarrow \text{solve for the two times in one period, subtract, count periods}$$

WORKED MODEL · JUNE 2024

$H = 10 + 12 \sin(kt + 18)^\circ$, $0 < k < 10$, $k = 6.41$. Maximum temperature and when.

1 $H_{\max} = 10 + 12 = 22^\circ\text{C}$

$$\sin = 1$$

2 $6.41t + 18 = 90 \Rightarrow t = 11.23 \text{ hours}$

$$\text{solve}$$

ANSWER 22°C at about 11:14 am

IDEA 03

Practise: Trig Models and Graphs

My questions first, then selected practice, then real past papers.

Q29**DR ESLAM**

A tide model

P2 • TOPIC 10

4 marks

$D = 6 + 2 \sin(30t)^\circ$, t hours after midnight. Find the greatest depth and the first time it occurs.

SELECTED PRACTICE

Q30**PRACTICE**

Practice · Medium

P2 · TOPIC 10**6 marks**

- 9 (a) A seagull sits on the surface of the sea and moves up and down as waves pass.

Its height, h metres, above its position in calm water is modelled by the function

$$h = \frac{1}{2} \sin(180t) \text{ where } t \text{ is the time in seconds after timing commences.}$$

Sketch a graph of h against t for $0 \leq t \leq 10$ showing the coordinates of the points of intersection with the t - axis

(2 marks)

- (b) How many times in the first minute after timing commences is the seagull 0.25 metres above its calm water position?

(1 mark)

- (c) Find the time at which the seagull is first 0.25 m above its calm water position **and moving downwards**. Give your answer to 3 significant figures.

(3 marks)

Q31**PRACTICE**

Practice · Hard

P2 · TOPIC 10 6 marks**9 (a)**

- (a) A seagull sits on the surface of the sea and moves up and down as waves pass.

Its height, h metres, above its position in calm water is modelled by the function

$$h = \frac{2}{5} \sin(180t)^\circ \text{ where } t \text{ is the time in seconds after timing commenced.}$$

Find the first time the seagull is 0.3 metres above its calm water position.

Give your answer to 2 decimal places.

(4 marks)**(b)**

- (b) How many times in the first minute after timing commences is the seagull 0.3 metres above its calm water position?

(2 marks)

Q32**PRACTICE**

Practice · Very Hard

P2 · TOPIC 10 7 marks

- 9 A seagull sits on the surface of the sea and moves up and down as waves pass.

Its height, h metres, above its position in calm water is modelled by the function

$$h = \frac{3}{5} \sin(90t)^\circ \quad \text{where } t \text{ is the time in seconds after timing commences.}$$

Find the amount of time the seagull is more than 0.5 metres above its calm water position in the first 20 seconds after timing commences.

Give your answer correct to 3 significant figures.

(7 marks)

PAST PAPERS

Q33 PAST PAPER Pearson IAL WMA12 · January 2024 · Q9

P2 · TOPIC 10 14 marks

9. In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for
- $0 \leq x < 360^\circ$
- , the equation

$$\sin x \tan x = 5$$

giving your answers to one decimal place.

(6)

- (ii)

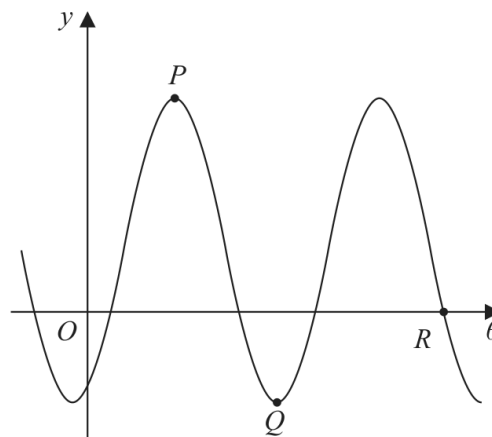


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = A \sin \left(2\theta - \frac{3\pi}{8} \right) + 2$$

where A is a constant and θ is measured in radians.The points P , Q and R lie on the curve and are shown in Figure 1.Given that the y coordinate of P is 7

- (a) state the value of
- A
- ,

(1)

- (b) find the exact coordinates of
- Q
- ,

(3)

- (c) find the value of
- θ
- at
- R
- , giving your answer to 3 significant figures.

(4)

(Total for Question 9 is 14 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June 2024 · Q8

P2 · TOPIC 10 12 marks

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < x \leq \pi$, the equation

$$5 \sin x \tan x + 13 = \cos x$$

giving your answer in radians to 3 significant figures.

(5)

- (ii) The temperature inside a greenhouse is monitored on one particular day.

The temperature, $H^\circ\text{C}$, inside the greenhouse, t hours after midnight, is modelled by the equation

$$H = 10 + 12 \sin(kt + 18)^\circ \quad 0 \leq t < 24$$

where k is a constant.

Use the equation of the model to answer parts (a) to (c).

Given that

- the temperature inside the greenhouse was 20°C at 6 am
- $0 < k < 20$

- (a) find all possible values for k , giving each answer to 2 decimal places.

(4)

Given further that $0 < k < 10$

- (b) find the maximum temperature inside the greenhouse,

(1)

- (c) find the time of day at which this maximum temperature occurs.

Give your answer to the nearest minute.

(2)

(Total for Question 8 is 12 marks)

Q35**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q5

P2 · TOPIC 10**6 marks****5.**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The depth of water, D metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after **midnight**.

(a) Show that the depth of water in the harbour at 2 am is just over 4 metres.

(1)

(b) Find, to the nearest minute, the first time after **midday** when the depth of water in the harbour is exactly 6 metres.

(5)

(Total for Question 5 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01** $\sin(ax + b) = k$ New range for X ; CAST; undo $ax + b$.
- 02** **Identities** One function only; factorise; reject $|\cdot| > 1$.
- 03** **Models** Max $D + A$, min $D - A$, period $\frac{360}{k}$.

BOTTOM LINE One function → new range → every value with CAST → back to x . Never divide by $\sin x$.

Answers

Trigonometric Identities and Equations · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · SOLVE $\sin(ax + b) = k$: NEW RANGE, THEN CAST

- Q1** Dr Eslam · Transformed angle
 $X = 60, 300, 420, 660$: $x = 45^\circ, 165^\circ, 225^\circ, 345^\circ$
 $\tan \theta = 3$: $\theta = 71.6^\circ, 251.6^\circ$
- Q2** Dr Eslam · Sine equals k co-sine
- Q3** Practice · Easy
 (a) 5 cm (b)(i) $\frac{5}{13}$ (ii) $\frac{12}{13}$ (iii) $\frac{5}{12}$
- Q4** Practice · Medium
 third side $\sqrt{39}$; smallest angle opposite 5:
 $\sin \theta = \frac{5}{8}$, $\cos \theta = \frac{\sqrt{39}}{8}$, $\tan \theta = \frac{5}{\sqrt{39}} = \frac{5\sqrt{39}}{39}$
- Q5** Practice · Very Hard
 $\sin x = \frac{8}{\sqrt{89}}$, $\cos x = -\frac{5}{\sqrt{89}}$, $\tan x = -\frac{8}{5}$ (x is obtuse)
- Q6** Pearson IAL WMA12 · October 2019 · Q9
 (i) $\theta = 14.7^\circ, 85.3^\circ$. (ii)(a) Proven.
 (ii)(b) $\alpha = 5.517$ radians
- Q7** Pearson IAL WMA12 · October 2021 · Q10
 (i) $x = -\frac{11\pi}{24}, -\frac{7\pi}{24}, \frac{\pi}{24}, \frac{5\pi}{24}$. (ii) $\theta = 53.1^\circ, 180^\circ, 233.1^\circ$
- Q8** Pearson IAL WMA12 · January 2022 · Q7
 (i) $x = -73.3^\circ, 16.7^\circ$. (ii) $\theta = 1.69, 4.59$
- Q9** Pearson IAL WMA12 · January 2023 · Q8
 (i) $x = -0.94, -0.17, 1.15, 1.92$. (ii) $\theta = 70.5^\circ, 289.5^\circ$
- Q10** Pearson IAL WMA12 · June 2025 · Q8
 (i) $x = \frac{29\pi}{60}, \frac{59\pi}{60}$. (ii) $\theta = 50.3^\circ, 309.7^\circ$
- Q11** Pearson IAL WMA12 · January 2026 · Q8
 (i) $\theta = 51.9^\circ, 68.1^\circ, 141.9^\circ, 158.1^\circ$. (ii) $x = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q12** Pearson IAL WMA12 · June 2026 · Q2
 (i) $\cos^2 x - \cos x - 2 = 0$: $\cos x = -1$: $x = \pi$ (ii)
 $\tan(\theta + 60^\circ) = 4$: $\theta = 16.0^\circ$

IDEA 02 · THE TWO IDENTITIES: MAKE ONE TRIG FUNCTION

- Q13** Dr Eslam · One function only $2 \cos^2 x + 3 \cos x - 2 = 0: \cos x = \frac{1}{2}: x = 60^\circ, 300^\circ$
- Q14** Dr Eslam · Never divide by sin $\sin x(\cos x - \frac{1}{2}) = 0: x = 0^\circ, 60^\circ, 180^\circ$
- Q15** Practice · Easy $\frac{\sin^2 x}{\sin^2 x / \cos^2 x} = \cos^2 x$
- Q16** Practice · Hard $2 \cos^2 \theta + \cos \theta - 1 = 0: \cos \theta = \frac{1}{2} \text{ or } -1: \theta = -180^\circ, -60^\circ, 60^\circ, 180^\circ$
- Q17** Practice · Very Hard (a) $\cos 2\theta = 0 \text{ or } \sin 2\theta = \frac{1}{2}: \theta = \frac{\pi}{12}, \frac{\pi}{4}, \frac{5\pi}{12}, \frac{3\pi}{4}, \frac{13\pi}{12}, \frac{5\pi}{4}, \frac{17\pi}{12}, \frac{7\pi}{4}$ (b) $\cos 4x = \frac{2}{3}: x = 0.210, 1.36, 1.78, 2.93$
- Q18** Pearson IAL WMA12 · January 2020 · Q7 (a) Proven. (b) $x = 9.74^\circ, 80.26^\circ$
- Q19** Pearson IAL WMA12 · January 2021 · Q6 (a) Proven. (b) $x = 0, 0.365$
- Q20** Pearson IAL WMA12 · June 2021 · Q8 (i) $\theta = 36.8^\circ, 216.8^\circ$. (ii) (a) $a = 3, b = -7, c = 2$. (ii) (b) $x = -\cos^{-1}\left(\frac{1}{3}\right), 0, \cos^{-1}\left(\frac{1}{3}\right)$
- Q21** Pearson IAL WMA12 · June 2022 · Q5 $\theta = -13^\circ, 107^\circ$
- Q22** Pearson IAL WMA12 · October 2022 · Q5 (a) Proven. (b) $x = -1.13, -0.438, 0.225, 1.35 \text{ radians}$
- Q23** Pearson IAL WMA12 · June 2023 · Q9 (a) Proven. (b) $x = 24.1^\circ, 155.9^\circ, 204.1^\circ, 335.9^\circ$
- Q24** Pearson IAL WMA12 · October 2023 · Q3 (a) $\theta = 131.8^\circ, 180^\circ, 228.2^\circ, 360^\circ$. (b) $x = 45.9^\circ$
- Q25** Pearson IAL WMA12 · June (R paper) 2024 · Q7 (i) $x = 1.318, 4.965$. (ii) $2\sqrt{2}$
- Q26** Pearson IAL WMA12 · October 2024 · Q9 (a) Proven. (b) $x = -2.50, -0.121, 0.645, 3.02$
- Q27** Pearson IAL WMA12 · January 2025 · Q10 (a) Proven. (b) $x = 3.48, 5.94$
- Q28** Pearson IAL WMA12 · October 2025 · Q5 (i) $\theta = 143.1^\circ, 180^\circ, 216.9^\circ, 360^\circ$. (ii) $x = 0.201, 2.94$

IDEA 03 · TRIG MODELS AND GRAPHS

- Q29** Dr Eslam · A tide model 8 m, when $30t = 90$: $t = 3$ (3 am)
- Q30** Practice · Medium (a) period 2 s, crosses at $t = 0, 1, 2, \dots, 10$ (b) 60 times (c) $180t = 150$: $t = 0.833$ s
- Q31** Practice · Hard (a) $\sin(180t) = 0.75$: $t = 0.27$ s (b) 60 times
- Q32** Practice · Very Hard $\sin(90t) > \frac{5}{6}$ for 0.746 s in each 4 s period:
 $5 \times 0.746 = 3.73$ s
- Q33** Pearson IAL WMA12 · January 2024 · Q9 (i) $x = 78.9^\circ, 281.1^\circ$. (ii)(a) $A = 5$. (ii)(b) $Q = \left(\frac{15\pi}{16}, -3\right)$. (ii)(c) $\theta = 5.51$
- Q34** Pearson IAL WMA12 · June 2024 · Q8 (i) $x = 1.91$ radians. (ii)(a) $k = 6.41$ or 17.59. (ii)(b) 22°C . (ii)(c) 11:14 am
- Q35** Pearson IAL WMA12 · June (R paper) 2024 · Q5 (a) $D(2) = 4.06 \dots$ m, just over 4 m. (b) 13:03, or 1:03 pm

DR ESLAM AHMED
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Applications of Differentiation

Stationary means the first derivative is zero. Its sign says up or down; the second derivative says maximum or minimum.

THE TOPIC ON ONE MAP

Applications of Differentiation

01 – Increasing, decreasing

$f'(x) > 0$ $f'(x) < 0$ for all x

02 – Stationary points

$f'(x) = 0$ f'' : max or min show that $k = \dots$

03 – Optimisation

one variable only max / min value justify with f''

04 – With areas

stationary point first then integrate

THE TWO DERIVATIVES

$$\frac{dy}{dx} = 0 \text{ stationary}$$

$$\frac{d^2y}{dx^2} < 0 \text{ maximum}$$

$$\frac{d^2y}{dx^2} > 0 \text{ minimum}$$

$$\frac{d^2y}{dx^2} > 0$$

rewrite first: $\sqrt{x} = x^{\frac{1}{2}}$, $\frac{1}{x^2} = x^{-2}$ · increasing: $\frac{dy}{dx} > 0$ decreasing: $\frac{dy}{dx} < 0$

01 Increasing and Decreasing Functions

Differentiate, then solve the inequality $f'(x) > 0$ (increasing) or $f'(x) < 0$ (decreasing). For a quadratic $f'(x)$, find its roots and sketch it. “Increasing for all x ”: show $f'(x)$ is always positive — complete the square.

WHEN YOU SEE → YOU DO

“find the values of x for which y is increasing”

→ $f'(x) > 0$: roots, sketch, read

“state the range of values of x for which y is decreasing”

→ $f'(x) < 0$

“show that f is increasing for all x ”

→ $f'(x) = a(x + p)^2 + q$ with $a, q > 0$

“prove that C has no turning points”

→ $f'(x) = 0$ has no real roots: $b^2 - 4ac < 0$

WORKED MODEL · OCTOBER 2019

$y = 2x^2(x - 5)$. Stationary points and where y is increasing.

1 $y = 2x^3 - 10x^2$, $\frac{dy}{dx} = 6x^2 - 20x = 2x(3x - 10)$

expand, differentiate

2 $\frac{dy}{dx} = 0$: $x = 0$, $\frac{10}{3}$

stationary

ANSWER increasing for $x < 0$ or $x > \frac{10}{3}$

IDEA 01

Practise: Increasing and Decreasing Functions

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Where is it increasing?

P2 · TOPIC 11 3 marks

Find the values of x for which $f(x) = x^3 - 6x^2 + 5$ is increasing.

Solve $f'(x) > 0$: sketch the quadratic.

Q2 **DR ESLAM** Increasing for all x

P2 · TOPIC 11 3 marks

Show that $f(x) = x^3 + 3x^2 + 6x$ is increasing for all x .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 11 6 marks

1 [i] Find an expression for $f'(x)$ when $f(x) = x^3 + x^2 - 5x$.

[ii] Solve the equation $3x^2 + 2x - 5 = 0$

[iii] Hence, or otherwise, find the values of x for which $f(x)$ is a decreasing function.

(6 marks)**Q4****PRACTICE**

Practice · Hard

P2 · TOPIC 11 5 marks

1 Find the values of x for which $f(x) = x^3 - 5x^2 + 3x - 2$ is a decreasing function.

(5 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11**4 marks**

- 2 Show that the function $f(x) = \sqrt{x} - \frac{7}{\sqrt{x}}$, $x > 0$, is increasing for all x in its domain.

(4 marks)**PAST PAPERS****Q6****PAST PAPER**

Pearson IAL WMA12 · October 2019 · Q1

P2 · TOPIC 11**6 marks**

1. A curve C has equation $y = 2x^2(x - 5)$
- (a) Find, using calculus, the x coordinates of the stationary points of C .
- (b) Hence find the values of x for which y is increasing.

(4)**(2)****(Total 6 marks)**

Q7**PAST PAPER**

Pearson IAL WMA12 · October 2019 · Q10

P2 · TOPIC 11**7 marks**

10. The curve C has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where a and b are constants.

Given that

- the point $(2, 5)$ lies on C
- the gradient of the curve at $(2, 5)$ is 7

(a) find the value of a and the value of b .

(4)

(b) Prove that C has no turning points.

(3)

Q8**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q6

P2 · TOPIC 11 9 marks

6.

$$f(x) = 4x^3 + px^2 + 8x + q$$

where p and q are constants.

Given that

- $(2x + 3)$ is a factor of $f(x)$
- $f(x)$ has a remainder of -5 when divided by $(x + 2)$

(a) (i) show that $p = 10$

(ii) find the value of q .

(5)

(b) Hence find the range of values of x for which $f(x)$ is decreasing.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 9 marks)

02 Stationary Points and Their Nature: $\frac{d^2y}{dx^2}$

Set $\frac{dy}{dx} = 0$ and solve for x ; substitute into y for the coordinates. Then put each x into $\frac{d^2y}{dx^2}$: negative is a maximum, positive is a minimum.

WHEN YOU SEE → YOU DO

“find the coordinates of the stationary point(s)”

→ $y' = 0$, then y

“determine the nature”, “justify”

→ sign of y'' at that x

“ P with $x = \frac{1}{2}$ is a stationary point: show $k = \dots$ ”

→ $y'(\frac{1}{2}) = 0$

“show that the x coordinates satisfy $2x^3 - 10x^2 + \dots = 0$ ”

→ $y' = 0$, clear the powers

“one turning point is at $x = 1$: find the others”

→ factor $(x - 1)$ out of the cubic

Rewrite

Differentiate

Set = 0 and solve

Second derivative

WORKED MODEL • JANUARY 2022

$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20$. Stationary point and its nature.

1 $\frac{dy}{dx} = \frac{27}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 27 = 3x$

multiply by $2x^{\frac{1}{2}}$

2 $x = 9$: $y = 81 - 27 - 20 = 34$

coordinates

ANSWER $(9, 34)$, $\frac{d^2y}{dx^2} < 0$: maximum

IDEA 02

Practise: Stationary Points and Their Nature:

$$\frac{d^2y}{dx^2}$$

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Find and classify

P2 · TOPIC 11**5 marks**

Find the stationary points of $y = 2x^3 - 3x^2 - 12x + 1$ and their nature.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 11 7 marks

2 (a) The curve C has equation $y = 3x^3 + 6x^2 - 5x + 1$.

Find expressions for $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(3 marks)

(b) [i] Evaluate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $x = \frac{1}{3}$.

[ii] What does your answer to part [b] tell you about curve C at the point where $x = \frac{1}{3}$?

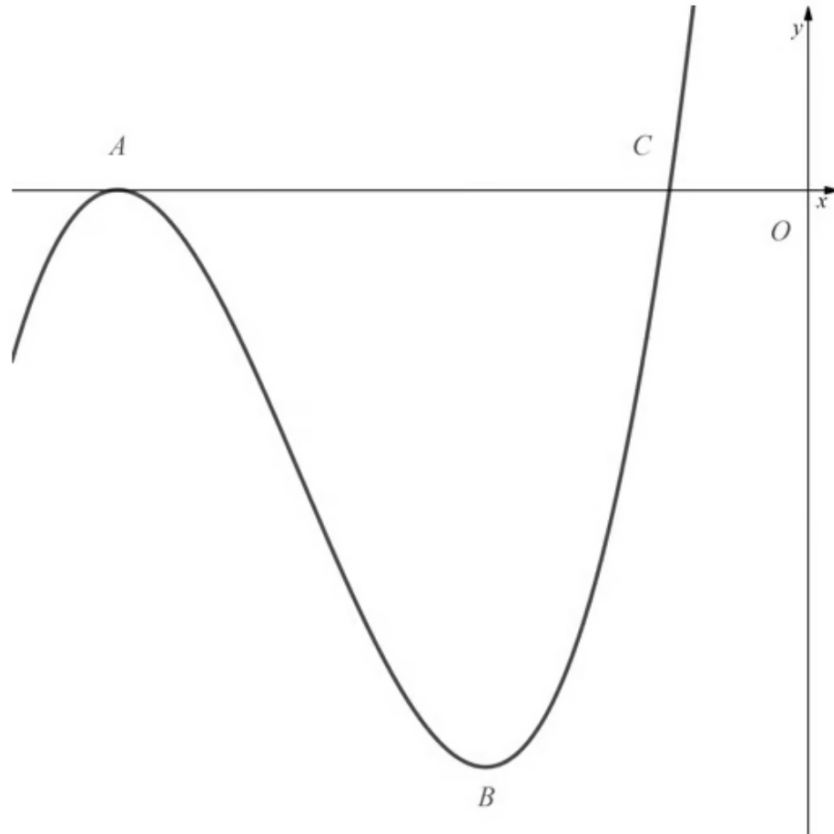
(4 marks)

Q11**PRACTICE**

Practice · Medium

P2 · TOPIC 11 7 marks

- 4 (a)** The diagram below shows part of the curve with equation $y = x^3 + 11x^2 + 35x + 25$. The curve touches the x -axis at A and cuts the x -axis at C . The points A and B are stationary points on the curve.



Using calculus, and showing all your working, find the coordinates of A and B .

(5 marks)

- (b)** Show that $(-1, 0)$ is a point on the curve and explain why those must be the coordinates of point C .

(2 marks)

Q12**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11 7 marks

3 (a) A curve is described by the equation $y = f(x)$, where $f(x) = 7 - 2x^2 + \sqrt{x}$, $x \geq 0$.

Find $f'(x)$ and $f''(x)$.

(3 marks)

(b) P is the stationary point on the curve.

Find the coordinates of P and determine its nature.

(4 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2020 · Q10

P2 · TOPIC 11 10 marks

10. A curve C has equation

$$y = 4x^3 - 9x + \frac{k}{x} \quad x > 0$$

where k is a constant.

The point P with x coordinate $\frac{1}{2}$ lies on C .

Given that P is a stationary point of C ,

(a) show that $k = -\frac{3}{2}$ (4)

(b) Determine the nature of the stationary point at P , justifying your answer. (2)

The curve C has a second stationary point.

(c) Using algebra, find the x coordinate of this second stationary point. (4)

Q14 **PAST PAPER** Pearson IAL WMA12 · January 2021 · Q2**P2 · TOPIC 11** **7 marks**

2. A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

- (a) Using calculus, find the x coordinates of the stationary points of the curve. (4)
- (b) Justify, by further calculus, the nature of all of the stationary points of the curve. (3)

(Total 7 marks)

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q2

P2 · TOPIC 11**8 marks**

2. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving each term in simplest form. (2)
- (b) Hence find the coordinates of the stationary point of C . (4)
- (c) Find $\frac{d^2y}{dx^2}$ and hence determine the nature of the stationary point of C . (2)

(Total 8 marks)

Q16**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q8

P2 · TOPIC 11 8 marks

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A curve has equation

$$y = 256x^4 - 304x - 35 + \frac{27}{x^2} \quad x \neq 0$$

(a) Find $\frac{dy}{dx}$

(3)

(b) Hence find the coordinates of the stationary points of the curve.

(5)

(Total 8 marks)

Q17 PAST PAPER Pearson IAL WMA12 · October 2022 · Q7

P2 · TOPIC 11 9 marks

7. The curve C has equation

$$y = \frac{12x^3(x - 7) + 14x(13x - 15)}{21\sqrt{x}} \quad x > 0$$

(a) Write the equation of C in the form

$$y = ax^{\frac{7}{2}} + bx^{\frac{5}{2}} + cx^{\frac{3}{2}} + dx^{\frac{1}{2}}$$

where a , b , c and d are fully simplified constants.

(3)

The curve C has three turning points.

Using calculus,

(b) show that the x coordinates of the three turning points satisfy the equation

$$2x^3 - 10x^2 + 13x - 5 = 0$$

(3)

Given that the x coordinate of one of the turning points is 1(c) find, using algebra, the exact x coordinates of the other two turning points.*(Solutions based entirely on calculator technology are not acceptable.)*

(3)

(Total for Question 7 is 9 marks)

Q18 PAST PAPER Pearson IAL WMA12 · June 2023 · Q7

P2 · TOPIC 11 8 marks

7. The height of a river above a fixed point on the riverbed was monitored over a 7-day period.

The height of the river, H metres, t days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \quad 0 \leq t \leq 7$$

Given that H has a stationary value at $t = \alpha$

- (a) use calculus to show that α satisfies the equation

$$5\alpha^2 - 18\alpha - 20 = 0 \quad (5)$$

- (b) Hence find the value of α , giving your answer to 3 decimal places. (1)

- (c) Use further calculus to prove that H is a maximum at this value of α . (2)

(Total for Question 7 is 8 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q4

P2 · TOPIC 11 7 marks

4. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ giving each term in simplest form. (2)
- (b) Hence find the x coordinate of the stationary point of C . (2)
- (c) (i) Find $\frac{d^2y}{dx^2}$ giving each term in simplest form.
- (ii) Hence determine the nature of the stationary point of C , giving a reason for your answer. (2)
- (d) State the range of values of x for which y is decreasing. (1)

(Total for Question 4 is 7 marks)

03 Optimisation: Build the Formula, Then Differentiate

A real object with a fixed volume (or area, or perimeter) and two unknown lengths. Use the fixed fact to write one length in terms of the other, substitute, and get a formula in one variable. Then $\frac{d}{dx} = 0$, and justify the minimum with the second derivative.

WHEN YOU SEE → YOU DO

“the volume is 972 cm^3 ”

→ height in terms of x

“show that $S = 6x^2 + \frac{2592}{x}$ ”

→ surface area, substitute the height

“open-topped”

→ leave out the top face

“use calculus to find the minimum value”

→ $S' = 0$, then S at that x

“justify that this is a minimum”

→ $S'' > 0$ at that x

WORKED MODEL · JANUARY 2023

$S = 6x^2 + \frac{2592}{x}$. Find the minimum surface area.

1 $S' = 12x - \frac{2592}{x^2} = 0 \Rightarrow x^3 = 216$

stationary

2 $x = 6; S'' = 12 + \frac{5184}{x^3} > 0$

minimum

ANSWER $S = 216 + 432 = 648 \text{ cm}^2$

× **TRAP** Differentiating before the formula has only one variable. Eliminate first.

IDEA 03

Practise: Optimisation: Build the Formula, Then Differentiate

My questions first, then selected practice, then real past papers.

Q20 DR ESLAM A box

P2 · TOPIC 11 5 marks

An open box has a square base of side x cm and volume 500 cm^3 . Show $A = x^2 + \frac{2000}{x}$ and find the minimum area.

Height from the volume, then the area in x only.

SELECTED PRACTICE

Q21**PRACTICE**

Practice · Medium

P2 · TOPIC 11 6 marks

- 5 (a)** A company manufactures food tins in the shape of cylinders which must have a constant volume of $150\pi \text{ cm}^3$. To lessen material costs the company would like to minimise the surface area of the tins.

By first expressing the height h of the tin in terms of its radius r , show that the surface area of the cylinder is given by $S = 2\pi r^2 + \frac{300\pi}{r}$.

(2 marks)

- (b)** Use calculus to find the minimum value for the surface area of the tins. Give your answer correct to 2 decimal places.

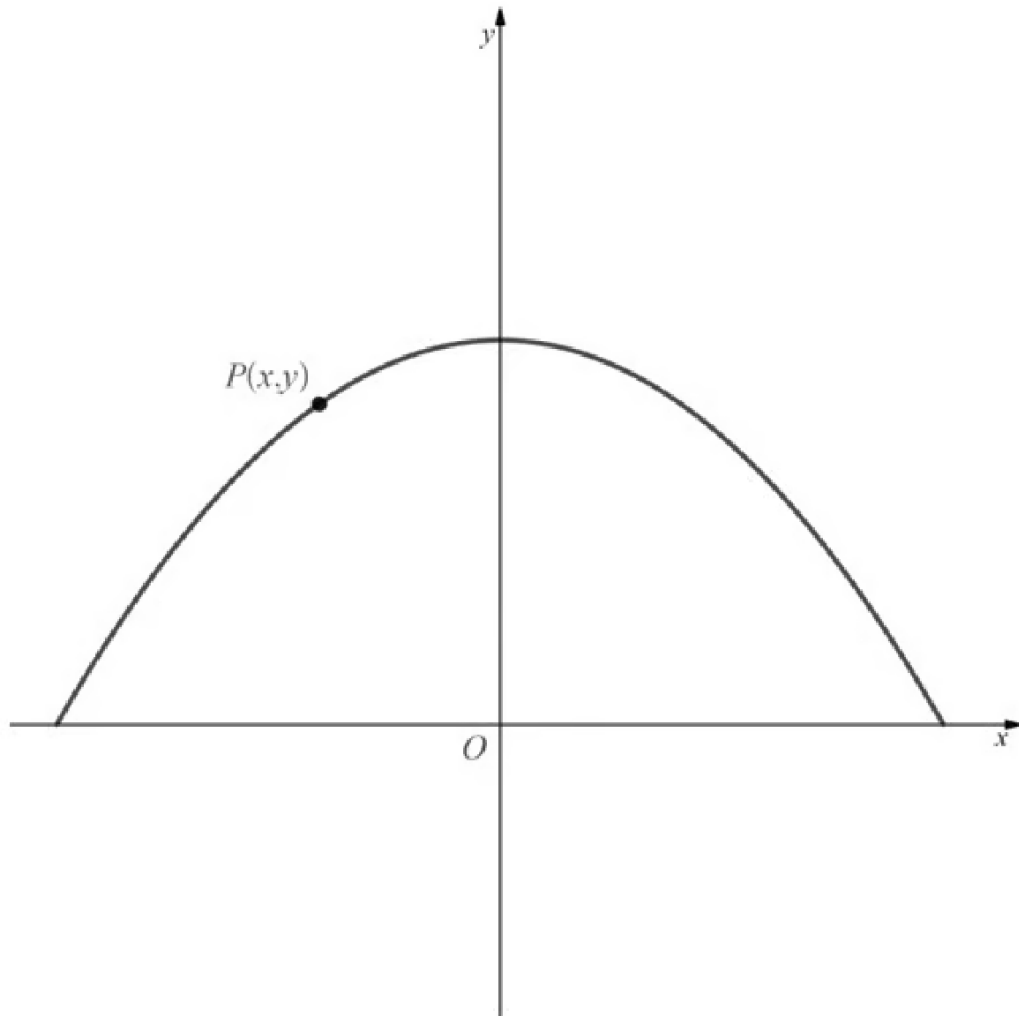
(4 marks)

Q22**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11 11 marks

- 4 (a) The diagram below shows the part of the curve with equation $y = 3 - \frac{1}{4}x^2$ for which $y > 0$. The marked point $P(x, y)$ lies on the curve. O is the origin.



Show that $OP^2 = 9 - \frac{1}{2}x^2 + \frac{1}{16}x^4$.

(3 marks)

- (b) Find the minimum distance from O to the curve, using calculus to prove that your answer is indeed a minimum.

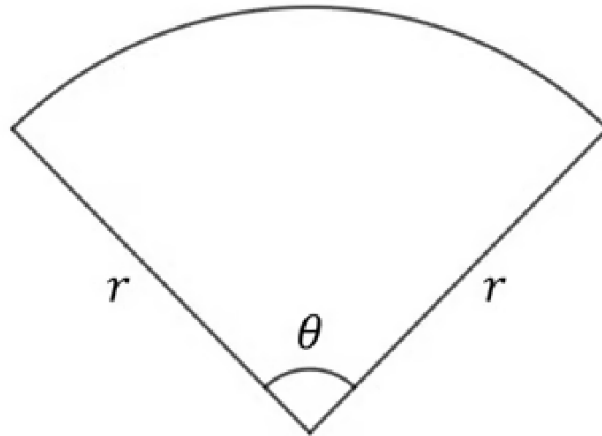
(8 marks)

Q23**PRACTICE**

Practice · Very Hard

P2 · TOPIC 11 9 marks

- 5 (a) The top of a patio table is to be made in the shape of a sector of a circle with radius r and central angle θ , where $0^\circ < \theta < 360^\circ$.



Although r and θ may be varied, it is necessary that the table have a fixed area of $A \text{ m}^2$.

Explain why $r > \sqrt{\frac{A}{\pi}}$.

(2 marks)

- (b) Show that the perimeter, P , of the table top is given by the formula

$$P = 2r + \frac{2A}{r}$$

(2 marks)

- (c) Show that the minimum possible value for P is equal to the perimeter of a square with area A . Be sure to prove that your value is a minimum.

(5 marks)

PAST PAPERS

Q24 PAST PAPER Pearson IAL WMA12 · June 2021 · Q9

P2 · TOPIC 11 10 marks

9.

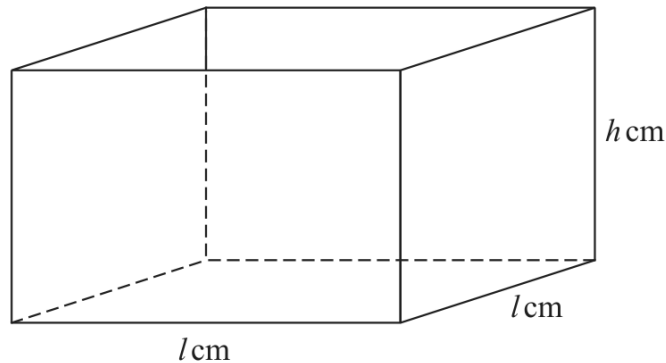


Figure 3

Figure 3 shows a sketch of a square based, open top box.

The height of the box is h cm, and the base edges each have length l cm.

Given that the volume of the box is $250\,000\text{ cm}^3$

(a) show that the external surface area, $S\text{ cm}^2$, of the box is given by

$$S = \frac{250\,000}{h} + 2000\sqrt{h} \quad (3)$$

(b) Use algebraic differentiation to show that S has a stationary point when $h = 250^k$ where k is a rational constant to be found.

(5)

(c) Justify by further differentiation that this value of h gives the minimum external surface area of the box.

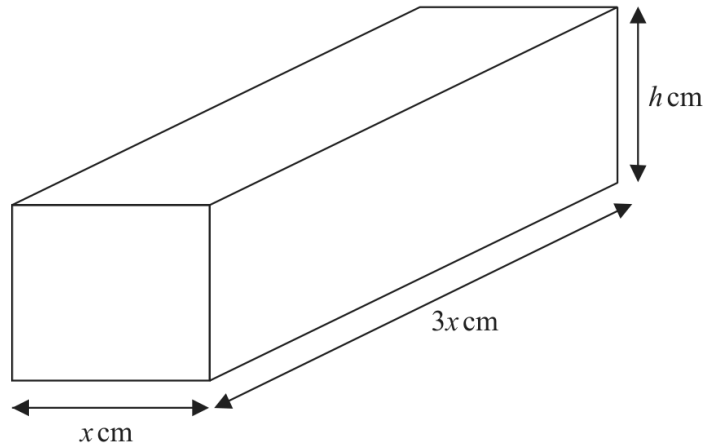
(2)

Q25**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q2

P2 · TOPIC 11 9 marks

2.

In this question you must show all stages of your working.**Solutions based entirely on calculator technology are not acceptable.****Figure 2**

A brick is in the shape of a cuboid with width $x \text{ cm}$, length $3x \text{ cm}$ and height $h \text{ cm}$, as shown in Figure 2.

The volume of the brick is 972 cm^3

(a) Show that the surface area of the brick, $S \text{ cm}^2$, is given by

$$S = 6x^2 + \frac{2592}{x} \quad (3)$$

(b) Find $\frac{dS}{dx}$ (1)

(c) Hence find the value of x for which S is stationary. (2)

(d) Find $\frac{d^2S}{dx^2}$ and hence show that the value of x found in part (c) gives the minimum value of S . (2)

(e) Hence find the minimum surface area of the brick. (1)

(Total for Question 2 is 9 marks)

Q26 PAST PAPER Pearson IAL WMA12 · January 2025 · Q3

P2 · TOPIC 11 10 marks

3.

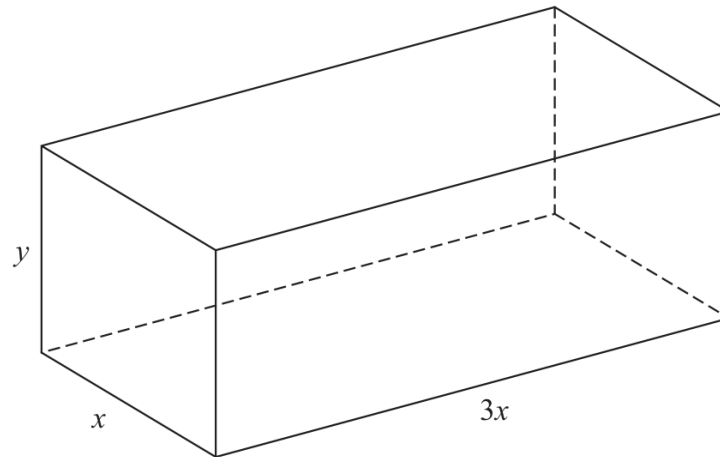
**Figure 1**

Figure 1 shows an open-topped container used for holding water.

The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle $3x$ metres by x metres.

The height of the container is y metres as shown in Figure 1.

Given that the capacity of the container is 120m^3

(a) show that the area $A\text{m}^2$ of the sheet metal used to make the container is given by

$$A = Px^2 + \frac{Q}{x}$$

where P and Q are positive constants to be found.

(4)

(b) Use calculus to find the value of x for which A has a stationary value, giving your answer to 3 significant figures.

(4)

(c) Find $\frac{d^2A}{dx^2}$ and hence show that the value of x found in part (b) gives the minimum value of A .

(2)

(Total for Question 3 is 10 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q9

P2 · TOPIC 11 11 marks

9:

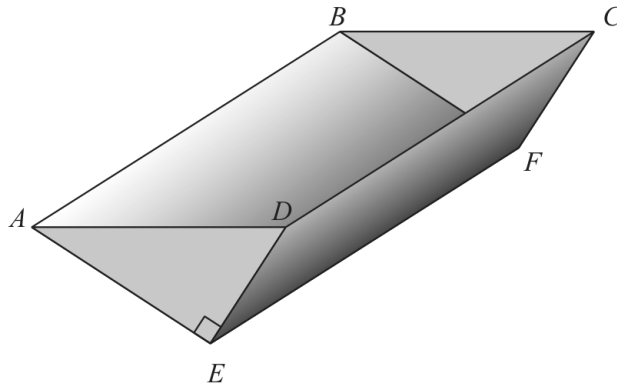
**Figure 2**

Figure 2 shows a sketch of an open container.

The sides $CDEF$ and $ABFE$ are rectangles.

The ends ADE and BCF are congruent (identical) right-angled triangles.

The container is made from metal of negligible thickness.

Given that

- $AE = BF = 3x$ metres
- $DE = CF = 2x$ metres
- $AB = DC = EF = L$ metres

and the capacity of the container is 12 m^3

(a) show that the area of metal used to make the container, $S \text{ m}^2$, is given by

$$S = Px^2 + \frac{Q}{x}$$

where P and Q are positive integers to be found.

(4)

Given that x can vary,

(b) use algebraic calculus to find the minimum value of S , giving your answer to one decimal place.

(5)

(c) Justify that the value of S found in part (b) is a minimum.

(2)

(Total for Question 9 is 11 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q9

P2 · TOPIC 11 11 marks

9.

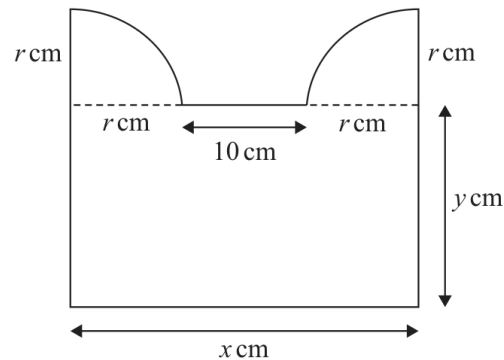
**Figure 4**

Figure 4 shows a sketch of a design for a logo.

The design consists of a rectangle of base x cm and height y cm with two quarter circles of equal radius r cm joined to the top edge with a gap of 10 cm between the nearest point of each quarter circle, as shown in Figure 4.

(a) Show that the area, A cm², of the design is given by

$$A = xy + \frac{\pi}{8}(x - 10)^2 \quad (2)$$

Given that

- the total area of the design is one square metre
- the perimeter, P cm, of the design is given by $P = 2y + \left(2 + \frac{\pi}{2}\right)x - 5\pi$

(b) show that

$$P = \frac{20000 - 25\pi}{x} + \left(2 + \frac{\pi}{4}\right)x \quad (4)$$

(c) Use algebraic differentiation to find the value of x that gives the minimum value for P , giving the answer to 3 significant figures.

(3)

By considering $\frac{d^2P}{dx^2}$

(d) justify that the value for x found in part (c) gives the minimum value for P .

(2)

(Total for Question 9 is 11 marks)

04 Stationary Points with Areas

These questions join this topic to Integration: find the stationary point first (it gives a limit or the height of a line), then integrate between the right limits. Areas below the x -axis come out negative.

WHEN YOU SEE → YOU DO

“the line l passes through M and is parallel to the x -axis”

→ $y =$ the y of the stationary point

“the region bounded by C , l and the y -axis”

→ line minus curve, or curve minus line

“area of R_1 equals area of R_2 ”

→ $\int_0^k y \, dx = 0$

“use algebraic integration”

→ show the integral and the substitution

WORKED MODEL · JUNE 2024

$y = 2x^{\frac{3}{2}}(4 - x)$. R_1 above the axis and R_2 below it, with equal areas. Find k .

$$1 \quad \int_0^k (8x^{\frac{3}{2}} - 2x^{\frac{5}{2}}) \, dx = \frac{16}{5}k^{\frac{5}{2}} - \frac{4}{7}k^{\frac{7}{2}} = 0$$

equal areas: total 0

ANSWER $k = \frac{28}{5}$

IDEA 04

Practise: Stationary Points with Areas

My questions first, then real past papers.

Q29**DR ESLAM**

Then the area

P2 · TOPIC 11**4 marks**

$y = x(4 - x)$. Find the stationary point and the area between the curve and the x -axis.

PAST PAPERS

Q30 PAST PAPER Pearson IAL WMA12 · October 2021 · Q8

P2 · TOPIC 11 10 marks

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

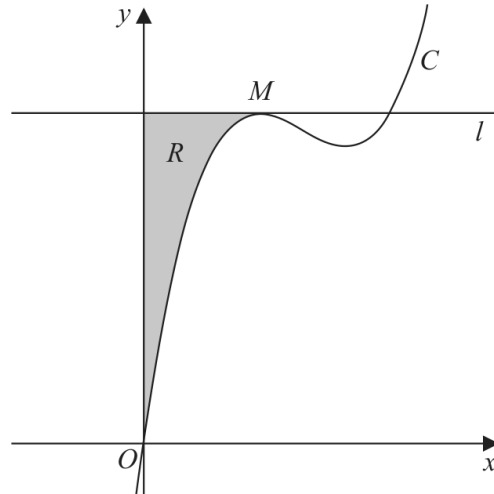


Figure 2

Figure 2 shows a sketch of part of the curve C with equation

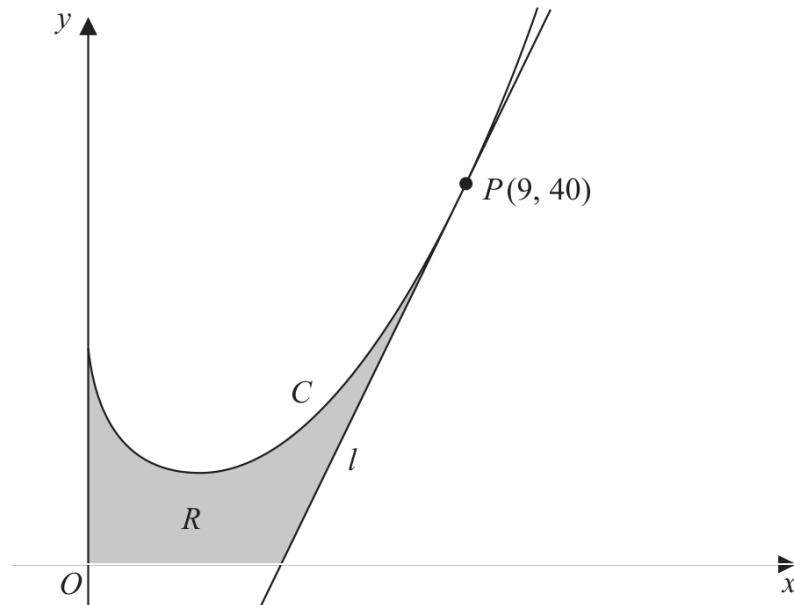
$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point M is the maximum turning point of C and is shown in Figure 2.Given that the x coordinate of M is 2(a) show that $k = 28$ (3)(b) Determine the range of values of x for which y is increasing. (2)The line l passes through M and is parallel to the x -axis.The region R , shown shaded in Figure 2, is bounded by the curve C , the line l and the y -axis.(c) Find, by algebraic integration, the exact area of R . (5)**(Total 10 marks)**

Q31 PAST PAPER Pearson IAL WMA12 · October 2023 · Q9

P2 · TOPIC 11 12 marks

9.

**Figure 3**

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve C with equation

$$y = \frac{2}{3}x^2 - 9\sqrt{x} + 13 \quad x \geq 0$$

(a) Find, using calculus, the range of values of x for which y is increasing.

(4)

The point P lies on C and has coordinates $(9, 40)$.

The line l is the tangent to C at the point P .

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the line l , the x -axis and the y -axis.

(b) Find, using calculus, the exact area of R .

(8)

(Total for Question 9 is 12 marks)

Q32**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q10

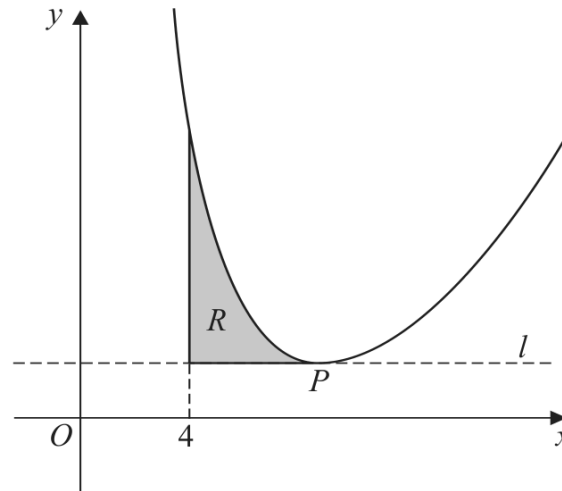
P2 · TOPIC 11 9 marks**10.****In this question you must show detailed reasoning.****Solutions relying entirely on calculator technology are not acceptable.****Figure 2**

Figure 2 shows a sketch of the curve with equation

$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \quad x > 0$$

The point P is the only stationary point on the curve.(a) Use calculus to show that the x coordinate of P is 9**(4)**The line l passes through the point P and is parallel to the x -axis.The region R , shown shaded in Figure 2, is bounded by the curve, the line l and the line with equation $x = 4$ (b) Use algebraic integration to find the exact area of R .**(5)****(Total for Question 10 is 9 marks)**

Q33**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q9

P2 · TOPIC 11 8 marks

9.

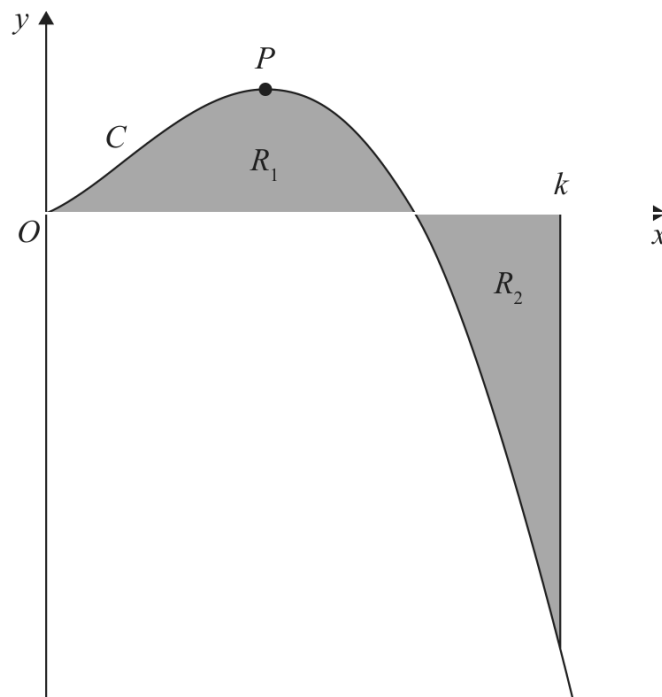
**Figure 1**

Figure 1 is a sketch of the curve C with equation

$$y = 2x^{\frac{3}{2}}(4 - x) \quad x \geq 0$$

The point P is the stationary point of C .

(a) Find, using calculus, the x coordinate of P .

(4)

The region R_1 , shown shaded in Figure 1, is bounded by C and the x -axis.

The region R_2 , also shown shaded in Figure 1, is bounded by C , the x -axis and the line with equation $x = k$, where k is a constant.

Given that the area of R_1 is equal to the area of R_2

(b) find, using calculus, the exact value of k .

(4)

(Total for Question 9 is 8 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June (R paper) 2024 · Q10

P2 · TOPIC 11 9 marks

10.

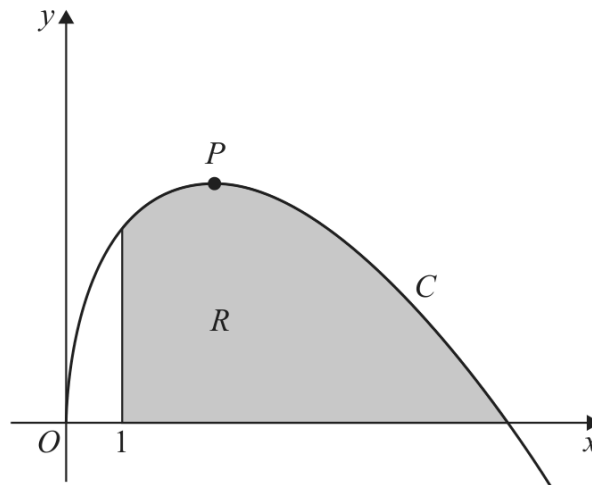


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{9x - x^2}{2\sqrt{x}} \quad x > 0$$

The point P is the stationary point on C .

(a) Find, using calculus, the x coordinate of P .

(4)

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 1$

(b) Using calculus, calculate the exact area of R .

(5)

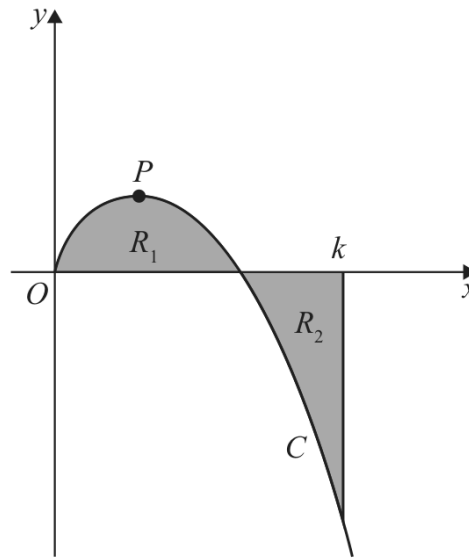
(Total for Question 10 is 9 marks)

Q35**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q10

P2 · TOPIC 11 8 marks

10.

**Figure 2**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 2 shows a sketch of the curve C with equation

$$y = \frac{\sqrt{x}(100 - x^2)}{40} \quad x \geq 0$$

The point P is a stationary point on C .

(a) Use algebraic differentiation to find the exact x coordinate of P .

(4)

The region R_1 , shown shaded in Figure 2, is bounded by C and the x -axis.

The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and the line with equation $x = k$, where k is a constant.

Given that the area of R_1 is equal to the area of R_2

(b) use algebraic integration to find the exact value of k .

(4)

(Total for Question 10 is 8 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------|---|
| 01 | Increasing | $f' > 0$; decreasing $f' < 0$; “for all x ”: complete the square. |
| 02 | Stationary | $f' = 0$; $f'' < 0$ max, $f'' > 0$ min. |
| 03 | Optimisation | One variable first; $f' = 0$; justify with f'' . |
| 04 | With areas | Stationary point first, then integrate. |

BOTTOM LINE Rewrite $\rightarrow$ differentiate $\rightarrow = 0 \rightarrow$ second derivative decides max or min.

Answers

Applications of Differentiation · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · INCREASING AND DECREASING FUNCTIONS

- Q1** Dr Eslam · Where is it increasing?
 $f'(x) = 3x(x - 4) > 0: x < 0 \text{ or } x > 4$
- Q2** Dr Eslam · Increasing for all x
 $f'(x) = 3(x + 1)^2 + 3 > 0$
- Q3** Practice · Easy
 (i) $3x^2 + 2x - 5$ (ii) $x = -\frac{5}{3}, 1$ (iii) $-\frac{5}{3} < x < 1$
- Q4** Practice · Hard
 $f'(x) = 3x^2 - 10x + 3 < 0: \frac{1}{3} < x < 3$
- Q5** Practice · Very Hard
 $f'(x) = \frac{1}{2}x^{-\frac{1}{2}} + \frac{7}{2}x^{-\frac{3}{2}} = \frac{x+7}{2x^{3/2}} > 0 \text{ for } x > 0$
- Q6** Pearson IAL WMA12 · October 2019 · Q1
 (a) $x = 0, \frac{10}{3}$. (b) $x < 0 \text{ or } x > \frac{10}{3}$
- Q7** Pearson IAL WMA12 · October 2019 · Q10
 (a) $a = \frac{4}{3}, b = \frac{1}{3}$. (b) Proven: C has no turning points
- Q8** Pearson IAL WMA12 · June (R paper) 2024 · Q6
 (a)(i) $p = 10$. (a)(ii) $q = 3$. (b) $-1 < x < -\frac{2}{3}$

IDEA 02 · STATIONARY POINTS AND THEIR NATURE: $\frac{d^2y}{dx^2}$

- Q9** Dr Eslam · Find and classify
 $x = -1: (-1, 8) \max (y'' = -18); x = 2: (2, -19) \min (y'' = 18)$
- Q10** Practice · Easy
 (a) $9x^2 + 12x - 5, 18x + 12$ (b)(i) 0 and 18 (ii) a stationary point, a minimum
- Q11** Practice · Medium
 (a) $A(-5, 0)$ maximum, $B\left(-\frac{7}{3}, -\frac{256}{27}\right)$ minimum
 (b) $y(-1) = 0$; the curve crosses the axis only once more
- Q12** Practice · Very Hard
 (a) $f'(x) = -4x + \frac{1}{2}x^{-\frac{1}{2}}, f''(x) = -4 - \frac{1}{4}x^{-\frac{3}{2}}$ (b) $P\left(\frac{1}{4}, \frac{59}{8}\right)$, maximum ($f'' < 0$)
- Q13** Pearson IAL WMA12 · January 2020 · Q10
 (a) $k = -\frac{3}{2}$. (b) local maximum. (c) $x = \frac{1}{\sqrt{2}}$
- Q14** Pearson IAL WMA12 · January 2021 · Q2
 (a) $x = -2, \frac{8}{3}$. (b) local maximum at $x = -2$ and local minimum at $x = \frac{8}{3}$
- Q15** Pearson IAL WMA12 · January 2022 · Q2
 (a) $\frac{dy}{dx} = \frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$. (b) (9, 34). (c) local maximum
- Q16** Pearson IAL WMA12 · June 2022 · Q8
 (a) $\frac{dy}{dx} = 1024x^3 - 304 - \frac{54}{x^3}$. (b) $\left(-\frac{1}{2}, 241\right), \left(\frac{3}{4}, -134\right)$
- Q17** Pearson IAL WMA12 · October 2022 · Q7
 (a) $y = \frac{4}{7}x^{7/2} - 4x^{5/2} + \frac{26}{3}x^{3/2} - 10x^{1/2}$. (b) Proven. (c) $x = 2 - \frac{\sqrt{6}}{2}$ and $x = 2 + \frac{\sqrt{6}}{2}$
- Q18** Pearson IAL WMA12 · June 2023 · Q7
 (a) Proven. (b) $\alpha = 4.491$. (c) H is a maximum at this value
- Q19** Pearson IAL WMA12 · October 2024 · Q4
 (a) $\frac{2}{\sqrt{x}} - \frac{9}{2x^{3/2}}$. (b) $x = \frac{9}{4}$. (c)(i) $-\frac{1}{x^{3/2}} + \frac{27}{4x^{5/2}}$. (c)(ii) Minimum. (d) $0 < x < \frac{9}{4}$

IDEA 03 · OPTIMISATION: BUILD THE FORMULA, THEN DIFFERENTIATE

- Q20** Dr Eslam · A box
 $A' = 2x - \frac{2000}{x^2} = 0: x = 10; A = 300 \text{ cm}^2$
 $(A'' = 6 > 0)$
- Q21** Practice · Medium
 $(a) h = \frac{150}{r^2}, S = 2\pi r^2 + 2\pi rh \quad (b) r = \sqrt[3]{75}: S_{\min} = 335.23 \text{ cm}^2$
- Q22** Practice · Very Hard
 $(a) OP^2 = x^2 + (3 - \frac{1}{4}x^2)^2 \quad (b) \frac{d}{dx}OP^2 = 0 \text{ at } x = \pm 2$
(second derivative > 0): $OP_{\min} = 2\sqrt{2}$
- Q23** Practice · Very Hard
 $(a) \theta < 2\pi \text{ so } A = \frac{1}{2}r^2\theta < \pi r^2 \quad (b) P = 2r + r\theta = 2r + \frac{2A}{r} \quad (c) P' = 0 \text{ at } r = \sqrt{A}: P = 4\sqrt{A}; P'' = \frac{4A}{r^3} > 0$
- Q24** Pearson IAL WMA12 · June 2021 · Q9
 $(a) \text{ Proven: } S = \frac{250000}{h} + 2000\sqrt{h}. \quad (b) h = 250^{2/3}, \text{ so } k = \frac{2}{3}. \quad (c) \text{ Minimum}$
- Q25** Pearson IAL WMA12 · January 2023 · Q2
 $(a) S = 6x^2 + \frac{2592}{x}. \quad (b) \frac{dS}{dx} = 12x - \frac{2592}{x^2}. \quad (c) x = 6. \quad (d) \text{ Minimum shown.}$
 $(e) 648 \text{ cm}^2$
- Q26** Pearson IAL WMA12 · January 2025 · Q3
 $(a) A = 3x^2 + \frac{320}{x}, \text{ so } P = 3 \text{ and } Q = 320. \quad (b) x = 3.76. \quad (c) \frac{d^2A}{dx^2} = 6 + \frac{640}{x^3}, \text{ minimum}$
- Q27** Pearson IAL WMA12 · June 2025 · Q9
 $(a) S = 6x^2 + \frac{20}{x}, \text{ so } P = 6 \text{ and } Q = 20. \quad (b) 25.3. \quad (c) \text{ Minimum justified}$
- Q28** Pearson IAL WMA12 · June 2026 · Q9
 $(a) r = \frac{x-10}{2}: xy + \frac{\pi}{2}r^2 \quad (b) y \text{ from } A = 10000$
 $(c) P' = 0: x = 84.6 \quad (d) P'' = \frac{2(20000-25\pi)}{x^3} > 0: \text{ minimum}$

IDEA 04 · STATIONARY POINTS WITH AREAS

- Q29** Dr Eslam · Then the area $(2, 4)$ max; $\int_0^4 (4x - x^2) dx = \frac{32}{3}$
- Q30** Pearson IAL WMA12 · October 2021 · Q8
 (a) Proven: $k = 28$. (b) $x < 2$ or $x > \frac{7}{2}$. (c) $\frac{40}{3}$
- Q31** Pearson IAL WMA12 · October 2023 · Q9
 (a) y is increasing for $x > \frac{9}{4}$. (b) Area = $\frac{857}{21}$
- Q32** Pearson IAL WMA12 · January 2024 · Q10
 (a) Proven. (b) $\frac{373}{3}$
- Q33** Pearson IAL WMA12 · June 2024 · Q9
 (a) $x = \frac{12}{5}$. (b) $k = \frac{28}{5}$
- Q34** Pearson IAL WMA12 · June (R paper) 2024 · Q10
 (a) $x = 3$. (b) $\frac{148}{5}$
- Q35** Pearson IAL WMA12 · January 2026 · Q10
 (a) $x = 2\sqrt{5}$. (b) $k = \frac{10\sqrt{21}}{3}$

DR ESLAM AHMED
eliteigcse.com

Integration: Areas and the Trapezium Rule

Integrate, top limit minus bottom. For an area: sketch, then top curve minus bottom curve. No formula? Trapezium rule.

THE TOPIC ON ONE MAP

Integration: Areas and the Trapezium Rule

01 — Definite integrals

$$[F(x)]_a^b$$

unknown limit k

unknown coefficient

02 — Areas

top - bottom

below the axis

curve and line

03 — Trapezium rule

$$h = \frac{b-a}{n}$$

over or under?

transform the answer

THE TWO TOOLS

$$\int_a^b f(x) dx = F(b) - F(a) \quad \int_a^b y dx \approx \frac{h}{2} \{y_0 + y_n + 2(y_1 + \dots + y_{n-1})\}$$

$h = \frac{b-a}{n}$ · n strips = $n+1$ ordinates (y values) · rewrite roots and fractions as powers first

01 Definite Integrals: Limits and Unknown Constants

Rewrite every term as ax^n , integrate (no $+c$), substitute the top limit and subtract the bottom. If a limit or a coefficient is unknown, the integral equals a number: that is your equation.

WHEN YOU SEE → YOU DO

“use calculus to find the value of

$$\int_4^9 \dots dx$$

→ rewrite, integrate, substitute

“given $\int_k^{2k} \dots dx = 14$, find k ”

→ an equation in k

“($x+2$) is a factor and $\int_3^5 f(x) dx = 176$ ”

→ two equations, solve for A, B

$$\frac{(x-k)^2}{\sqrt{x}}$$

→ expand the top, divide each term by $x^{\frac{1}{2}}$

WORKED MODEL · JANUARY 2026

$$\int_k^{2k} \left(\frac{12}{x^2} + 4 \right) dx = 14, k > 0. \text{ Find } k.$$

$$1 \quad \left[-\frac{12}{x} + 4x \right]_k^{2k} = \frac{6}{k} + 4k$$

integrate, substitute

$$2 \quad \frac{6}{k} + 4k = 14 \Rightarrow 2k^2 - 7k + 3 = 0$$

equation

$$\text{ANSWER} \quad k = \frac{1}{2} \text{ or } k = 3$$

IDEA 01

Practise: Definite Integrals: Limits and Unknown Constants

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Evaluate

P2 · TOPIC 12 3 marks

Evaluate $\int_1^4 \left(3\sqrt{x} - \frac{2}{x^2} \right) dx$.

Q2 DR ESLAM Find the limit

P2 · TOPIC 12 4 marks

Given $\int_1^k (2x + 1) dx = 18$, $k > 1$, find k .

Integrate, substitute, then solve the quadratic in k .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 12 4 marks

1 Evaluate

[i] $\int_1^2 4x \, dx,$

[ii] $\int_0^3 (9x^2 + 4x) \, dx.$

(4 marks)

Q4**PRACTICE**

Practice · Hard

P2 · TOPIC 12**5 marks**

- 1 Use calculus to find the value of

$$\int_4^9 \frac{x^2 + 1}{\sqrt{x}} dx$$

(5 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 12 5 marks

2 Given

$$\int_q^{4q} 5x\sqrt{x} \, dx = 15\,066$$

find the value of the constant q .**(5 marks)**

PAST PAPERS

Q6 **PAST PAPER** Pearson IAL WMA12 · June 2022 · Q7**P2 · TOPIC 12** **7 marks**

7. $f(x) = Ax^3 + 6x^2 - 4x + B$

where A and B are constants.

Given that

- $(x + 2)$ is a factor of $f(x)$

- $\int_3^5 f(x) dx = 176$

find the value of A and the value of B .

(7)

(Total 7 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q4

P2 · TOPIC 12 5 marks

4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Given that k is a positive constant and

$$\int_k^{2k} \left(\frac{12}{x^2} + 4 \right) dx = 14$$

find the possible values of k .

(5)

(Total for Question 4 is 5 marks)

02 Areas: Curves, Lines and Below the Axis

Sketch and find where the curves meet — those are the limits. The area between two graphs is $\int(\text{top} - \text{bottom})dx$. An integral below the x-axis is negative: find each part separately and add the sizes.

WHEN YOU SEE → YOU DO

“the region bounded by C and the line l ”

→ meeting points, then top - bottom

“bounded by the curve, the y-axis and $y = 17$ ”

→ rectangle - area under the curve

a region part above, part below the axis

→ split at the root, add the sizes

“area of R_1 equals area of R_2 ”

→ the signed integral is 0

“use algebraic integration to find the exact area”

→ exact: fractions, surds, π

a triangle or a trapezium in the region

→ use the shape formula, then add or subtract

Sketch

Meeting points

Top - bottom

Integrate and add

WORKED MODEL · JUNE 2025

$y = 3x + \frac{16}{x^2} - 8$; $P(1, 11)$ and $Q(4, 5)$. Area between C and the segment PQ .

1 trapezium under PQ : $\frac{1}{2}(11 + 5)(3) = 24$

the line

2 under C : $\left[\frac{3}{2}x^2 - \frac{16}{x} - 8x\right]_1^4 = \frac{21}{2}$

the curve

ANSWER $24 - \frac{21}{2} = \frac{27}{2}$

IDEA 02

Practise: Areas: Curves, Lines and Below the Axis

My questions first, then selected practice, then real past papers.

Q8 **DR ESLAM** Curve and line

P2 · TOPIC 12 5 marks

Find the area enclosed by $y = x^2 - 4x + 6$ and $y = x + 2$.

Q9 **DR ESLAM** Below the axis

P2 · TOPIC 12 3 marks

Find the area between $y = x^2 - 4$ and the x-axis.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 12 5 marks

- 3 (a)** The area bounded by the curve with equation $y = 9 - x^2$, the x -axis and the vertical lines with equations $x = 1$ and $x = 2$ is to be found.

Write down an integral that would find this area.

(2 marks)

- (b)** Evaluate your integral from part [a] and hence find the area described above.

(3 marks)

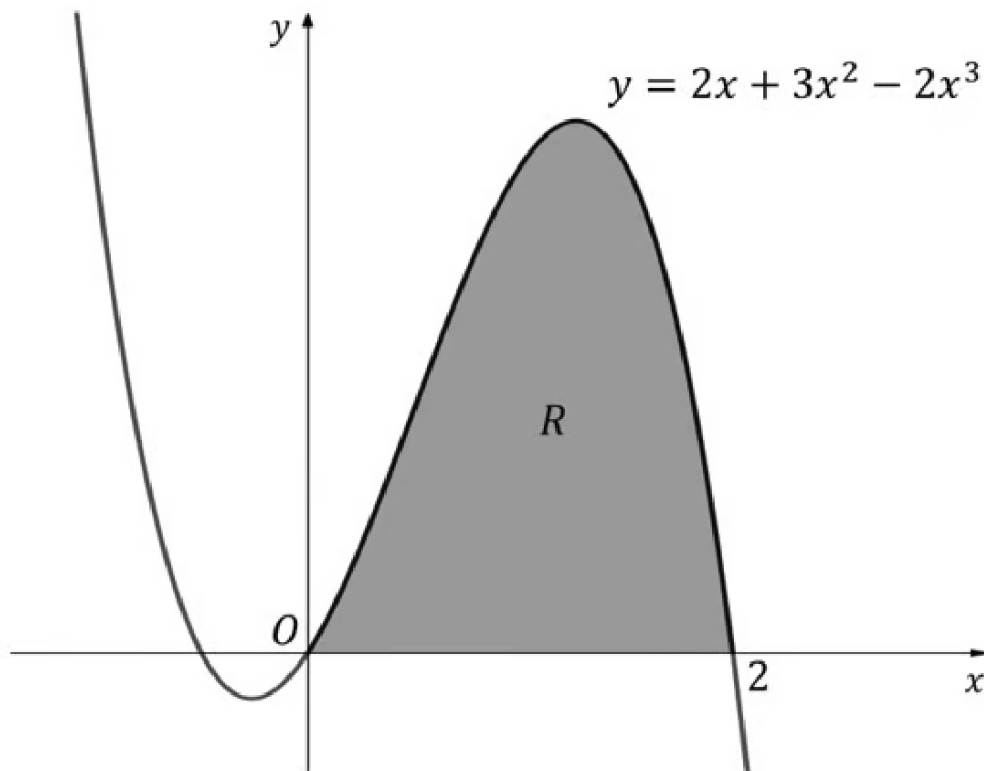
Q11**PRACTICE**

Practice · Hard

P2 · TOPIC 12

4 marks

- 3 The diagram below shows part of the graph of $y = 2x + 3x^2 - 2x^3$.



Find the area of the shaded region labelled R .

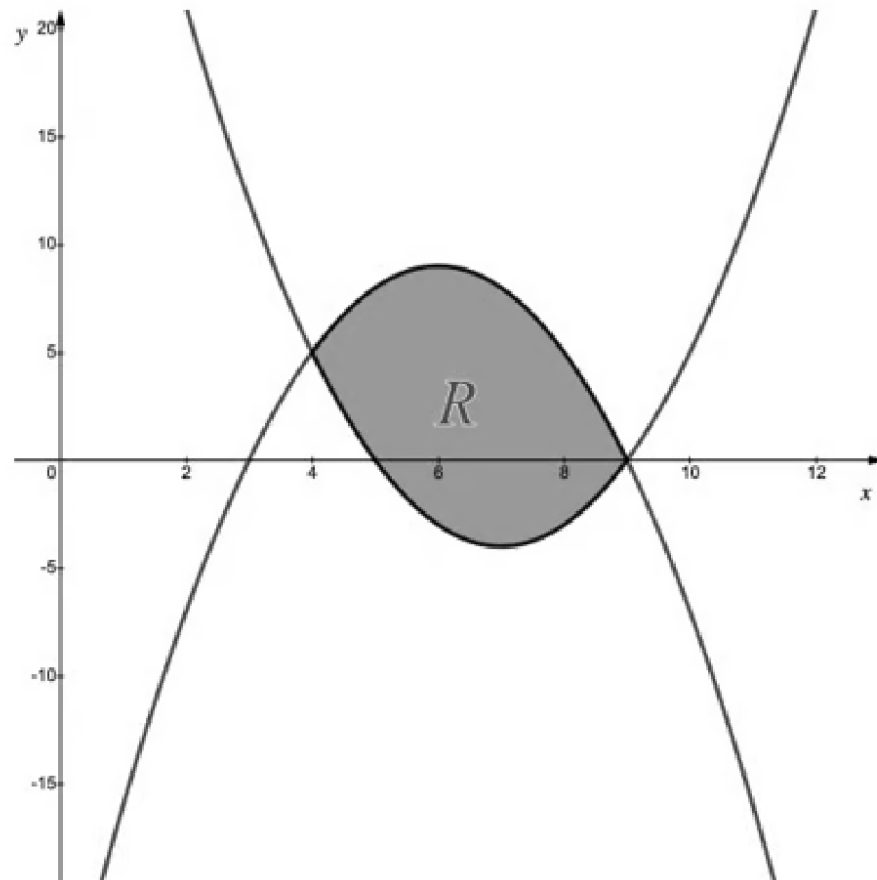
(4 marks)

Q12**PRACTICE**

Practice · Very Hard

P2 · TOPIC 12**8 marks**

7 The diagram below shows the graphs of $y = 12x - x^2 - 27$ and $y = x^2 - 14x + 45$.



Find the area of the shaded region, R .

(8 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · October 2019 · Q8

P2 · TOPIC 12 9 marks

8. Solutions relying on calculator technology are not acceptable in this question.

(i)

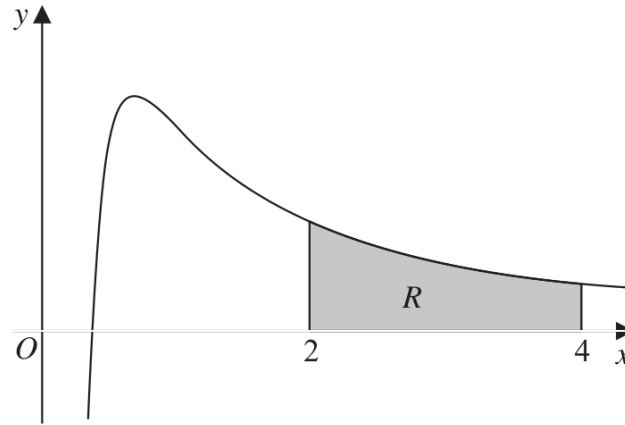
**Figure 2**

Figure 2 shows a sketch of part of a curve with equation

$$y = \frac{8\sqrt{x} - 5}{2x^2} \quad x > 0$$

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 4$

Find the exact area of R .

(5)

(ii) Find the value of the constant k such that

$$\int_{-3}^6 \left(\frac{1}{2}x^2 + k \right) dx = 55$$

(4)

(Total 9 marks)

Q14 PAST PAPER Pearson IAL WMA12 · January 2020 · Q4

P2 · TOPIC 12 6 marks

4.

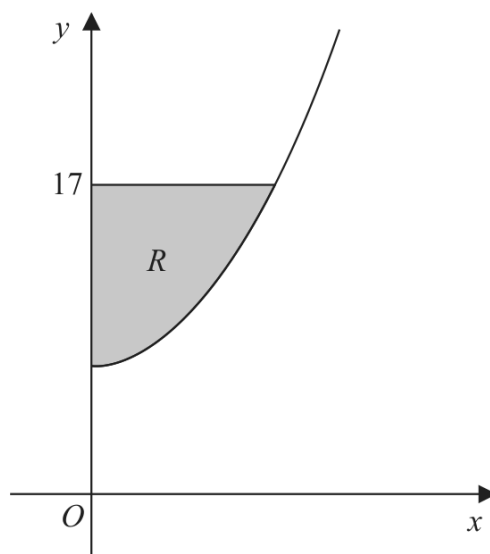
**Figure 1**

Figure 1 shows a sketch of the curve with equation

$$y = 2x^2 + 7 \quad x \geq 0$$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the y -axis and the line with equation $y = 17$

Find the exact area of R .

(6)**(Total 6 marks)**

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q9

P2 · TOPIC 12 10 marks

9.

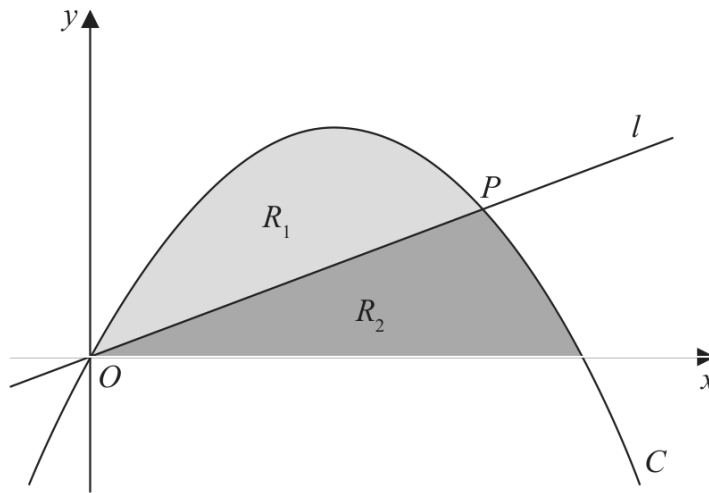
**Figure 2**

Figure 2 shows

- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

The line and the curve intersect at the origin O and at the point P .

- (a) Find, in terms of
- m
- , the coordinates of
- P
- .

(2)The region R_1 , shown shaded in Figure 2, is bounded by C and l .

- (b) Show that the area of
- R_1
- is

$$\frac{(1-m)^3}{6}$$

(5)The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and l .Given that the area of R_1 is equal to the area of R_2

- (c) find the exact value of
- m
- .

(3)**(Total 10 marks)**

Q16**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q9

P2 · TOPIC 12**8 marks**

9. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

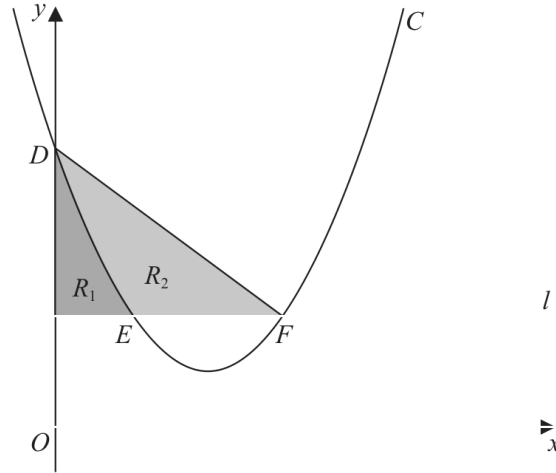


Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y -axis at the point D .

(a) Write down the coordinates of D .

(1)

The curve C intersects the line l at the points E and F , as shown in Figure 3.

(b) Find the x coordinate of E and the x coordinate of F .

(2)

Shown shaded in Figure 3 is

- the region R_1 which is bounded by C , l and the y -axis
- the region R_2 which is bounded by C and the line segments EF and DF

Given that $\frac{\text{area of } R_1}{\text{area of } R_2} = k$, where k is a constant,

(c) use algebraic integration to find the exact value of k , giving your answer as a simplified fraction.

(5)

(Total for Question 9 is 8 marks)

Q17**PAST PAPER**

Pearson IAL WMA12 · June 2023 · Q10

P2 · TOPIC 12 10 marks

10. The curve C has equation

$$y = \frac{(x - k)^2}{\sqrt{x}} \quad x > 0$$

where k is a **positive** constant.

(a) Show that

$$\int_1^{16} \frac{(x - k)^2}{\sqrt{x}} \, dx = ak^2 + bk + \frac{2046}{5}$$

where a and b are integers to be found.

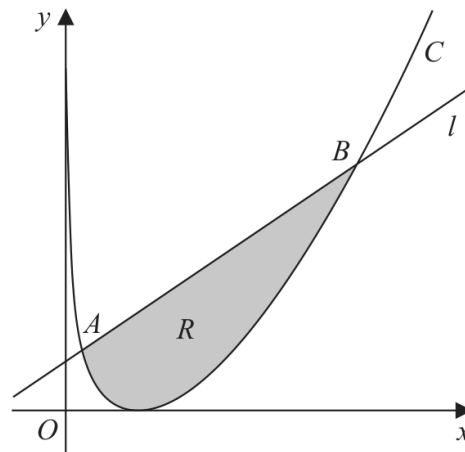
(5)

Figure 1

Figure 1 shows a sketch of the curve C and the line l .

Given that l intersects C at the point $A(1, 9)$ and at the point $B(16, q)$ where q is a constant,

(b) show that $k = 4$

(2)

The region R , shown shaded in Figure 1, is bounded by C and l

Using the answers to parts (a) and (b),

(c) find the area of region R

(3)

(Total for Question 10 is 10 marks)

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

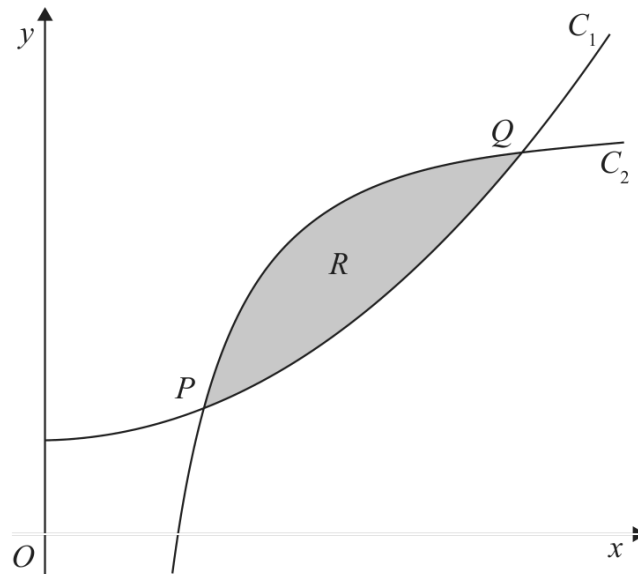


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation

$$y = x^2 + 3 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 13 - \frac{9}{x^2} \quad x > 0$$

The curves C_1 and C_2 intersect at the points P and Q as shown in Figure 1.

(a) Use algebra to find the x coordinate of P and the x coordinate of Q .

(4)

The finite region R , shown shaded in Figure 1, is bounded by C_1 and C_2

(b) Use algebraic integration to find the exact area of R .

(4)

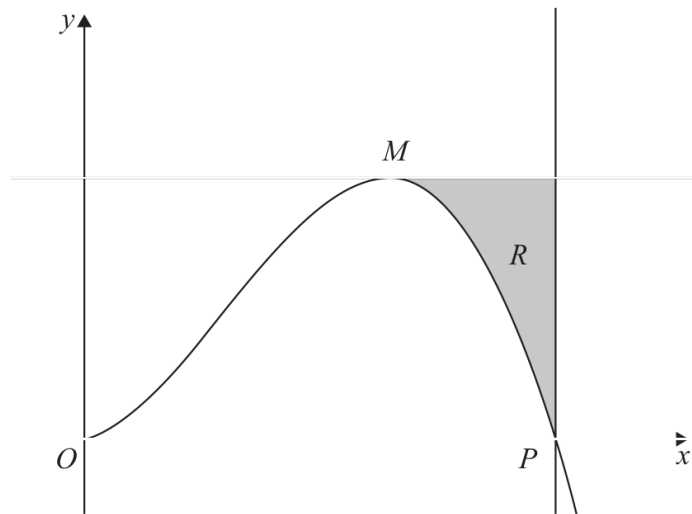
(Total for Question 8 is 8 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q9

P2 · TOPIC 12 12 marks

9.

**Figure 3**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

(a) Using calculus, find the coordinates of M .

(5)

The curve crosses the x -axis at the point P , as shown in Figure 3.

(b) Use algebra to find the x coordinate of P .

(2)

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line through M parallel to the x -axis and the line through P parallel to the y -axis.

(c) Use algebraic integration to find the area of R , giving your answer to one decimal place.

(5)

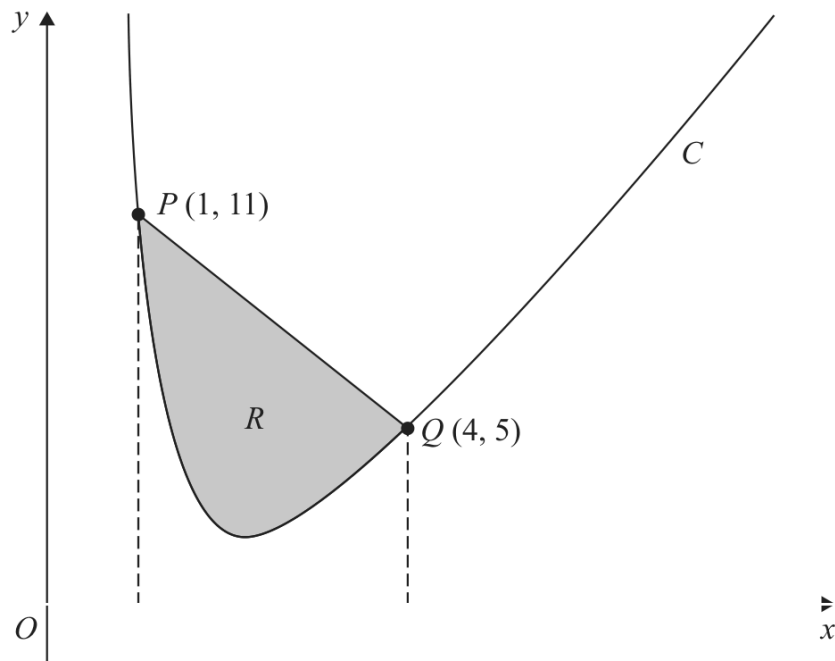
(Total for Question 9 is 12 marks)

Q20**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q3

P2 · TOPIC 12 5 marks

3:

**Figure 1**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 1 shows a sketch of the curve C with equation

$$y = 3x + \frac{16}{x^2} - 8 \quad x > 0$$

The points $P(1, 11)$ and $Q(4, 5)$ lie on C and are shown in Figure 1.

The region R , shown shaded in Figure 1, is bounded by C and line segment PQ .

Use algebraic integration to find the area of R .

(5)

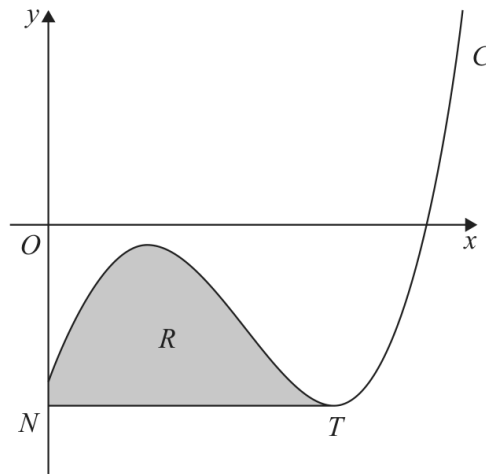
(Total for Question 3 is 5 marks)

Q21**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q7

P2 · TOPIC 12 10 marks

7.

**Figure 3**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of part of the curve C with equation

$$y = x^3 - 4x^{\frac{5}{2}} - kx^{\frac{1}{2}} + 28x - 44 \quad x \geq 0$$

where k is a positive constant.

(a) Find $\frac{dy}{dx}$ in simplest form.

(2)

The point T , shown in Figure 3, is a minimum stationary point on C .

Given that the x coordinate of T is 9

(b) show that $k = 6$

(2)

The line through T parallel to the x -axis meets the y -axis at the point N .

The finite region R , shown shaded in Figure 3, is bounded by C , the y -axis and the line segment NT .

(c) Use algebraic integration to find the area of R , giving the answer to 3 significant figures.

(6)

(Total for Question 7 is 10 marks)

8.

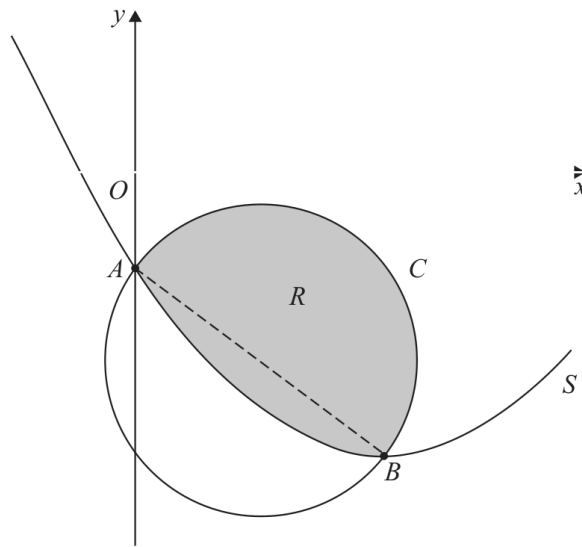


Figure 3

Figure 3 shows a sketch of a circle C and a parabola S .

The circle C has equation $x^2 + y^2 - 8x + 12y + 27 = 0$

(a) Find

- (i) the coordinates of the centre of C
- (ii) the radius of C

(3)

The parabola S

- has equation $y = ax^2 - \frac{3}{2}x - 3$
- intersects C on the y -axis at the point $A(0, -3)$
- intersects C again at the point B where AB is a diameter of C

Given that B is the minimum point of S ,

(b) show that $a = \frac{3}{32}$

(3)

The finite region R is bounded by S and C and is shown shaded in Figure 3.

(c) Use algebraic integration to find the exact area of R .

(7)

(Total for Question 8 is 13 marks)

03 The Trapezium Rule: Estimate, Then Transform

Find h , list every y value (complete the table first), and use the formula with the first and last y once and all the middle ones twice. Then use the answer: a constant multiple multiplies the answer, and $+k$ adds $k \times$ the width.

WHEN YOU SEE → YOU DO

“complete the table”

→ substitute each x — to the accuracy asked

“using all the y values”

→ h is the gap between the x values

$$\int_a^b (3f(x) - 2) dx$$

→ $3 \times \text{estimate} - 2(b - a)$

$\log_2(4x^2) = 2 + 2 \log_2 x$ **style**

→ split with log laws into the known integral

“overestimate or underestimate?”

→ curve bends up (convex) → over;
bends down → under

“how could the estimate be improved?”

→ more strips (smaller h)

WORKED MODEL · JANUARY 2023

$x = -1, -0.5, 0, 0.5, 1$; $y = 2.287, 4.470, 6.719, 7.291, 2.834$. Estimate $\int_{-1}^1 f(x) dx$.

1 $h = 0.5$

gap

2 $\frac{0.5}{2} \{2.287 + 2.834 + 2(4.470 + 6.719 + 7.291)\}$

ends once, middle twice

ANSWER ≈ 10.52 ; $\int_{-1}^1 \{f(x) - 2\} dx \approx 10.52 - 4 = 6.52$

× **TRAP** Using $n + 1$ for the number of strips. Five y values means four strips.

IDEA 03

Practise: The Trapezium Rule: Estimate, Then Transform

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Trapezium rule

P2 · TOPIC 12 3 marks

Use the trapezium rule with 4 strips to estimate $\int_0^2 \sqrt{1+x^2} dx$.

Q24 DR ESLAM Use the estimate

P2 · TOPIC 12 2 marks

Given $\int_0^2 f(x) dx \approx 5.4$, estimate $\int_0^2 \{3f(x) + 2\} dx$.

SELECTED PRACTICE

Q25

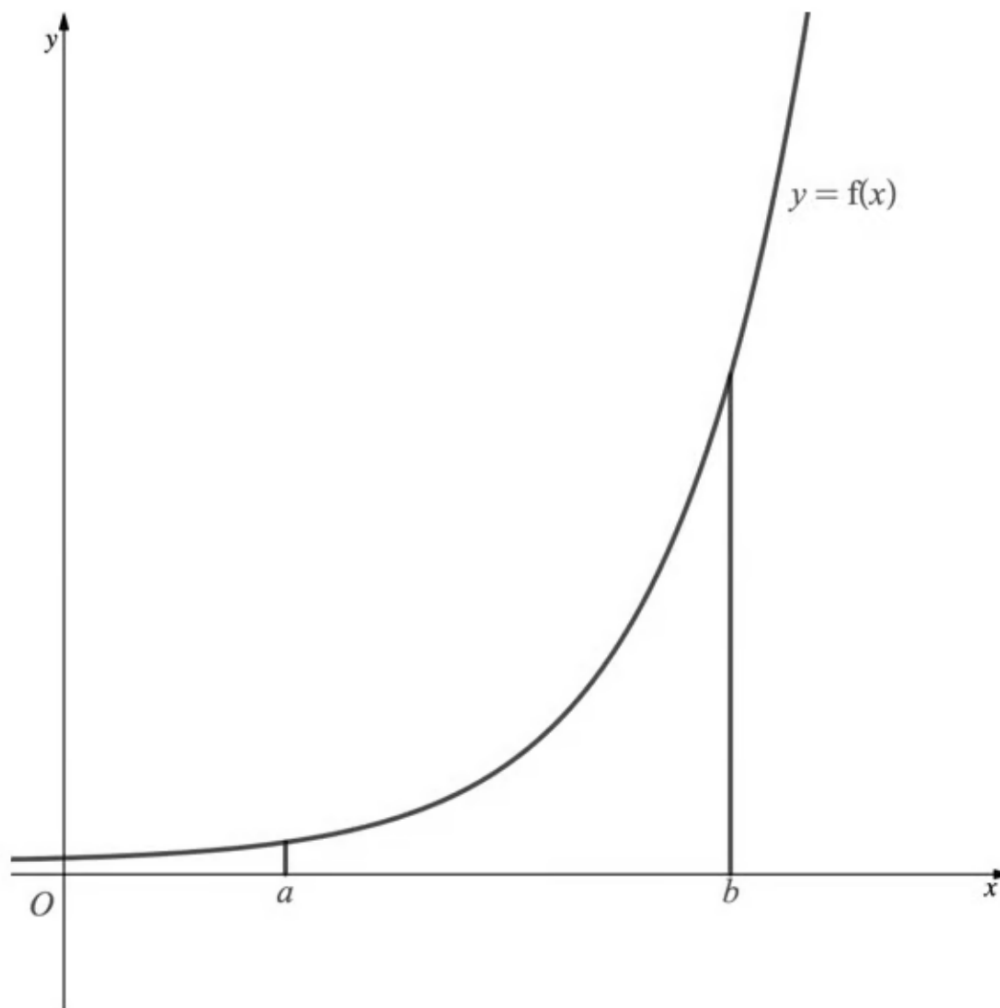
PRACTICE

Practice · Easy

P2 · TOPIC 12 2 marks

- 7 A student is estimating the area bounded by the curve $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$.

The student intends to estimate the area by using trapezia of equal width.



Add to the diagram above to show how the student can use 4 trapezia to estimate the area.

(2 marks)

Q26**PRACTICE**

Practice · Medium

P2 · TOPIC 12 8 marks**11 (a)** The trapezium rule is to be used to estimate the integral

$$\int_{-2}^0 x^3 + 8 \, dx$$

By completing the table of values below, use the trapezium rule to estimate the integral given above.

x	-2	-1.5	-1	-0.5	0
$y = x^3 + 8$					

(5 marks)

(b) i] By finding the exact value of the integral above, find the percentage error of your estimate from part [a].

ii] Explain how you could increase the accuracy of your trapezium rule estimate.

(3 marks)

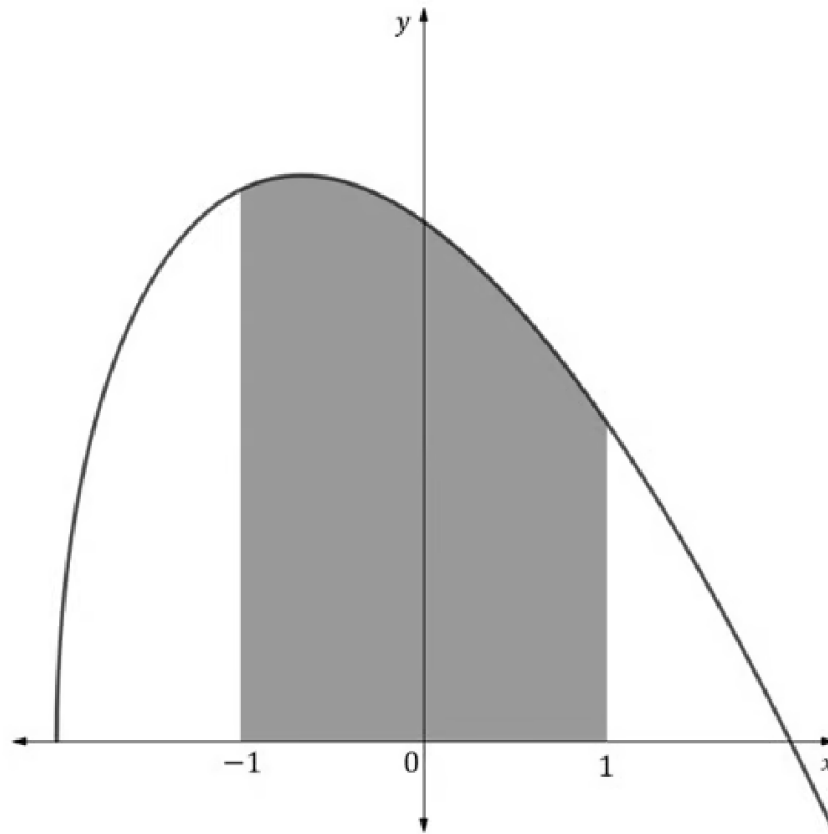
Q27**PRACTICE**

Practice · Very Hard

P2 · TOPIC 12 6 marks

9 (a) The diagram below shows the graph of

$$y = \frac{4 - x^2}{\sqrt{x + 2}}, \quad x > -2.$$



Use the trapezium rule with six strips to find an estimate of the integral

$$\int_{-1}^1 \left(\frac{4 - x^2}{\sqrt{x + 2}} \right) dx$$

to five significant figures.

(4 marks)

(b) By using your calculator to find the exact value of the integral to five significant figures, find the percentage error of your estimate from part [a].

(2 marks)

PAST PAPERS

Q28 PAST PAPER Pearson IAL WMA12 · October 2019 · Q5

P2 · TOPIC 12 7 marks

5. (a) Given
- $0 < a < 1$
- , sketch the curve with equation

$$y = a^x$$

showing the coordinates of the point at which the curve crosses the y-axis.

(2)

x	2	2.5	3	3.5	4
y	4.25	6.427	9.125	12.34	16.06

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$

The values of y are given to 4 significant figures as appropriate.

Using the trapezium rule with all the values of y in the given table,

- (b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x \right) dx$

(3)

Using your answer to part (b) and making your method clear, estimate

- (c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x \right) dx$

(2)

(Total 7 marks)

Q29**PAST PAPER**

Pearson IAL WMA12 · January 2020 · Q1

P2 · TOPIC 12 7 marks

1. The table below shows corresponding values of x and y for $y = \log_2(2x)$

The values of y are given to 2 decimal places as appropriate.

x	2	5	8	11	14
y	2	3.32	4	4.46	4.81

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for $\int_2^{14} \log_2(2x) dx$, giving your answer to one decimal place. (3)

Using your answer to part (a) and making your method clear, estimate

- (b) (i) $\int_2^{14} \frac{\log_2(4x^2)}{5} dx$
 (ii) $\int_2^{14} \log_2\left(\frac{2}{x}\right) dx$ (4)

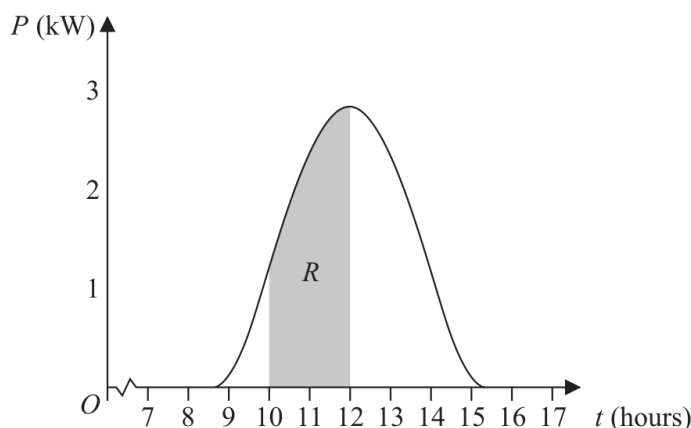
(Total 7 marks)

Q30**PAST PAPER**

Pearson IAL WMA12 · January 2021 · Q7

P2 · TOPIC 12**6 marks**

7.

**Figure 1**

Solar panels are installed on the roof of a building.

The power, P , produced on a particular day, in kW, can be modelled by the equation

$$P = 0.95 + 2^{t-12} + 2^{12-t} - (t-12)^2 \quad 8.5 \leq t \leq 15.2$$

where t is the time in hours after midnight. The graph of P against t is shown in Figure 1.

A table of values of t and P is shown below, with the values of P given to 4 significant figures where appropriate.

Time, t (hours)	10	10.5	11	11.5	12
Power, P (kW)		1.882	2.45		2.95

- (a) Use the given equation to complete the table, giving the values of P to 4 significant figures where appropriate.

(2)

The amount of energy, in kWh, produced between 10:00 and 12:00 can be found by calculating the area of region R , shown shaded in Figure 1.

- (b) Use the trapezium rule, with all the values of P in the completed table, to find an estimate for the amount of energy produced between 10:00 and 12:00. Give your answer to 2 decimal places.

(4)**(Total 6 marks)**

Q31 PAST PAPER Pearson IAL WMA12 · June 2021 · Q5

P2 · TOPIC 12 10 marks

5.

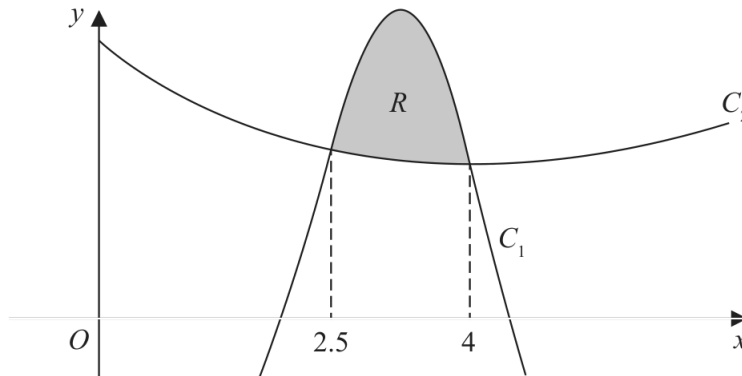
**Figure 2**

Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

- (a) find, to 2 decimal places, an estimate for the area bounded by the curve C_1 , the line with equation $x = 2.5$, the x -axis and the line with equation $x = 4$
- (4)

The curve C_2 has equation

$$y = x^{\frac{3}{2}} - 3x + 9 \quad x > 0$$

- (b) Find $\int \left(x^{\frac{3}{2}} - 3x + 9 \right) dx$
- (3)

The region R , shown shaded in Figure 2, is bounded by the curves C_1 and C_2

- (c) Use the answers to part (a) and part (b) to find, to one decimal place, an estimate for the area of the region R .
- (3)

(Total 10 marks)

Q32**PAST PAPER**

Pearson IAL WMA12 · October 2021 · Q3

P2 · TOPIC 12 8 marks

3.

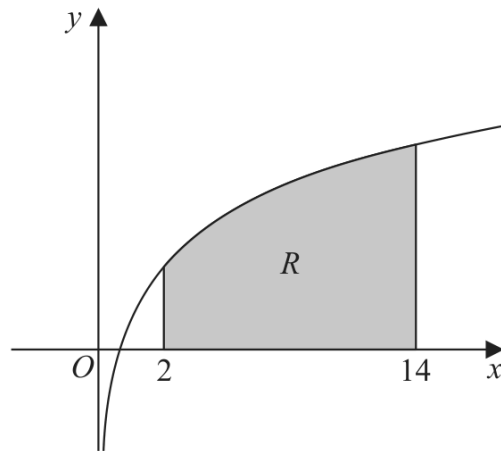
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

(a) show that the area of R is approximately 10.10 (3)

(b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R . (1)

(c) Using the answer to part (a) and making your method clear, estimate the value of

(i) $\int_2^{14} \log_{10} \sqrt{x} \, dx$

(ii) $\int_2^{14} \log_{10} 100x^3 \, dx$

(4)**(Total 8 marks)**

Q33**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q1

P2 · TOPIC 12 7 marks

1. The table below shows corresponding values of x and y for

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for

$$\int_5^7 2^{5-\sqrt{x}} dx$$

giving your answer to 2 decimal places.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

(i) $\int_5^7 2^{6-\sqrt{x}} dx$

(ii) $\int_5^7 (3 + 2^{5-\sqrt{x}}) dx$

(4)

(Total 7 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June 2022 · Q2

P2 · TOPIC 12 8 marks

2.

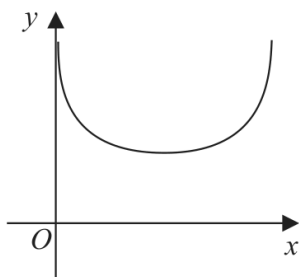
**Figure 1**

Figure 1 shows the graph of

$$y = 1 - \log_{10}(\sin x) \quad 0 < x < \pi$$

where x is in radians.The table below shows some values of x and y for this graph, with values of y given to 3 decimal places.

x	0.5	1	1.5	2	2.5	3
y	1.319		1.001		1.223	1.850

(a) Complete the table above, giving values of y to 3 decimal places.

(2)

(b) Use the trapezium rule with all the y values in the completed table to find, to 2 decimal places, an estimate for

$$\int_{0.5}^3 (1 - \log_{10}(\sin x)) dx$$

(3)

(c) Use your answer to part (b) to find an estimate for

$$\int_{0.5}^3 (3 + \log_{10}(\sin x)) dx$$

(3)

(Total 8 marks)

Q35 PAST PAPER Pearson IAL WMA12 · October 2022 · Q6

P2 · TOPIC 12 7 marks

6. The curve C_1 has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below, with the y values rounded to 4 decimal places where appropriate.

x	0	0.5	1	1.5	2
y	3	2.6833	2.4	2.1466	1.92

- (a) Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_0^2 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

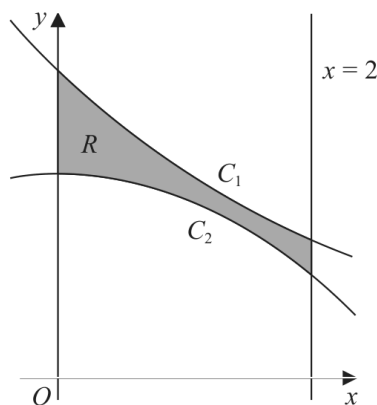


Figure 1

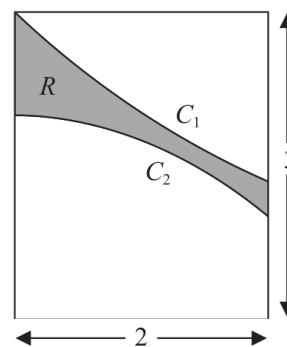


Figure 2

The region R , shown shaded in Figure 1, is bounded by

- the curve C_1
- the curve C_2 with equation $y = 2 - \frac{1}{4}x^2$
- the line with equation $x = 2$
- the y -axis

The region R forms part of the design for a logo shown in Figure 2.

The design consists of the shaded region R inside a rectangle of width 2 and height 3

Using calculus and the answer to part (a),

- (b) calculate an estimate for the percentage of the logo which is shaded.

(4)

(Total for Question 6 is 7 marks)

Q36**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q1

P2 · TOPIC 12 6 marks

1.

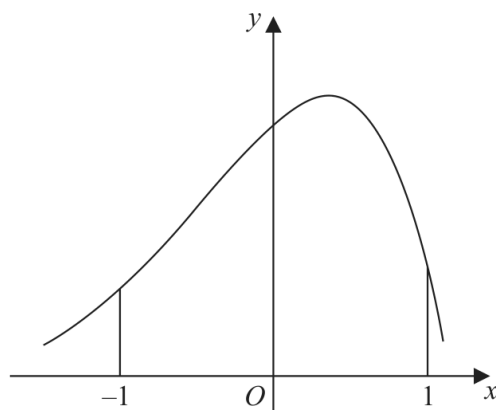
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$

The table below shows some corresponding values of x and y for this curve.

The values of y are given to 3 decimal places.

x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_{-1}^1 f(x) \, dx$$

giving your answer to 2 decimal places.

(3)

(b) Use your answer to part (a) to estimate

(i) $\int_{-1}^1 (f(x) - 2) \, dx$

(ii) $\int_1^3 f(x-2) \, dx$

(3)

(Total for Question 1 is 6 marks)

Q37 PAST PAPER Pearson IAL WMA12 · June 2023 · Q1

P2 · TOPIC 12 3 marks

1. The continuous curve C has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	4.0	4.2	4.4	4.6	4.8	5.0
y	9.2	8.4556	3.8512	5.0342	7.8297	8.6

Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_4^5 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

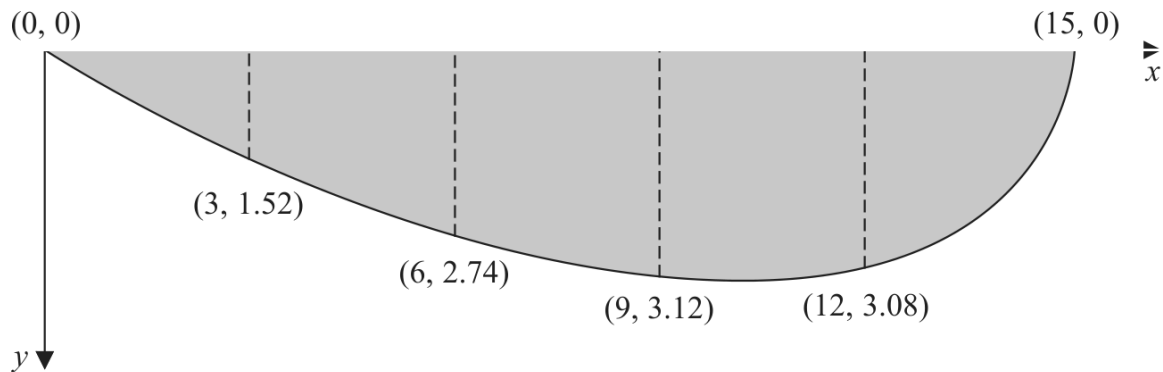
(Total for Question 1 is 3 marks)

Q38**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q6

P2 · TOPIC 12**6 marks**

6.

**Figure 1**

A river is being studied.

At one particular place, the river is 15 m wide.

The depth, y metres, of the river is measured at a point x metres from one side of the river.

Figure 1 shows a plot of the cross-section of the river and the coordinate values (x, y)

- (a) Use the trapezium rule with all the y values given in Figure 1 to estimate the cross-sectional area of the river.

(3)

The water in the river is modelled as flowing at a constant speed of 1.5 m s^{-1} across the whole of the cross-section.

- (b) Use the model and the answer to part (a) to estimate the volume of water flowing through this section of the river each minute, giving your answer in m^3 to 2 significant figures.

(2)

Assuming the model,

- (c) state, giving a reason for your answer, whether your answer for part (b) is an overestimate or an underestimate of the true volume of water flowing through this section of the river each minute.

(1)

(Total for Question 6 is 6 marks)

Q39**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q4

P2 · TOPIC 12 9 marks

4. (a) Sketch the curve with equation

$$y = a^{-x} + 4$$

where a is a constant and $a > 1$

On your sketch show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote to the curve.

(3)

x	-4	-1.5	1	3.5	6	8.5
y	13	6.280	4.577	4.146	4.037	4.009

The table above shows corresponding values of x and y for $y = 3^{-\frac{1}{2}x} + 4$

The values of y are given to four significant figures, as appropriate.

Using the trapezium rule with all the values of y in the table,

- (b) find an approximate value for

$$\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx$$

giving your answer to two significant figures.

(3)

- (c) Using the answer to part (b), find an approximate value for

(i) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} \right) dx$

(ii) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx + \int_{-8.5}^4 \left(3^{\frac{1}{2}x} + 4 \right) dx$

(3)

(Total for Question 4 is 9 marks)

Q40**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q6

P2 · TOPIC 12**9 marks**

6. (a) Sketch the curve with equation

$$y = a^x + 4$$

where a is a positive constant greater than 1

On your sketch, show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote of the curve

(3)

x	2	2.3	2.6	2.9	3.2	3.5
y	0	0.3246	0.8629	1.6643	2.7896	4.3137

The table shows corresponding values of x and y for

$$y = 2^x - 2x$$

with the values of y given to 4 decimal places as appropriate.Using the trapezium rule with all the values of y in the given table,(b) obtain an estimate for $\int_2^{3.5} (2^x - 2x) dx$, giving your answer to 2 decimal places.**(3)**

(c) Using your answer to part (b) and making your method clear, estimate

(i) $\int_2^{3.5} (2^x + 2x) dx$

(ii) $\int_2^{3.5} (2^{x+1} - 4x) dx$

(3)**(Total for Question 6 is 9 marks)**

Q41**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q2

P2 · TOPIC 12 5 marks

2. The table shows corresponding values of x and y for a continuous curve with equation $y = f(x)$ between $x = -4$ and $x = 5$, where a is a constant.

x	-4	-2.5	-1	0.5	2	3.5	5
y	4.16	2.91	a	1.73	1.37	1.43	2.28

The trapezium rule is used with all the y values in the table to find an approximation for

$$\int_{-4}^5 f(x) \, dx$$

Given that the value of this approximation is 19.3

- (a) find the value of the constant a to 3 significant figures.

(3)

- (b) Use the given answer of 19.3 to find an approximate value for

$$\int_{-4}^5 (2f(x) - 3) \, dx$$

(2)

(Total for Question 2 is 5 marks)

Q42**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q1

P2 · TOPIC 12 5 marks

1. A continuous curve has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	0.5	1.75	3	4.25	5.5
y	3.479	6.101	7.448	6.823	5.182

Using the trapezium rule with all the values of y in the given table,

- (a) find an estimate for

$$\int_{0.5}^{5.5} f(x) \, dx$$

giving your answer to one decimal place.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

$$\int_{0.5}^{5.5} (f(x) + 4x) \, dx$$

(2)

(Total for Question 1 is 5 marks)

Q43**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q5

P2 · TOPIC 12 6 marks**5:**

x	-2	-0.5	1	2.5	4	5.5	7
y	12	4.243	1.5	0.530	0.188	0.066	0.023

The table above shows corresponding values of x and y for

$$y = 3\left(\frac{1}{2}\right)^x$$

The values of y are given to 3 decimal places as appropriate.

- (a) Using the trapezium rule with all the values of y in the given table, obtain an estimate for

$$\int_{-2}^7 3\left(\frac{1}{2}\right)^x dx$$

giving the answer to one decimal place.

(3)

Using the answer to part (a) and making your method clear, estimate

(b) (i) $\int_{-2}^7 3\left(\frac{1}{2}\right)^{x+2} dx$

(ii) $\int_{-2}^7 (2^{-x} + 2x) dx$

(3)

(Total for Question 5 is 6 marks)

Q44 PAST PAPER Pearson IAL WMA12 · October 2025 · Q3

P2 · TOPIC 12 7 marks

3.

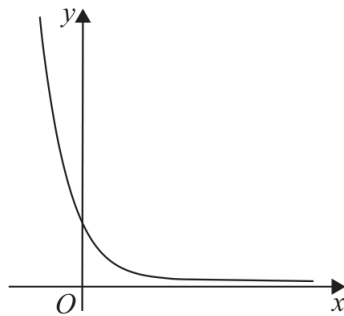


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve with equation $y = 3 \times 2^{-x}$

The point $P(k, 300\,000)$ lies on the curve.

(a) Use logarithms to find the value of k to 2 decimal places.

(2)

x	-0.5	1	2.5	4.0	5.5	7
y	4.243	1.5	0.530	0.188	0.066	0.023

The table shows corresponding values of x and y for $y = 3 \times 2^{-x}$

The values of y are given to 3 decimal places where appropriate.

(b) (i) Use the trapezium rule, with all the values of y from the table, to find an approximate value, to 2 decimal places, for

$$\int_{-0.5}^7 3 \times 2^{-x} \, dx$$

(3)

(ii) Use your answer to part (b)(i) to estimate

$$\int_{-0.5}^7 2^{-x} \, dx + \int_{-7}^{0.5} 2^x \, dx$$

(2)

(Total for Question 3 is 7 marks)

Q45**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q2

P2 · TOPIC 12**6 marks**

2.

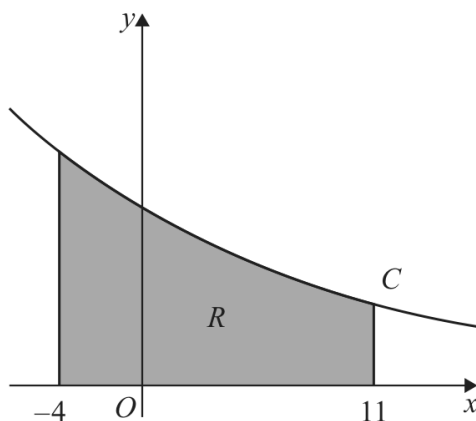
**Figure 1**

Figure 1 shows a sketch of part of the curve C with equation $y = 2^{-0.1x}$

The table below shows corresponding values of x and y for $y = 2^{-0.1x}$

The y values are given to 4 decimal places.

x	-4	-1	2	5	8	11
y	1.3195	1.0718	0.8706	0.7071	0.5743	0.4665

The region R , shown shaded in Figure 1, is bounded by C , the x -axis and the lines with equations $x = -4$ and $x = 11$

- (a) Use the trapezium rule with all the values of y in the table to find an estimate for the area of R . Give the answer to 2 decimal places.

(3)

- (b) State how you would use the trapezium rule to get a more accurate estimate for the true area of R .

(1)

Using the answer to part (a) and showing your working,

- (c) estimate the value of

$$\int_{-4}^{11} 2^{3-0.1x} dx$$

(2)

(Total for Question 2 is 6 marks)

Q46**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q5

P2 · TOPIC 12 7 marks

5. The table below shows values of x and y for the curve C with equation $y = f(x)$.

x	0.9	1.5	2.1	2.7	3.3	3.9	4.5	5.1
y	1.5	0.48	0.30	0.26	0.26	0.30	0.48	1.5

- (a) Using the trapezium rule with all the y values in the table, find an estimate for the area of the region bounded by C and the x -axis between the lines with equations $x = 0.9$ and $x = 5.1$

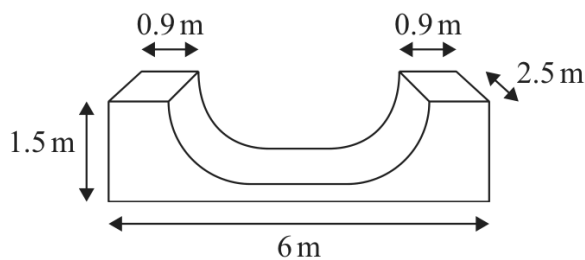
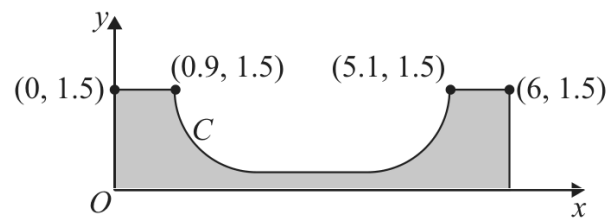
(3)**Figure 1****Figure 2**

Figure 1 shows a sketch of a solid concrete skateboard ramp.

Figure 2 shows a graph of the cross-section of the same ramp.

The ramp

- has a total length of 6 m
- has a platform at each end at a height of 1.5 m and with a length of 0.9 m
- has a curved section C with equation $y = f(x)$, where x and y are in metres
- has a uniform cross-section and is 2.5 m wide

as shown in Figure 1 and/or Figure 2.

- (b) Using the answer to part (a), find an estimate for the volume of concrete in the **whole** of the ramp.

(3)

- (c) State, giving a reason for your answer, whether the answer for part (b) is an overestimate or an underestimate of the true volume of concrete in the whole of the skateboard ramp.

(1)

(Total for Question 5 is 7 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|--|
| 01 | Definite integrals | Rewrite, integrate, top – bottom; unknown → equation. |
| 02 | Areas | Meeting points; top – bottom; below the axis is negative. |
| 03 | Trapezium rule | $\frac{h}{2}\{\text{ends} + 2(\text{middle})\}$; convex → over. |

BOTTOM LINE Sketch first • limits from the meeting points • no formula → trapezium rule.

Answers

Integration: Areas and the Trapezium Rule · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · DEFINITE INTEGRALS: LIMITS AND UNKNOWN CONSTANTS

- Q1** Dr Eslam · Evaluate $\left[2x^{\frac{3}{2}} + \frac{2}{x}\right]_1^4 = (16 + \frac{1}{2}) - (2 + 2) = 12.5$
- Q2** Dr Eslam · Find the limit $k^2 + k - 2 = 18 \Rightarrow (k + 5)(k - 4) = 0: k = 4$
- Q3** Practice · Easy (i) 6 (ii) 99
- Q4** Practice · Hard $\int_4^9 (x^{\frac{3}{2}} + x^{-\frac{1}{2}}) dx = \left[\frac{2}{5}x^{\frac{5}{2}} + 2x^{\frac{1}{2}}\right]_4^9 = \frac{432}{5}$
- Q5** Practice · Very Hard $2x^{\frac{5}{2}}: 2q^{\frac{5}{2}}(32 - 1) = 15066 \Rightarrow q^{\frac{5}{2}} = 243: q = 9$
- Q6** Pearson IAL WMA12 · June 2022 · Q7 $A = \frac{1}{2}, B = -28$
- Q7** Pearson IAL WMA12 · January 2026 · Q4 $k = \frac{1}{2} \text{ or } k = 3$

IDEA 02 · AREAS: CURVES, LINES AND BELOW THE AXIS

- Q8** Dr Eslam · Curve and line meet at $x = 1, 4$: $\int_1^4 \{(x+2) - (x^2 - 4x + 6)\} dx = \frac{9}{2}$
- Q9** Dr Eslam · Below the axis $\int_{-2}^2 (x^2 - 4) dx = -\frac{32}{3}$: area $\frac{32}{3}$
- Q10** Practice · Easy (a) $\int_1^2 (9 - x^2) dx$ (b) $\frac{20}{3}$
- Q11** Practice · Hard $\int_0^2 (2x + 3x^2 - 2x^3) dx = 4$
- Q12** Practice · Very Hard meet at $x = 4$ and 9 : $\int_4^9 (-2x^2 + 26x - 72) dx = \frac{125}{3}$
- Q13** Pearson IAL WMA12 · October 2019 · Q8 (i) $4\sqrt{2} - \frac{37}{8}$. (ii) $k = \frac{29}{18}$
- Q14** Pearson IAL WMA12 · January 2020 · Q4 $\frac{20\sqrt{5}}{3}$
- Q15** Pearson IAL WMA12 · January 2022 · Q9 (a) $P = (1-m, m(1-m))$. (b) Proven: $\frac{(1-m)^3}{6}$.
(c) $m = 1 - \frac{1}{\sqrt[3]{2}}$
- Q16** Pearson IAL WMA12 · January 2023 · Q9 (a) $D = (0, 5)$. (b) $x_E = 1, x_F = 3$. (c) $k = \frac{8}{19}$
- Q17** Pearson IAL WMA12 · June 2023 · Q10 (a) $a = 6, b = -84$. (b) $k = 4$. (c) Area $= \frac{1683}{10}$
- Q18** Pearson IAL WMA12 · October 2024 · Q8 (a) $x_P = 1, x_Q = 3$. (b) $\frac{16}{3}$
- Q19** Pearson IAL WMA12 · January 2025 · Q9 (a) $M\left(16, \frac{2304}{5}\right)$. (b) $x_P = 25$. (c) 1312.7
- Q20** Pearson IAL WMA12 · June 2025 · Q3 $\frac{27}{2}$
- Q21** Pearson IAL WMA12 · October 2025 · Q7 (a) $3x^2 - 10x^{3/2} - \frac{k}{2\sqrt{x}} + 28$. (b) Proven. (c) 248
- Q22** Pearson IAL WMA12 · June 2026 · Q8 (a)(i) $(4, -6)$ (ii) 5 (b) $B(8, -9)$ is the minimum:
 $\frac{dy}{dx} = 16a - \frac{3}{2} = 0$ (c) $\frac{25\pi}{2} + 8$

IDEA 03 · THE TRAPEZIUM RULE: ESTIMATE, THEN TRANSFORM

- Q23** Dr Eslam · Trapezium rule
 $h = 0.5$: $y = 1, 1.118, 1.414, 1.803, 2.236$:
 ≈ 2.98
 $3(5.4) + 2(2) = 20.2$
- Q24** Dr Eslam · Use the estimate
 four trapezia of equal width between a and b , tops joining the curve
- Q25** Practice · Easy
 $(a) y = 0, 4.625, 7, 7.875, 8$: 11.75 (b)(i) exact 12: error 2.08% (ii) more (narrower) strips
- Q26** Practice · Medium
 $(a) 5.3364$ (b) exact 5.3544: error $\approx 0.336\%$
- Q27** Practice · Very Hard
 (a) A decreasing exponential curve crossing the y -axis at $(0, 1)$. (b) 19.0. (c) 1.0
- Q28** Pearson IAL WMA12 · October 2019 · Q5
 $(a) 45.6$. (b)(i) 18.2. (b)(ii) -21.6
- Q29** Pearson IAL WMA12 · January 2020 · Q1
 $(a) P(10) = 1.20, P(11.5) = 2.821$. (b) 4.61 kWh
- Q30** Pearson IAL WMA12 · January 2021 · Q7
 $(a) 12.33$. (b) $\frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x + c$. (c) 4.6
- Q31** Pearson IAL WMA12 · June 2021 · Q5
 $(a) 10.10$. (b) Use more strips. (c)(i) 5.05. (c)(ii) 54.30
- Q32** Pearson IAL WMA12 · October 2021 · Q3
 $(a) 11.79$. (b)(i) 23.57. (b)(ii) 17.79
- Q33** Pearson IAL WMA12 · January 2022 · Q1
 $(a) y(1) = 1.075, y(2) = 1.041$. (b) 2.96. (c) 7.04
- Q34** Pearson IAL WMA12 · June 2022 · Q2
 $(a) 4.845$. (b) 25.2%
- Q35** Pearson IAL WMA12 · October 2022 · Q6
 $(a) 10.52$. (b)(i) 6.52. (b)(ii) 10.52
- Q36** Pearson IAL WMA12 · January 2023 · Q1
 6.814
- Q37** Pearson IAL WMA12 · June 2023 · Q1
 $(a) 31.38 \text{ m}^2$. (b) $2.8 \times 10^3 \text{ m}^3$. (c) Under-estimate
- Q38** Pearson IAL WMA12 · October 2023 · Q6
 $(a) y$ -intercept $(0, 5)$ and asymptote $y = 4$.